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Robust Ischemic Lesion Segmentation in Neonatal Brain MRI

Cross-Cohort Generalization and Weak Supervision
for Hypoxic-Ischemic Encephalopathy

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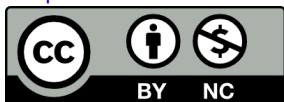


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Abstract

Neonatal Hypoxic-Ischemic Encephalopathy (HIE) affects 1–3 per 1000 term newborns and is a major cause of neonatal mortality and long-term neurological disability. Early, accurate lesion localization lets clinicians anticipate likely symptoms and begin interventional therapy before developmental problems become visible. The aim of this thesis is to build a robust automated segmentation pipeline that localizes HIE lesions across the variation found in real clinical practice. This is challenging because lesions are small, diffuse, and multifocal; variability across scanners, protocols, and treatment limits cross-cohort generalization; regional differences in normal diffusion complicate the interpretation of raw diffusion values; and annotated data is scarce, slow to produce, and subject to inter-annotator disagreement.

These four challenges are addressed as follows. Robust baseline models, tuned through cross-validation, segment the full range of lesions. These models then transfer frozen pretrained weights to new scanner and domain settings for cross-cohort generalization. Regional diffusion is made interpretable through a per-patient normative reference of healthy tissue, derived from the patient’s own data, against which each voxel’s deviation is quantified. Label scarcity is addressed through a human-in-the-loop scheme that enables weak-supervision based segmentation. The approaches are benchmarked against the agreement between an experienced and an inexperienced rater, giving a clinically grounded reference.

Performance varied with lesion type and cohort homogeneity, and the pretrained baseline transferred across sites with a benefit that narrowed as cohorts diverged. The normative prior raised segmentation well above raw diffusion, increasing mean Dice from 0.64 to 0.72 under both full supervision and weak supervision. The median Dice of 0.73 was substantially higher than the cohort’s inter-rater agreement of 0.56, and the model’s agreement with the experienced annotator was higher than the inexperienced rater’s agreement with the same annotator, under both full and weak supervision.

In conclusion, this thesis addresses each of the four challenges. Strong baselines handled the full range of lesions, pretrained transfer extended segmentation to new cohorts, the normative prior accounted for regional diffusion variation, and weak supervision reduced the annotation burden while approaching full-supervision performance with substantially less effort. Weak supervision in particular is a promising direction for the field. Because these findings rest on small cohorts, future work should validate the methods on larger datasets. Furthermore, the weak-annotation interface should be optimized in collaboration with clinical users.

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List of Abbreviations

HII	Hypoxic-ischemic injury
HIE	Hypoxic-ischemic encephalopathy
TH	Therapeutic hypothermia
DWI	Diffusion-weighted imaging
ADC	Apparent diffusion coefficient
CV	Cross-Validation
FP	False Positive
TP	True Positive
FN	False Negative
TN	True Negative
GT	Ground Truth
SOTA	State-of-the-art
IQR	Interquartile Range

Introduction

1.1 Introduction

Hypoxic–ischemic injury (HII) to the brain is a major cause of neonatal mortality and long-term neurological disability [13]. The resulting clinical syndrome is known as hypoxic–ischemic encephalopathy (HIE). Studies estimate that approximately 1–3 per 1000 full-term newborns experience perinatal asphyxia leading to HII [19, 21]. Historically, among affected infants, 15–20% died during the early post-partum period, while an additional 25% developed permanent neuropsychological impairments [39]. Since then, a major advance in neonatal care has been the successful incorporation of therapeutic hypothermia (TH) into the standard care of perinatal asphyxia. [34]. Clinical evidence demonstrates that TH significantly improves survival without disability into infancy and childhood, while also reducing the risk of the most severe neuropsychological impairments, such as cerebral palsy [42]. Since the introduction of TH as standard care for moderate-to-severe HIE in infants ≥ 36 weeks gestational age, mortality has fallen by more than half, from around 27% in the non-cooled cases of the TOBY trial [1] to 10.9% in more recent population data [34].

Long-term neurological complications range from mild cognitive and behavioral deficits to severe outcomes, including epilepsy, cerebral palsy, and profound intellectual disability [19]. As such, early and accurate lesion localization and quantification are critical for treatment planning and prognosis.

Magnetic resonance imaging (MRI) plays a central role in the assessment of these injuries. More specifically, diffusion-weighted imaging (DWI) is widely used to identify lesion location. It is an MRI sequence that highlights variations in water diffusion, which becomes restricted in tissue affected by HII. DWI is usually interpreted together with apparent diffusion coefficient (ADC) maps, which are computed by fitting the DWI signal decay across multiple diffusion weightings (b-values) to yield a quantitative map of local water diffusivity. In ADC maps, injured tissue appears as regions of reduced ADC. The two are complementary and can highlight different regions, as lesions appear bright on DWI while corresponding to low signal on the ADC map. Imaging findings on DWI and ADC depend on multiple factors, including brain maturity, the duration and severity of the hypoxic–ischemic insult, and the timing and modality of imaging acquisition [13].

Accurate and early identification of lesions in infants with HII is essential for predicting long-term outcomes [13]. As such, the development of efficient and reliable automated techniques is crucial. However, lesions resulting from HIE are often diffuse and small, making full volumetric segmentation both technically challenging and time-intensive. In addition, manual annotations often exhibit substantial inter-rater variability, which is reflected by relatively low inter-observer Dice similarity coefficients [11].

Additionally, algorithms that perform well on larger, focal

pathologies such as brain tumors or acute stroke often perform poorly when applied directly to MRI scans of HIE patients [4]. Even architectures specifically optimized for the only publicly available fully annotated HIE dataset [4] struggle to exceed Dice scores of 0.6 on held-out test data. These findings underscore the intrinsic difficulty of HIE lesion segmentation.

Concretely, automated HIE lesion segmentation faces several, connected challenges. Lesions are small, diffuse, and multifocal, often less than 1% of the brain volume in modern TH-treated patient cohorts, yet frequently much larger in older cohorts or severely affected patients. This wide range of lesion burden and location means a model must learn the challenging task of detecting both subtle, scattered lesions and large, widespread injuries, which is especially difficult given the limited data available.

Furthermore, normal diffusion values vary substantially across brain regions, so lesions often show limited intensity separation from healthy tissue and raw ADC/DWI intensities cannot be interpreted in isolation. As such, a region-aware reference is needed to distinguish genuine injury from normal regional variation. Normative ADC atlases address this by capturing the voxel-wise mean and standard deviation of healthy diffusion, against which a patient’s ADC can be scored. The resulting per-patient Z_{ADC} maps quantify, at each voxel, how many standard deviations the patient’s ADC deviates from the normative mean [28].

Also, differences in scanners, acquisition protocols, and treatment (e.g., with or without TH) introduce substantial variability between datasets. Particularly problematic for the use of ADC is that normative ADC values are scanner-dependent and can vary across hospitals. As a result, models trained on a single cohort can struggle to generalize to data acquired elsewhere.

Together, these factors make not only the training of robust, generalizable models challenging, but also the manual annotation needed to build large, representative datasets. Lesion location and boundaries are ambiguous and slow to annotate, often leading to high inter-annotator disagreement. As a result, fully annotated datasets are scarce and often inconsistent, with only one public dataset currently offering per-patient voxel-level annotations [4]. This scarcity limits both the training and evaluation of robust HIE segmentation models. For this reason, building a model and pipeline that produces reliable priors from sparse inputs could be a crucial next step for the field: by turning annotation into a correction task, it could lower the barrier to building new annotated datasets and support clinical studies, trials, and eventually clinical use.

This thesis addresses these challenges and makes the following three contributions:

1. We develop a pretrained segmentation baseline that serves as the foundation for adaptation to new cohorts. It is the basis for our weak-supervision approach, which

collapses without it, and additionally provides a competitive, data-efficient method for full-supervision transfer across scanners and sites. We report direct-inference, from-scratch, and fine-tuned segmentation-performance, finding all three to be valuable strategies for cross-scanner and cross-domain transfer.

2. We introduce a pipeline for constructing pseudo-normative spatial priors (Z_{ADC} maps), for both fully supervised and weakly supervised settings, that quantify per-voxel deviation from healthy diffusion without requiring a separate healthy cohort. The resulting region-aware prior greatly improves segmentation performance over using raw ADC alone.
3. We extend the pipeline to weak supervision through a human-in-the-loop annotation scheme (clicks and lassos), showing that it surpasses naive thresholding and approaches full-supervision performance while substantially reducing annotation effort.

Our contributions are guided by a single overarching research question:

How can we develop a segmentation pipeline that achieves robust HIE lesion segmentation across domains and scanners, while reducing the need for time-intensive full manual annotation?

This question is addressed through three sub-questions:

- RQ1** *Can a pretrained baseline, used to initialize training on a new cohort, achieve robust cross-scanner and cross-domain segmentation, and how does it compare to direct-inference and from-scratch training?*
- RQ2** *Can we build cohort-level normative atlases, and corresponding per-patient pseudo-normative spatial priors, without an external healthy cohort, that provide a reliable region-aware prior?*
- RQ3** *Can weak supervision, provided through a human-in-the-loop scheme of clicks and lassos, approach full-supervision performance and improve significantly over naive thresholding?*

The remainder of this thesis is organized as follows. Chapter 2 reviews the clinical background of neonatal HIE, the role of MRI imaging, core related work on automated lesion segmentation in HIE, including background on normative ADC atlases and Z_{ADC} maps, and a background on weakly-supervised segmentation methods. Chapter 3 describes the three cohorts used in this research (D3T-B, D1.5T-B, D1.5T-U) and their differences in acquisition and lesion characteristics. Chapter 4 details the segmentation models used, the pipeline for constructing per-patient pseudo-normative spatial priors, and the training and evaluation setup, including the background of the weak-supervision experiments. Chapter 5 reports the results across the baseline, cross-scanner, cross-domain, and weak-supervision experiments. Chapter 6 then interprets these results, providing an analysis of their meaning and concluding with limitations and possible future work. Chapter 7 concludes the research by answering the research questions.

Related Work

This chapter is divided into four parts. Section 2.1 covers the clinical background of neonatal HIE. It discusses how the injury develops, the role of TH, and why lesion location and burden matter for prognosis. Section 2.2 covers the neuroimaging background. It explains the role of MRI in neonatal HIE, giving background on the DWI and ADC maps used to identify injury. Furthermore, it highlights the variability in acquisition timing, field strength, and protocol that makes cross-scanner analysis harder. Section 2.3 covers existing automated lesion segmentation background for neonatal HIE. Here, existing fully supervised methods and the BONBID-HIE benchmark are reviewed, alongside background on the normative ADC atlases and corresponding pseudo-normative spatial priors that provide a region-aware reference. Section 2.4 covers weak supervision, reviewing methods from related fields that are used to train segmentation models, since there is no prior work in HIE.

2.1 Neonatal HIE

Neonatal HIE is a brain injury caused by reduced oxygen and glucose supply to the brain during the perinatal period [21, 33]. It is associated with high mortality and long-term neurological impairments, including cerebral palsy, epilepsy, and cognitive deficits [19, 33]. Following the initial HII, brain injury evolves over time, beginning with metabolic disturbances and primary neuronal death. A delayed phase of secondary energy failure can occur hours to days later, characterized by excitotoxicity, oxidative stress, and inflammatory processes that contribute to further neuronal injury [9, 21].

Injury patterns are characterized by the location and extent of brain lesions. These patterns are clinically relevant because the exact location and severity are associated with specific neurological outcomes, and they vary with brain maturity [25].

Therapeutic hypothermia is the standard treatment for moderate to severe HIE and reduces secondary brain injury [9, 21, 33, 42]. It acts by limiting secondary energy failure, delayed neuronal death, and inflammatory injury, and is typically maintained for approximately 72 hours [9, 42]. By limiting these secondary injury processes, TH reduces the resulting tissue damage: in a sub-study of the TOBY trial, cooled infants had fewer lesions in the basal ganglia, thalamus, and white matter, and were more likely to have normal scans than non-cooled infants [31]. The same effect is seen in modern cohorts, where routine, early cooling has reduced the presence of the most severe lesions [37]. This reduction in lesion burden translates into clinical benefits, with trials showing that TH lowers the risk of death or major neuro-developmental disability and increases survival with normal neurological function in infants with moderate to severe HIE [36, 42].

2.2 MRI in neonatal HIE

Magnetic resonance imaging (MRI) is the primary modality for assessing brain injury in neonatal HIE due to its high sensitivity to structural and metabolic abnormalities [3]. MRI enables localization and characterization of injury patterns and is widely used in clinical trials to evaluate treatment effects and neurological outcomes [4, 43]. Consequently, MRI-derived biomarkers are commonly used for both clinical prognosis and computational analysis of HIE-related brain injury.

2.2.1 Diffusion-weighted imaging

As introduced in Chapter 1, DWI captures local water diffusion by acquiring images at different diffusion weightings (b -values, in s/mm^2). ADC maps can be derived from this signal. Diffusion is measured by comparing a non-diffusion-weighted image ($b = 0$) with a diffusion-weighted image ($b > 0$). At each voxel, the ADC is computed from the logarithmic ratio of the two, yielding a quantitative measure of water diffusion [17, 32]. Therefore, ADC is derived from DWI rather than being an independent sequence.

In patients who have suffered HII, cytotoxic edema restricts water diffusion, producing hyperintensity on DWI and correspondingly reduced values on ADC maps [3]. ADC maps provide a quantitative representation of diffusion. Consequently, they are commonly used for lesion identification and automated segmentation [4, 26], with radiologists typically interpreting them jointly with DWI to identify regions of abnormal diffusion.

2.2.2 Acquisition timing, field strength, and protocol variability

MRI appearance in neonatal HIE depends strongly on the timing of image acquisition. Diffusion abnormalities may not be detectable within the first 24 hours after injury and can pseudonormalize after approximately 10–12 days [3]. As observed in T1w-, T2w-, DWI-, and ADC images, HIE-related abnormalities evolve rapidly and overlap with normal neonatal brain maturation, which complicates lesion detection and expert annotation [3, 43].

MRI signal intensities are scanner-dependent and expressed in arbitrary units. Although ADC maps provide quantitative diffusion measurements (mm^2/s), measured ADC values can still vary across scanners and acquisition protocols due to differences in hardware, sequence implementation, and noise characteristics [12]. Despite this variability, little work has explicitly investigated cross-scanner training or generalization for automated HIE lesion segmentation.

2.3 Automated lesion segmentation in neonatal HIE

2.3.1 Fully supervised segmentation methods

BONBID-HIE is currently the only publicly available dataset with voxel-wise lesion annotations for neonatal HIE [4]. It was deployed in a world-wide challenge at the MICCAI conference in 2023. General-purpose segmentation models perform poorly on it, and although task-specific challenge submissions do better, absolute scores remain modest. This reflects the difficulty of segmenting the wide size range of diffuse, multi-focal lesions typical of HIE, compared to focal pathologies such as tumors.

A range of architectural, trainer, and loss-design strategies have been explored in and since the challenge. Chandran and Reghunath [7] propose a Residual U-Net with a 2.5D input that stacks neighboring ADC and Z_{ADC} slices, paired with a class-weighted loss, reporting a Dice score of 0.75. Rao et al. [30] use a 2D UNet++ with an Inception-v4 encoder and squeeze-and-excitation attention, taking ADC, Z_{ADC} , and thresholded Z_{ADC} as spatial priors under a hybrid boundary-aware loss, reporting a Dice of 0.61. Kim [18] instead feed the whole volume in a single pass with a full-volume 3D U-Net using Z_{score} -derived masks, reaching a challenge-validated Dice of 0.63. In a different test, Usman et al. [38] fix a simple U-Net architecture and vary the loss, showing that compound overlap- and boundary-aware losses beat standalone ones.

A limitation common to all these reported scores is that they are computed on test-cohorts mixing hypothermia-treated and untreated neonates. As therapeutic hypothermia is now standard care for moderate-to-severe HIE, these benchmarks do not reflect the segmentation performance to be expected in current clinical practice.

2.3.2 Normative ADC atlases and Z-score maps

As introduced in Chapter 1, normal ADC values vary from one brain region to another, so a region-aware reference is needed to interpret local values. Normative ADC atlases provide this reference: they capture the voxel-wise mean and standard deviation of healthy diffusion, against which a patient’s ADC can be scored to produce a Z_{ADC} map [4, 28]. Conventionally, such an atlas is built from a cohort of healthy neonates acquired on a similar scanner and protocol, making it directly comparable for registration of unhealthy cases [4, 28]. No existing work in the HIE setting addresses the scenario where an atlas must be constructed without such healthy scans.

Both atlas construction and per-patient scoring depend on registration, which is difficult in neonates: brains vary in size and shape, ADC offers little anatomical contrast to align on, and slight misalignments can displace the small lesions that define HIE injury.

Registration can be performed by learning-based methods, which train a network to predict the deformation field [2, 45], or by deterministic methods that estimate the deformation directly for each case. Once trained, learning-based methods are fast, but reaching the voxel-level precision that small HIE lesions

demand remains difficult. In contrast, deterministic methods are slower, but they optimize each registration explicitly, making the required precision easier to reach with less data. They accordingly form the basis of current normative-atlas pipelines for HIE [4, 28].

For HIE specifically, the relevant deterministic method is DRAMMS [27], which is the registration backbone of the BONBID-HIE pipeline [4]. It combines voxel-wise and feature-based registration through two mechanisms. First, it describes each voxel with a collection of multi-scale, multi-orientation Gabor attributes. These provide anatomical discrimination that is largely modality-independent, also reducing voxel-matching ambiguity. Second, a continuously-valued mutual-saliency map weights each voxel by how uniquely it can be matched to the target. This automatically down-weights regions that cannot be matched reliably, such as lesions and other outlier tissue. Additionally, pathological voxels can also be excluded explicitly through supplied masks. Together, these properties make DRAMMS well suited to working with cohorts containing pathological tissue, both when building an atlas with its groupwise variant Popdramms and when registering individual cases to it with standard pairwise DRAMMS.

2.4 Weak supervision for lesion segmentation

To our knowledge, weak-supervision has not been studied for neonatal HIE lesion segmentation specifically. Existing work, described in Section 2.3.1, is fully supervised.

Weakly-supervised segmentation approaches train on sparse annotations, which are cheaper to produce than dense masks, such as single points per object [5], scribbles loosely tracing each region [22], or bounding boxes from which pixel-level masks can be recovered [8]. The tightness of a box is informative: a tight box closely bounds the object, and pixels outside it are likely negatives. Some approaches integrate this tightness prior directly into model training [41]. A common approach supervises only the annotated voxels and leaves the unlabeled majority to an auxiliary signal, such as a pseudo-label derived from the model’s own predictions [20, 24].

A separate line of work uses segmentation models not as an end in themselves but as a way to produce full annotations more cheaply: the model generates an initial segmentation that a human corrects, and the corrected output becomes the fully annotated label. This human-in-the-loop approach aims to keep an expert in the loop while reducing manual annotation effort [6]. A similar approach, DeepIGeoS [40], produces an automatic segmentation and then corrects it with a separate refinement network driven by a few annotator clicks marking incorrectly segmented regions. Here, the user can add clicks iteratively until satisfied. In relation to this, nnInteractive [16] is a promptable 3D segmentation model that generates full segmentations from point, scribble, box, or lasso prompts across diverse modalities. Across these approaches, the model serves as an annotation prior that a clinician guides or corrects, lowering the effort of producing full annotations.

CHAPTER 3

Data

In this research, we utilize data from multiple sources to train the segmentation models in our pipeline. We test in-scanner, cross-scanner and cross-domain segmentation performance. For in-scanner and cross-scanner tests we use the publicly available BONBID dataset from Boston [4], while cross-domain performance is tested through a private dataset from UMC Utrecht [26].

In this chapter, we discuss these datasets, detailing acquisition properties and lesion characteristics.

3.1 Boston

3.1.1 Original Dataset

BONBID-HIE is the first publicly available neonatal HIE MRI dataset with expert voxel-wise lesion annotations, released to support lesion detection and segmentation HIE. Brain lesions in the dataset are typically small, diffuse, and multi-focal, often occupying less than 1% of the total brain volume [4]. That said, some cases exhibit much larger lesion burdens. Such extreme cases are mostly from before TH was introduced in standard care. Consequently, the range across the datasets discussed below is wide (Figure 3.1), increasing the challenge of building a generally robust pipeline.

Following the dataset description of Bao et al. [4], the cohort consists of 133 retrospectively collected neonatal MRI scans from Massachusetts General Hospital (MGH). The study included term-born neonates with clinically confirmed HIE who were treated at MGH between 2001 and 2018, had no major comorbidities such as hydrocephalus or congenital syndromes, and underwent a high-quality MRI within the first 14 days after birth. Cases were excluded if the imaging contained excessive motion artifacts or missing data, or if the primary pathology was perinatal stroke, focal arterial ischemic stroke, or hemorrhage.

The dataset focuses on diffusion MRI, in particular apparent diffusion coefficient (ADC) maps, which are clinically important for identifying acute HIE-related injury. As described by Bao et al. [4], imaging was acquired on two different scanners: a GE 1.5T Signa scanner ($N = 52$, acquired between 2001–2012) and Siemens 3T TrioTim or PrismaFit scanners ($N = 81$, acquired between 2012–2018). Diffusion tensor sequences across scanners used a diffusion weighting of $b = 1000$ s/mm².

The raw DICOM MRI scans underwent preprocessing as described in [4], including N4 bias correction, field-of-view normalization, multi-atlas skull stripping for the ADC maps, and deformable registration with DRAMMS to a normative neonatal ADC atlas.

Binary lesion masks were produced by multi-expert consensus on the ADC maps, with a final mask established through majority agreement [4]. Per patient, the dataset provides a skull-

stripped ADC map, a derived Z_{ADC} map (Section 3.1.2), an expert lesion annotation mask, and associated clinical variables. Notably, the raw T1w, $b=0$, and $b=1000$ volumes are not released, meaning they are not available for the reconstruction of the normative atlas. This is a limitation, as ADC alone is poorly suited to registration.

3.1.2 Z_{ADC} Maps

In addition to ADC maps and lesion annotations, BONBID-HIE provides a derived Z_{ADC} map for each patient [4]. These maps are obtained by comparing the patient’s ADC values to a normative neonatal ADC atlas constructed from 13 healthy neonates, which defines the voxel-wise mean and standard deviation of normal ADC values. For each voxel, a Z-score is computed as

$$Z_x = \frac{I_x - \mu(D(x))}{\sigma(D(x))}$$

where I_x is the patient ADC value and $\mu(D(x))$ and $\sigma(D(x))$ are the atlas mean and standard deviation at the corresponding anatomical location after registration [4]. The resulting Z_{ADC} map quantifies how strongly diffusion at each voxel deviates from the expected normative value. In this thesis, the provided Z_{ADC} maps are used as an additional input representation.

3.1.3 Subset Used in This Study

The BONBID-HIE dataset is originally divided into a training and validation set ($N = 89$), and a held-out test set ($N = 44$), with both splits containing cases from two scanner types and a similar distribution of lesion patterns [4]. However, only the training/validation portion of the dataset is publicly accessible, and the test set is therefore not available for this study.

To align the cohort with our research goals, we further filter the 89 publicly available cases according to two criteria: Firstly, whether the patient had received TH, and secondly, whether the scan was acquired within seven days of birth. The resulting cohort is summarized in Table 3.1. Cases scanned more than seven days after birth are excluded, as imaging beyond this window reflects substantially different pathological dynamics. Since most 1.5T scans come from patients who did not receive TH, the 48 TH-treated 3T cases scanned within seven days form the only subset large enough to filter on both criteria. We retain this group as the pretraining cohort, referred to from here on as D3T-B. For cross-scanner evaluation we additionally define D1.5T-B: the 28 1.5T cases scanned within seven days, combining TH-treated (4) and untreated (24) patients. The 1.5T cohort is too small to filter on TH as well, so both are kept. The D3T-B model serves as the initialization for fine-tuning on D1.5T-B, allowing us to assess cross-scanner and cross-field-strength generalization within a single institution.

Table 3.1: BONBID-2023 cohort breakdown by scanner field strength and TH treatment status.

Scanner	TH	n	Scanned ≤ 7 d	Scanned > 7 d
1.5 T	yes	8	4	4
1.5 T	no	25	24	1
3 T	yes	53	49 ¹	4
3 T	no	3	3	0
Total		89	80	9

¹ One additional 3T TH-treated case scanned within seven days (the only 3T case in the four-case validation set) was unintentionally omitted during dataset setup. As this was noticed too late to rerun the experiments, we retained the 48-case cohort throughout for continuity, rather than the 49 cases the table permits.

3.2 Utrecht Dataset

For the second phase of our research, we used the Utrecht dataset for fine-tuning and evaluation of our weakly supervised learning approach, comparison with full supervision, and development of the pipeline for building per-patient pseudo-normative spatial priors. This cohort is referred to from here on as D1.5T-U.

MRI data for the Utrecht cohort were acquired at the Wilhelmina Children’s Hospital, University Medical Center Utrecht, between 2005 and 2011 [26]. In this work we use the annotated subset (Dataset A), which consists of 20 subjects scanned on a 1.5T Philips Achieva MRI scanner. Scanning took place between day 2 and day 7 after birth. Diffusion-weighted images were acquired in three orthogonal directions ($b = 1000$), together with a non-diffusion-weighted image ($b = 0$).

None of the Utrecht cases received TH. This makes the cohort well suited for evaluating cross-hospital and cross-scanner generalization, while also enabling assessment of how robustly models trained on TH-treated patients transfer to untreated ones.

3.3 Cross-Dataset Comparison

3.3.1 Acquisition Properties

The acquisition geometries of the three final datasets are summarized in Table 3.2. Datasets D3T-B and D1.5T-B are highly heterogeneous. In contrast, D1.5T-U is fully homogeneous, with all cases sharing identical voxel spacing, in-plane resolution, and slice count.

Datasets D3T-B and D1.5T-B differ in the structure of their heterogeneity. On the one hand, D3T-B is dominated by two acquisition protocols: a near-isotropic configuration and a thick-slice, higher in-plane resolution configuration, with only a few outliers within this range. On the other hand, D1.5T-B is more dispersed: its cases are distributed across several spacing groups without a single dominant protocol, and it spans a wider range of in-plane spacings than D3T-B.

Slice thickness and volume size follow a consistent pattern across all three properties: the Boston cohorts extend beyond

Table 3.2: Imaging characteristics per cohort. Voxel spacings are reported as (in-plane $\times$ in-plane $\times$ slice thickness) in mm; cases with similar acquisition geometries are grouped.

Property	Value
D3T-B, $N = 48$	
<i>Voxel spacing (mm)</i>	
$2 \times 2 \times 2$	21 cases
$1.31\text{--}1.38 \times 1.31\text{--}1.38 \times 6$	22 cases
$1.37\text{--}1.38 \times 1.37\text{--}1.38 \times 4.8$	2 cases
Other (≈ 2 mm slice)	3 cases
<i>In-plane (X, Y)</i>	$128 \times 128 - 160 \times 160$
<i>Slices (Z)</i>	16 – 64
D1.5T-B, $N = 28$	
<i>Voxel spacing (mm)</i>	
$0.86 \times 0.86 \times 2.2\text{--}2.4$	12 cases
$0.86 \times 0.86 \times 4.5\text{--}6.0$	4 cases
$1.56 \times 1.56 \times 4.0$	4 cases
$1.25 \times 1.25 \times 4.0\text{--}6.0$	2 cases
$0.78 \times 0.78 \times 2.2\text{--}4.5$	3 cases
Other ($\approx 0.6\text{--}1.5$ in-plane)	3 cases
<i>In-plane (X, Y)</i>	$128 \times 128 - 256 \times 256$
<i>Slices (Z)</i>	16 – 60
Utrecht (D1.5T-U), $N = 20$	
<i>Voxel spacing (mm)</i>	
$0.70 \times 0.70 \times 4.0$	20 (all)
<i>In-plane (X, Y)</i>	256×256
<i>Slices (Z)</i>	25 (all)

the Utrecht values at both the thin- and thick-slice ends, and vary substantially in slice count, while Utrecht’s fixed geometry sits comfortably within the range they cover. In in-plane resolution, Utrecht matches the upper bound of D1.5T-B and exceeds that of D3T-B.

3.3.2 Lesions Characteristics

All together, these factors suggest that D3T-B will be the most difficult cohort to train on ADC alone, due to the combination of small lesions and high intensity overlap, both of which favor false-positive-heavy predictions. D1.5T-B and D1.5T-U are more comparable overall, but D1.5T-B’s small-lesion subset and somewhat higher intensity overlap are likely to make it the harder of the two.

The three cohorts differ substantially in their lesion characteristics, as summarized in Table 3.3 and Figure 3.1. All three span a wide range of lesion burden, but the shape of each distribution is considerably different. D3T-B exhibits severe class imbalance, with the large majority of patients having a lesion fraction below roughly 2.5%. D1.5T-U is more evenly distributed, with most cases falling in a moderate range. D1.5T-B occupies both extremes, combining a cluster of very low-burden cases with a second cluster of high-burden cases.

As a simple measure of how separable lesion and healthy tissue are on raw ADC, we threshold each cohort at the single ADC value that maximizes its mean Dice, and examine how

Table 3.3: Cohort-level summary (top) and per-patient FP : TP voxel ratio at each cohort’s optimal-Dice ADC threshold (bottom subtable), on lesioned patients only. Bottom columns span the per-patient ratio distribution (lower is better) from best (min) to worst (max). Lesion burden in Figure 3.1.

Metric					D3T-B (N=48)			D1.5T-B (N=28)		D1.5T-U (N=20)		
Lesion fraction (%): min / median / max					0 / 0.20 / 73.6			0 / 19.8 / 82.6		0.52 / 2.52 / 40.0		
Lesion / non-lesion ADC overlap: mean $\pm$ SD					36.7 % $\pm$ 15.8			30.1 % $\pm$ 13.4		20.5 % $\pm$ 7.2		
Cohort	Threshold	min	p10	p20	p30	p40	p50	p60	p70	p80	p90	max
D3T-B (N=46)	950	0.1	2.0	3.6	7.0	11.5	18.5	27.6	47.4	70.0	154.0	686
D1.5T-B (N=25)	1000	0.0	0.1	0.2	0.3	0.4	0.5	1.0	1.8	8.5	26.8	1025
D1.5T-U (N=20)	900	0.1	0.1	0.4	0.6	1.6	2.5	3.3	4.4	5.8	7.0	9

many false-positive voxels this incurs per true-positive voxel. Since this optimal threshold is similar but not identical across cohorts, each is evaluated at its own threshold rather than a shared one. The bottom subtable of Table 3.3 reports the resulting per-patient false-positive (FP) to true-positive (TP) voxel ratio, and shows three clearly distinct regimes. D3T-B has substantially higher ratios than the other two cohorts across the entire distribution, reflecting both its small lesion fractions and its high lesion/non-lesion ADC overlap. D1.5T-U and D1.5T-B behave similarly through the lower percentiles, but D1.5T-B diverges sharply in the upper tail, driven by its subset of extreme small-lesion cases visible in Figure 3.1.

Inspection of the per-patient overlap between lesion and non-lesion ADC values reveals differences between cohorts (see Figure A.1 in Appendix A for a detailed per-patient visualization). In D1.5T-U, lesion and healthy distributions are clearly offset within almost every patient, with lesion IQRs sitting well below the healthy IQR. D1.5T-B shows broadly similar per-patient separation, but with more variability across cases and somewhat larger lesion IQR spreads. In contrast, D3T-B is considerably less separable: lesion IQRs overlap with healthy IQRs in the majority of patients, and lesion medians frequently sit just below the healthy IQR of other patients

These patterns are reflected in the population-level overlap statistics in Table 3.3. D3T-B shows the highest mean overlap (36.7%), nearly twice that of D1.5T-U (20.5%), while D1.5T-B lies between the two (30.1%).

Absolute ADC values also shift substantially across patients within each cohort. This effect is most visibly in D1.5T-B, where healthy medians span roughly $1000\text{--}1800 \times 10^{-6} \text{ mm}^2/\text{s}$, limiting the usefulness of any single global intensity threshold (Figure A.1, Appendix A).

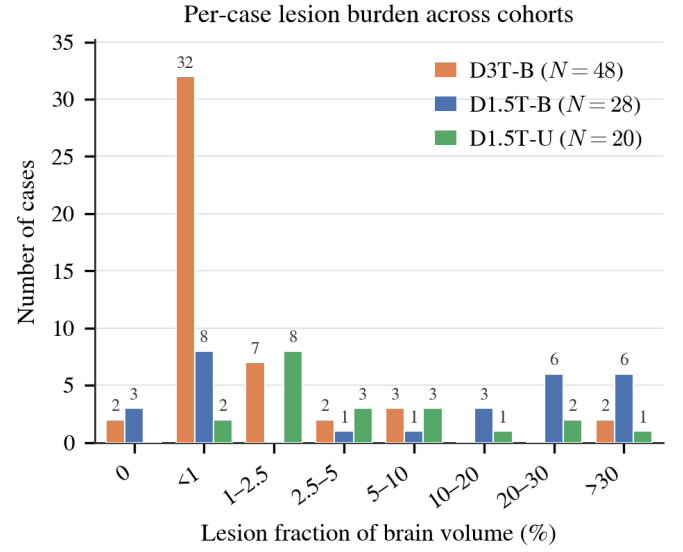


Figure 3.1: Per-case lesion burden distribution across the three cohorts. Bars show the number of cases per bucket, each bucket spanning a range of lesion volume as a percentage of total brain volume. D3T-B concentrates at <1%, with a few severe outliers. D1.5T-B spans both extremes: a low-burden cluster below 1% and a second cluster in the higher-burden bins. D1.5T-U has a wide range, but most lesions are moderate, falling within roughly 1–10%.

Methodology

The aim of this thesis, as stated in Chapter 1, is to develop a HIE segmentation pipeline that achieves robust lesion segmentation across scanners and domains while reducing the need for time-intensive full manual annotation. Achieving this is challenging due to several connected reasons. Lesions are diffuse, span size ranges from tiny to large, and are multi-focal. Normal ADC values vary regionally, meaning raw ADC cannot be interpreted in isolation. On top of this, variability across scanners, protocols, and treatment limits cross-cohort generalization. Finally, the dense annotations needed to train and evaluate robust models are scarce and inconsistent across annotators. This chapter describes the full pipeline designed to address these difficulties.

Figure 4.1 gives a schematic overview of the full pipeline, showing three goals that mirror the research questions stated in Chapter 1.

Goal 1 is to establish a robust and transferable baseline. We develop and tune 3D U-Net and 2.5D SegResNet segmentation models, assessing both their use as standalone models and their transfer, via pretrained weights, to new scanners and sites. Tuning is done with the fully annotated D3T-B cohort under 3-fold cross-validation (CV), where within-scanner generalization is reported on the held-out test folds. The frozen design is then retrained on all D3T-B cases to produce a pretrained checkpoint, which serves as the initialization for cross-scanner and cross-domain fine-tuning and as the backbone for weak supervision (Figure 4.1, Goal 1).

Goal 2 is to build a pipeline that enables the construction of reference atlases of healthy neonatal tissue for new cohorts and the construction of corresponding per-patient Z_{ADC} maps, in both a supervised and unsupervised setting. The pipeline enables normative modeling from a typical neonatal cohort, in which all or a large portion of patients carry HIE lesions. The pipeline is designed to accommodate normative modeling in the absence of separately curated healthy cohort for comparison. It can derive a pseudo-normative atlas directly from the patient cohort itself, masking suspected lesion voxels and treating the remaining tissue as the healthy reference. Such an atlas captures the voxel-wise spatial distribution of normative diffusion characteristics, against which each patient can be scored to produce per-patient voxel-level standard-deviation Z_{ADC} maps. The pipeline reuses the existing DRAMMS groupwise registration backbone [27] to enable this normative modeling. The result is a standardized, region-aware reference to support consistent comparison across brain-regions, without the need for a separate healthy population (Figure 4.1, Goal 2).

Goal 3 is to reduce the amount of manual annotation required, by enabling weak supervision that surpasses naive thresholding and approaches full-supervision performance We design a sparse clicks-and-lassos scheme intended to mimic a quick clinical annotation pass, investigating how closely it approaches full-supervision accuracy while

reducing annotation effort. As true weak labels are unavailable, we adapt the dense ground-truth labels artificially into the sparse clicks-and-lassos scheme. The frozen baseline then acts as both initialization and teacher in a teacher-student setup. The Z_{ADC} maps used for weak-supervision modeling must be built without the need for full GT-annotations. The provided Z_{ADC} maps for D3T-B and D1.5T-B are possibly constructed with GT interference. Consequently, our weak-supervision experiments are performed on D1.5T-U, because this is the only dataset with the $b=0$ and $b=1000$ DWI scans required to build the GT-free Z_{ADC} maps that the weak-supervision setting depends on (Figure 4.1, Goal 3).

In the remainder of this chapter, we describe this pipeline in detail. Section 4.1 presents the baseline architectures and their training setup. Section 4.2 introduces the groupwise registration procedure and the construction of the normative atlas. Section 4.3 describes the full-supervision fine-tuning and weak-supervision approaches, together with the evaluation protocol.

4.1 Architectures

In this section we describe the baseline architectures and their training setup. Three models were tested: nnU-Net, a 3D U-Net, and a 2.5D SegResNet. Initial testing was done with nnU-Net, but as it lacked robustness across all three of our datasets, it was discarded in favor of the other two. We therefore describe it only briefly, and the 3D U-Net and 2.5D SegResNet in depth. For both these models, we describe the architecture, training set-up, and augmentations used. These models are the basis for all training and evaluation in this work, across both Goal 1 and Goal 3. Both models have their distinct advantages. The pipeline uses both together to improve robustness.

4.1.1 nnU-Net

Initial experiments were conducted with the nnU-Net framework [15], selected as an initial candidate due to its strong performance across a wide range of biomedical segmentation tasks.

We considered it as a reliable starting point, due to its self-configuring framework that automatically adapts its architecture, preprocessing, and training schedule to a new dataset. It trains on patches sampled from each volume rather than the full 2D image or 3D volume.

nnU-Net was evaluated under two input configurations to assess the importance of normative modeling: a single-channel setup using only the ADC map, and the three-channel setup using the additional Z_{ADC} channels. Comparing the two isolates the effect of the region-aware reference.

While its general benefits make nnU-Net a reliable starting point, some properties of our datasets work poorly with its de-

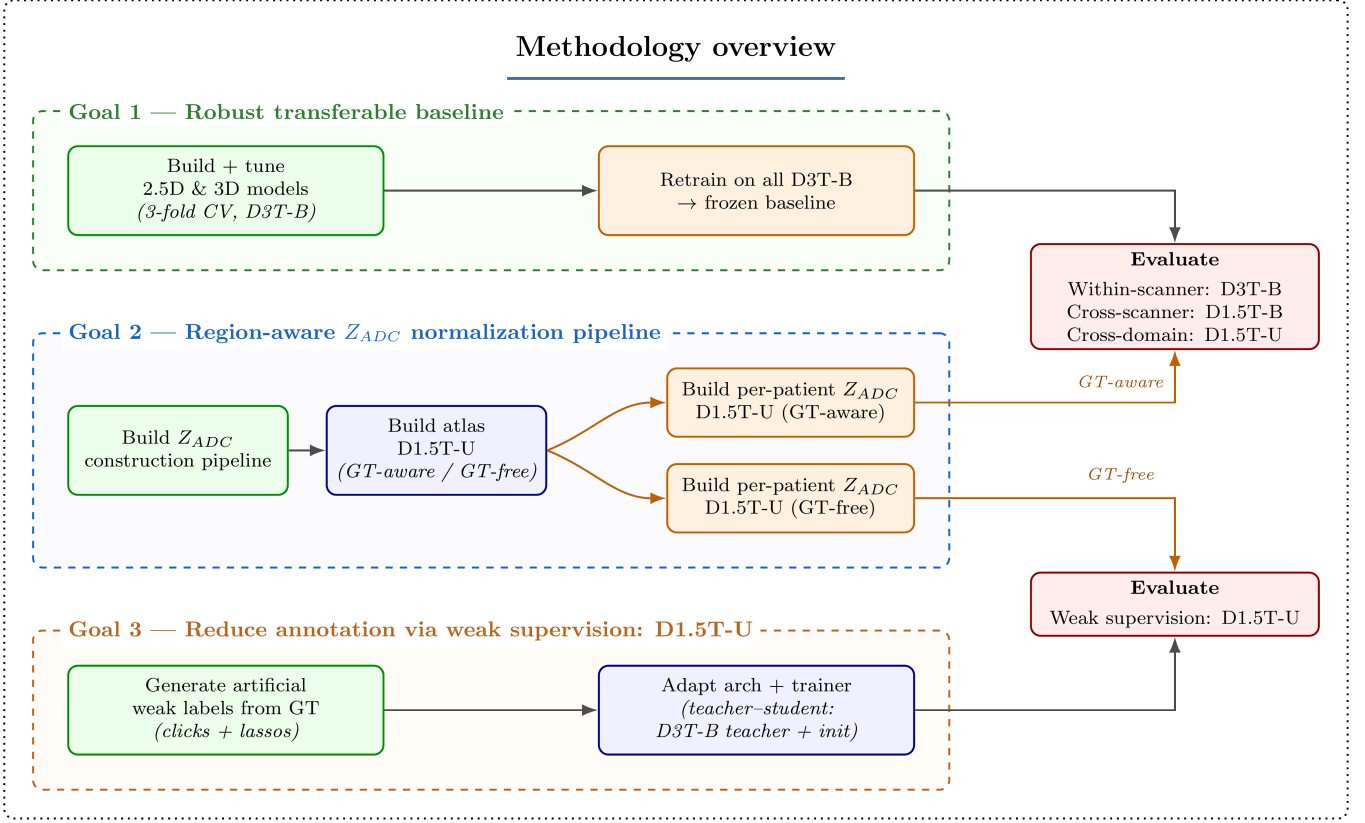


Figure 4.1: Methodology overview, structured around the three research goals: a robust transferable baseline (Goal 1); a pipeline that builds normative reference atlases and corresponding per-patient pseudo-normative spatial priors (Z_{ADC} maps), in GT-aware and GT-free variants (Goal 2); a teacher-student weak-supervision scheme initialized from the baseline (Goal 3). The evaluation (red boxes) uses the GT-aware maps for the within-, cross-scanner, and cross-domain comparison, while the GT-free maps are used for the weak-supervision evaluation.

sign. As discussed in Section 3.1, lesion and healthy tissue overlap heavily in intensity, particularly in D3T-B. Local intensity alone is therefore not enough to tell them apart. Instead, models need global context to learn which ADC values are normal for each region. This is where patch-based training falls short, as patches restrict the global context available during training and inference. This effect is amplified by the high learning capacity of nnU-Net. It is built to fit local intensity patterns, but here those patterns are unreliable.

For these reasons, we moved away from nnU-Net in favor of models better suited to capturing global context and exploiting the Z_{ADC} maps.

4.1.2 Full-Volume 3D U-Net

The BONBID dataset [4] was used in a recent MICCAI challenge. The top-performing methods reported in this challenge achieve state-of-the-art segmentation results for HIE MRI data, and therefore provide a useful performance target when developing our own baseline model. We identified the publicly released trained model of one of the top-performers from this challenge [18]. This model provided an important starting point for our baseline development.

Although the publicly available implementation provides

the model architecture, the original training pipeline and trainer configuration were not released. As a result, we needed to implement a custom training framework for this architecture. In addition, we tweaked some architectural design choices through an ablation study.

3D U-Net Architecture

A key design choice of the architecture is that the network processes the entire MRI volume in a single forward pass, rather than relying on patch-based training. Providing the full volume as input allows the network to capture global anatomical context that may be lost when training on smaller patches. In addition, the single-pass strategy significantly reduces training and inference time compared to the multi-stage patch-based nnU-Net pipeline.

The architecture itself is a 3D U-Net. The full-volume processing strategy enables the network to efficiently model global spatial relationships within the brain volume. This is particularly relevant for data like ours, where local intensity values do not reliably distinguish lesion from healthy tissue; the network can rely more heavily on global anatomical context and Z_{score} -derived inputs.

An additional component of the model is the inclusion of a

3D Residual U-Net

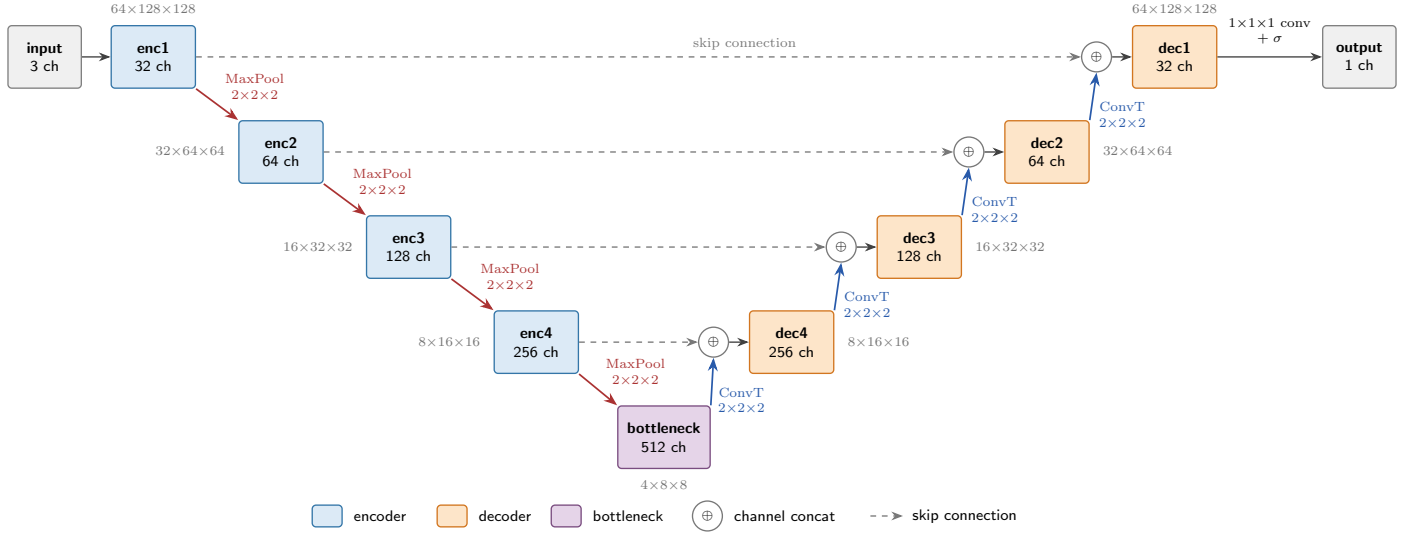


Figure 4.2: 3D Residual U-Net architecture. Channels double on the way down and halve on the way up; MaxPool 2^3 downsamples and ConvT 2^3 upsamples. At each level encoder features pass forward by a skip connection (dashed) and are concatenated ($\oplus$) with the upsampled decoder features. The final $1 \times 1 \times 1$ conv + sigmoid yields the segmentation map. Each colored block is a residual block (Figure 4.3).

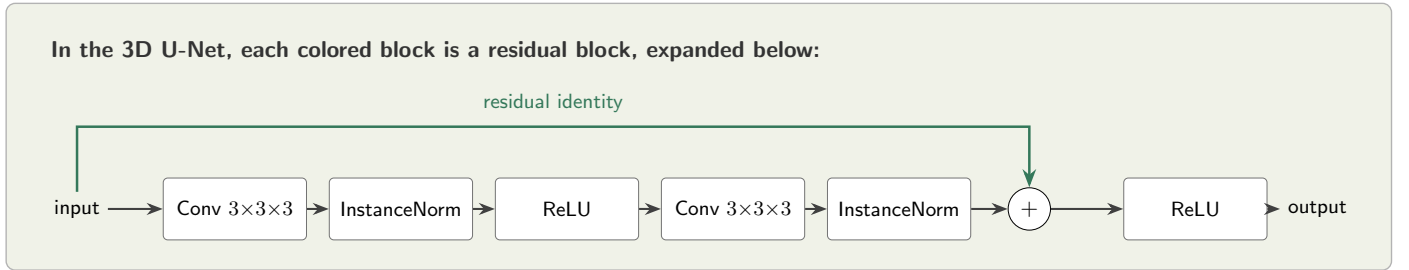


Figure 4.3: Residual block structure used throughout the 3D U-Net. Two Conv $3 \times 3 \times 3$ + InstanceNorm + ReLU stages with an identity shortcut (a $1 \times 1 \times 1$ conv when the channel count changes) added before the final ReLU.

binary mask derived from the pseudo-normative spatial priors. This mask is added as an additional input channel alongside the MRI modalities. The intention of this mask is to highlight regions with extreme deviations from normative statistics, mostly corresponding to pathological tissue. The architecture is visualized schematically in Figure 4.2

3D U-Net Training Setup

As an initial starting point for the training setup, we adopted the basic training settings and loss functions used by nnU-Net, which have been shown to be robust across a wide range of segmentation tasks.

From this starting point an extensive ablation study was done to slightly tweak architecture design, while fully optimizing trainer design and hyperparameters. These ablations are shown in Appendix B: the loss-function sweep in Appendix B.2, the architecture and trainer sweep in Appendix B.3, and the augmentation sweep in Appendix B.4. The final architecture and inputs are summarized in Table 4.1.

The model takes three input channels: the ADC map, the

Table 4.1: 3D U-Net: architecture and inputs.

Input channels	3 (ADC , Z_{ADC} , Binary mask)
Patch size [†]	$64 \times 128 \times 128$ (Boston)
Family	3D residual U-Net
Depth	4 enc/dec + bottleneck
Initial features	32
Normalization	InstanceNorm (affine)
Activation	LeakyReLU
Output	$1 \times 1 \times 1$ conv (logits)

[†] auto-derived per fold as next_pow2 of the smallest bounding box fitting all cohort brains.

Z_{ADC} map, and a binary mask. The ADC maps provides absolute intensity information, while the per-patient Z_{ADC} maps express how far each voxel deviates from a normative mean, giving the region-aware reference. The third channel is a binary mask derived from the Z_{ADC} map, thresholded at $Z_{ADC} < K$. As the Z_{ADC} scores represent the number of standard deviations from the mean, a lower K threshold flags voxels that deviate more strongly from normal diffusion. The mask therefore acts as an additional prior, marking likely candidate lesion voxels so the network can focus on those regions. The value of K

follows the masking approach and threshold value set in [18]. A sweep over a range of thresholds confirmed this value as a robust choice (Appendix Table B.1).

The network always sees the global anatomical context in a single forward pass. The crop size is derived per cohort as the smallest brain bounding box across all cases, rounded up to the next power of two ($64 \times 128 \times 128$ for Boston).

The trainer optimization and loss choices are summarized in Table 4.2. We kept the trainer relatively simple, as more complex designs quickly led to overfitting on the small datasets we use. The batch size is set to 4, but since we use InstanceNorm, this is not critical and was sometimes adjusted to fit available compute. The 5-epoch warm-up is an important addition: without it, poor initialization sometimes left the model stuck in local minima. Given the strong foreground/background imbalance, we expected to need loss functions like Tversky or foreground upweighting, and accordingly tuned several variations. Interestingly, none gave consistent improvements, so we settled on the simple and reliable Dice + BCE loss. This loss tends to favor overprediction, which we prefer over missing lesions entirely.

Table 4.2: 3D U-Net: optimization and loss.

Loss	$0.5 \cdot \text{Dice} + 0.5 \cdot \text{BCE}$ (on logits)
Optimizer	SGD, Nesterov, momentum 0.99
Learning rate	3×10^{-3}
Weight decay	3×10^{-5}
LR schedule	Cosine, 5-ep linear warmup
Gradient clip	1.0 (L2 norm)
Epochs	200
Batch size	4

Augmentation and EMA settings are shown in Table 4.3. We tried extensive augmentation, but found that for our one-pass 3D U-Net simple augmentation works best. It clearly helps over no augmentation, but heavier recipes distracted the model from the actual data, given the limited dataset size. We therefore kept only flipping, rotation, and light gamma and noise. From the rest of nnU-Net’s default augmentation pipeline, the only piece retained here was the 4-flip TTA at inference.

A final important addition was EMA model saving. We observed large inter-run variance, with model performance peaking at different stages across folds and cohorts. With EMA, weights from earlier in training continue to contribute to the final predictions, which improved stability across runs. This is not cherry-picking: the EMA decay is set by a fixed rule relative to the training length, so the averaging window is always the same.

Table 4.3: 3D U-Net: augmentation, EMA, and inference.

Flip (D, H, W)	$p=0.5$ per axis
Rot90 in-plane	$p=0.5$
Gamma	$0.7-1.5$, $p=0.1$ (V channel)
Gaussian noise	$\sigma \in [0, 0.05]$, $p=0.1$ (V channel)
EMA decay	$0.1^{1/N}$, N = steps in last 150 epochs
Saved model	EMA shadow only
TTA	4 flips (id, H, W, H+W), avg in prob. space
Threshold	Binarization threshold: 0.5

4.1.3 2.5D SegResNet

The second model used in our pipeline is a 2.5D representation that combines properties of full-volume 3D and 2D approaches: stacking a few neighboring slices along the z-axis gives the network some 3D context while keeping full in-plane resolution. Since lesions are often small in our data, a few stacked slices are usually enough to capture lesion extent along the z-axis. As such, this model trades only a little of the 3D model’s global awareness for stronger local sensitivity.

Our 2.5D baseline is adapted from the SegResNet-based model of Chandran and Reghunath [7], which combines residual blocks with a U-Net style encoder-decoder and uses stacked neighboring slices from ADC and Z-ADC maps as input. No source code was released, so we reconstructed the architecture and hyperparameters from the paper as a starting point. Tuning of the augmentation pipeline and learning rate was needed to make the model competitive. One notable deviation from the original is the crucial addition of the binary mask, also used in our 3D U-Net, as a 7th input channel (ablated in Appendix B.5).

The full model architecture is shown in Fig. 4.4.

2.5D Inputs and Architecture

The final architecture and inputs are summarized in Table 4.4. The model uses the same three input types as the 3D U-Net (Section 4.1.2): the ADC map, the Z_{ADC} map, and the binary $Z_{ADC} < K$ mask. However, in contrast to the 3D U-Net, it packs them into seven channels: three adjacent slices each from the ADC and Z_{ADC} map and a single binary mask channel. Stacking neighboring slices gives the model the most important global context, while still allowing for more local variation during training by shifting the stack position. This 2.5D approach also strengthens 2D in-plane learning compared to the 3D model.

Table 4.4: 2.5D SegResNet: architecture and inputs.

Input channels	7 ($3 \times \text{ADC}$, $3 \times Z_{ADC}$, K)
Input size [†]	128×128 (Boston)
Patch size [†]	96×96 (Boston, random crop)
FG oversample	0.5
Family	2.5D SegResNet
Depth	3 enc/dec + bottleneck
Initial features	64
Normalization	BatchNorm
Activation	ReLU
Output	1×1 conv + sigmoid

[†] input size = $\text{next_pow2}(\text{max bbox})$; patch = $\frac{3}{4} \cdot \text{input}$.

The input size is set per cohort as the next power of two above the maximum brain bounding box. During training, a random $\frac{3}{4}$ -size patch is cropped from the input (128×128 input, 96×96 patch for Boston). Unlike the 3D U-Net, this model does not rely on full-volume context. The slice stack provides mainly local context across the slice axis, so cropping removes less global information. Here, the patch adds sampling variation. The per-slice-stack design exposes the model to more inputs per epoch than the full-volume model. Foreground oversampling at

2.5D Residual U-Net

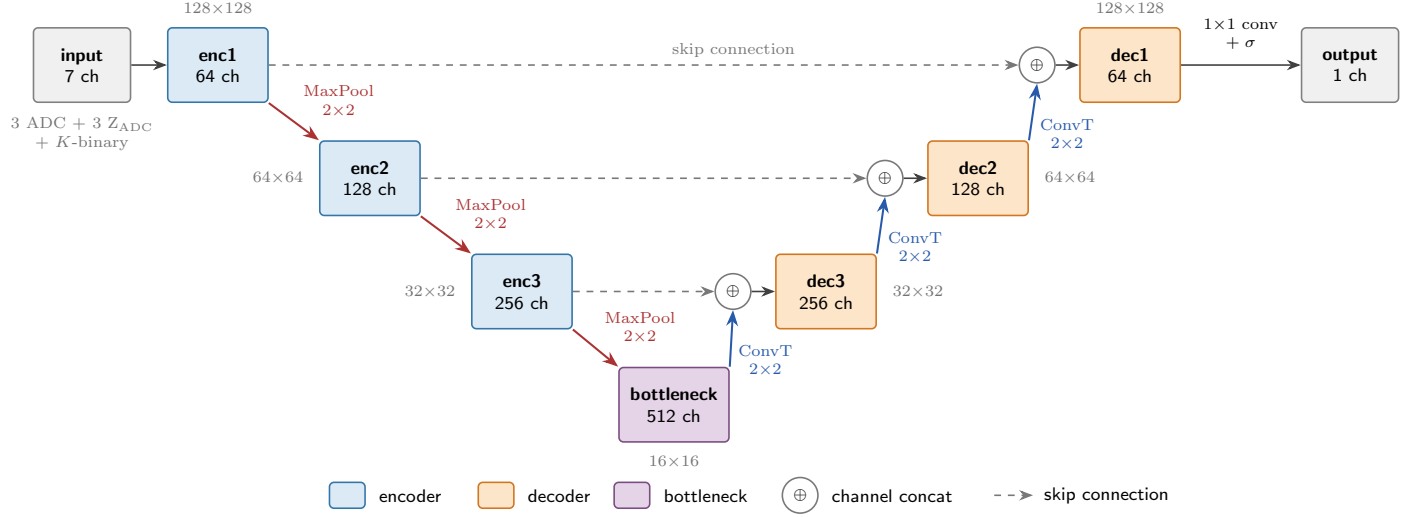


Figure 4.4: 2.5D Residual U-Net architecture. Three encoder blocks (blue) and three decoder blocks (orange). Channels double on downsampling (MaxPool 2^2) and halve on upsampling (ConvT 2^2). The input stacks three adjacent slices each from ADC and Z_{ADC} plus a K-binary mask (7 channels total), letting a 2D network see local 3D context. Encoder features are passed forward by skip connections (dashed) and concatenated ($\oplus$) with upsampled decoder features. A final 1×1 conv + sigmoid yields the segmentation map. Each colored block is a residual block (Figure 4.5).

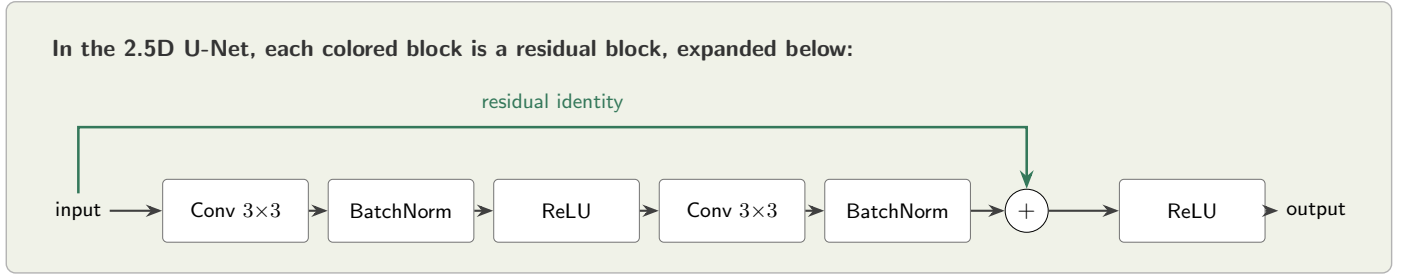


Figure 4.5: Residual block structure used throughout the 2.5D U-Net. Two Conv 3×3 + BatchNorm + ReLU stages, with an identity shortcut adding the input back before the final ReLU. Spatial dimensions are preserved inside blocks, while only the channel count changes.

$p = 0.5$ taken from the original paper, ensures that at least half of the training patches contain lesion voxels. This helps with the strong class imbalance. Most other choices are copied unchanged.

Trainer and Loss

The trainer optimization and loss choices are summarized in Table 4.5. Since the original paper provided concrete directions here, less tuning was required, meaning we could focus on other areas of the project. As in the 3D U-Net, we added a linear warmup to prevent early crashes and set the epoch budget to 200.

The paper is ambiguous about the loss function, discussing both a Hausdorff-Dice loss and a simple BCE + Dice loss. In our experiments, the difference in loss functions across the 3-fold CV was within noise. As such, we opted for BCE + Dice to keep both baseline models aligned.

Table 4.5: 2.5D SegResNet: optimization and loss.

Loss	$0.5 \cdot \text{Dice} + 0.5 \cdot \text{BCE}$
Optimizer	AdamW
Learning rate	5×10^{-5}
Weight decay	3×10^{-5}
LR schedule	Cosine, 10-ep linear warmup
Gradient clip	1.0 (L2 norm)
Epochs	200
Batch size	4

Augmentation, EMA, and Inference

Augmentation, EMA, and inference settings are shown in Table 4.6. In contrast to the 3D model, copying the much heavier nnU-Net style augmentation is beneficial for this model. The per-slice nature of the model allows for more variation without losing anatomical coherence. Additionally, each epoch sees many more inputs through variations of stacks, instead of full volumes. We therefore adopted most of the nnU-Net augmen-

tation pipeline: in-plane rotation and scaling, flipping, triplet z-flipping of the slice stacks, gamma, noise, blur, low-resolution simulation, brightness, and contrast. EMA model saving and 4-flip TTA at inference are identical to those of the 3D model.

Table 4.6: 2.5D SegResNet: augmentation, EMA, and inference.

Rotation	$\pm 30^\circ$, $p=0.5$
Scale	$0.85-1.25$, $p=0.3$
Flip H, W	$p=0.5$ each
Triplet z-flip	$p=0.5$ (ADC and Z_{ADC} stacks)
Gamma	$0.7-1.5$, $p=0.3$
Gaussian noise	$\sigma \in [0, 0.1]$, $p=0.15$
Blur	$\sigma \in [0.5, 1.0]$, $p=0.2$
Low-res	zoom $\in [0.5, 1.0]$, $p=0.25$
Brightness	$\times [0.75, 1.25]$, $p=0.15$
Contrast	factor $\in [0.75, 1.25]$, $p=0.15$
EMA decay	$0.1^{1/N}$, N = steps in last 150 epochs
Saved model	EMA shadow only
TTA	4 flips (none, H, W, H+W), averaged in probability space
Threshold	0.5

4.2 Per-patient Z_{ADC} Map Construction

4.2.1 Pipeline Overview

As discussed in Section 2.3.2, raw ADC alone is often insufficient for accurate lesion localization. Since normal ADC varies regionally, a region-aware reference is needed [4, 28, 29, 35]. Constructing such a reference requires a population-level atlas, defined by a voxel-wise mean and standard deviation of healthy-brain ADC against which each patient is scored. Measured ADC is scanner- and protocol-specific. As such, this reference must be rebuilt for each cohort or scanner on which HIE segmentation is performed, rather than reused across sites.

The standard approach [4, 28, 29, 35] builds this atlas from a separate cohort of healthy neonates. This dependency is costly for two reasons. First, it requires someone to manually select scans that are free of lesions, a time-intensive effort that must be repeated for every new cohort. Second, this selection is itself unreliable: in neonatal cohorts with a potential mix of lesioned and non-lesioned patients, even a scan judged as healthy may contain very small or subtle lesions. Cases mistakenly included as healthy then contaminate the normative statistics. Curating a clean healthy set is therefore both an annotation burden and prone to error.

We address this through a pipeline that constructs (pseudo-)normative atlases and corresponding per-patient Z_{ADC} maps without the need for a separately curated healthy cohort. Instead, it approximates the normative distribution from the patient data itself. This is viable because HIE lesions generally occupy only a small fraction of the brain, leaving most tissue usable.

The normative-atlas mean (μ) and standard deviation (σ) are computed per voxel across the cohort, excluding lesion voxels in two steps so they do not contaminate the statistics. First, a per-subject lesion mask flags which voxels to exclude, in one of two

modes. In the full-supervision setting, ground-truth (GT) lesion annotations are available and used directly as the per-subject masks. Otherwise, when neither GT annotations nor healthy controls are available, a heuristic is used to flag and suppress suspected lesion voxels from each subject’s contribution. The heuristic deliberately over-masks, removing some healthy tissue in order to capture most lesion voxels. This relies on statistics being aggregated across the cohort: since lesions fall in different locations across patients, enough subjects should remain unmasked at each atlas voxel to estimate μ and σ reliably. In contrast, missing lesion voxels is more damaging, as it biases cohort statistics and reduces registration accuracy. Second, a voxel-wise outlier rejection, applied in both modes, removes remaining lesion voxels that escape the masks before the final aggregation. The effectiveness of these pseudo-normative spatial priors, built entirely from unhealthy patient data, is evaluated in Sections 5 and 6.

The pipeline is organized into five phases, visualized in Figure 4.6. Phase 0 reorients the per-subject inputs and builds mode-dependent per-subject masks. Phase 1 constructs a population-level $b=0$ atlas. Phase 2 derives atlas voxel-wise ADC μ and σ . Phase 3 scores each patient against the atlas to construct per-patient Z_{ADC} maps. Phase 4 post-processes the Z_{ADC} maps to reduce registration noise.

4.2.2 Phase 0: Input Preprocessing

The pipeline takes three skull-stripped per-subject inputs in native space: paired $b=0$ and $b=1000$ DWI, and an ADC volume. Additionally, in the GT-aware mode, ground-truth lesion annotations are required.

All three modalities are then resaved with a fixed affine (direction = $(-1, -1, 1)$, origin = $(0, 0, 0)$) so they share one physical-space convention. Otherwise, inconsistent orientation tags are mistaken for anatomical mismatch and subjects are unnecessarily dropped during atlas construction.

Lesion Masking The atlas-construction phase takes a per-subject mask, either GT-free or GT-aware, flagging which voxels contribute to registration and to the aggregation of cohort statistics. Voxels are excluded if they lie outside the brain or are suspected of being a lesion. Both modes share the same brain mask, $\text{brain} = (b_0 > 0)$, and differ only in how lesion voxels are flagged.

GT-free Lesion-masking This part of the pipeline is designed to enable atlas construction in a fully unsupervised setting: building the normative atlas with no separately curated healthy cohort, and no ground-truth annotations to mask the lesioned cases that contribute their remaining unlesioned tissue.

A logical thought might be to then mask based on the direct-inference predictions of a pre-trained model. We do not, for structural reasons: our pre-trained baseline requires the Z_{ADC} map as an input channel, yet constructing that map is precisely what the masking serves. The model, therefore, cannot be used to mask the data needed to build the maps it depends on.

Instead, we propose a purely DWI/ADC-based heuristic that

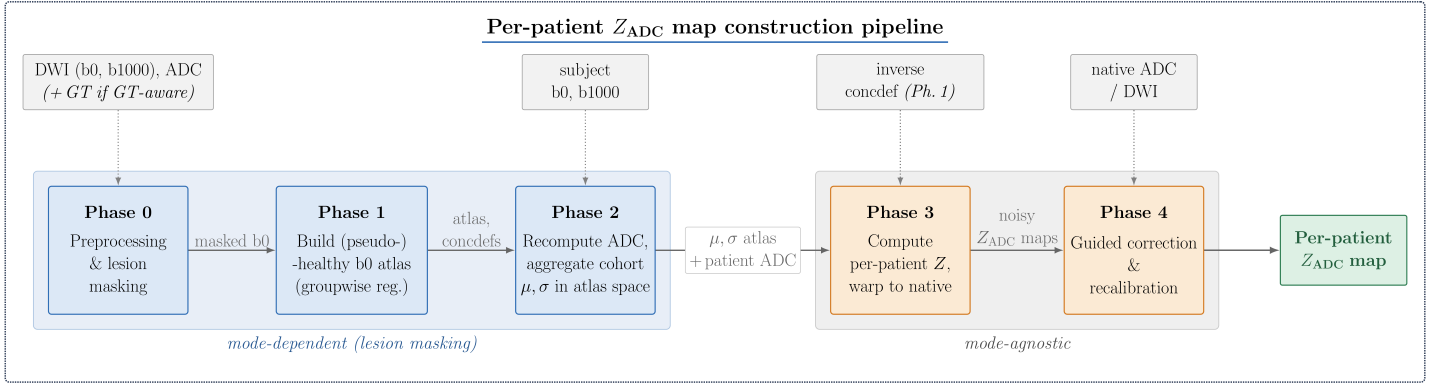


Figure 4.6: Overview of the per-patient Z_{ADC} map construction pipeline. **Phase 0** preprocesses per-subject inputs ($b=0$, $b=1000$ DWI, skull-stripped ADC) and builds lesion masks. **Phase 1** constructs a population-level $b=0$ atlas with per-subject deformation fields via groupwise registration (Popdramms). **Phase 2** recomputes ADC in atlas space from $b=0$ and $b=1000$ DWI and aggregates voxel-wise cohort μ and σ with outlier rejection. **Phase 3** computes per-patient Z -scores and warps them back to native space. **Phase 4** applies guided correction and recalibration. Phases 0–2 are lesion-mask dependent and differ only in the mask source: GT-aware uses ground-truth annotations, GT-free DWI/ADC-based heuristics. Phases 3–4 are mode-agnostic.

avoids this circularity while remaining fully unsupervised. No single criterion separates lesion from healthy tissue across the full range of HIE lesions, so we combine two complementary ones, each targeting a different lesion type: an absolute-ADC threshold for large or severe lesions, and a per-voxel confidence score for the smaller, localized lesions the threshold misses. Their individual effectiveness is reported across a range of hyperparameter settings, highlighting why we combine both (Section 5.1.3). Both criteria use the patient’s own $b=0$ and $b=1000$ DWI, with ADC recomputed from the DWI pair (not the provided ADC) for consistency with Phase 2.

Relying on these heuristics rather than full annotations or a separately curated healthy cohort is a deliberate trade-off, tied to the same motivation as the weak-supervision setting: missed lesion voxels and over-masked healthy voxels both degrade the Z_{ADC} maps, but the approach saves the substantial time that curation or annotation would cost.

Absolute threshold. The first criterion thresholds raw diffusivity, $ADC(\mathbf{x}) < 1 \mu\text{m}^2/\text{ms}$. It catches large or severe lesions, where ADC is strongly reduced, but also flags a considerable number of healthy voxels with low normal ADC. As discussed, this over-inclusion is tolerated: a lesion voxel that escapes the mask and enters the atlas biases the cohort statistics more than the loss of a few healthy voxels.

Confidence score. The second criterion is a per-voxel confidence score $c(\mathbf{x}) \in [0, 1]$ that combines a local-ADC anomaly $a(\mathbf{x})$ and a global DWI deviation $z(\mathbf{x})$ into a single suspect-lesion value:

$$c(\mathbf{x}) = \sigma\left((-a(\mathbf{x}) \cdot \max(z(\mathbf{x}), 0) - 2.0) \cdot 2.0\right). \quad (4.1)$$

The score only rises substantially where both signals agree: a low-ADC anomaly ($a(\mathbf{x}) < 0$) co-occurring with a high-DWI deviation ($z(\mathbf{x}) > 0$). When only one signal is present, the other stays neutral, keeping the score low, and the sigmoid offset ensures only strong anomalies in both maps reach high confidence.

The ADC-based anomaly $a(\mathbf{x})$ scores each voxel $\mathbf{x}$ against its own neighborhood rather than the whole brain, since a global reference would hide region-relative drops and largely flag the

same voxels as the absolute-ADC threshold:

$$a(\mathbf{x}) = \frac{ADC(\mathbf{x}) - m(\mathbf{x})}{\max(0.1 m(\mathbf{x}), 1)}, \quad m(\mathbf{x}) = \text{median}_{C_{11}(\mathbf{x})}(ADC), \quad (4.2)$$

where $C_{11}(\mathbf{x})$ is the $11 \times 11 \times 11$ cube centered on $\mathbf{x}$ and $m(\mathbf{x})$ is its median ADC, computed over brain voxels only. Normalizing by 10% of the cube median expresses $a(\mathbf{x})$ as a drop relative to normal local ADC, so a given fractional drop scores roughly the same whether the region’s baseline ADC is high or low. The floor of 1 on the denominator is a numerical guard against unstable medians at low-signal edge voxels.

In contrast, the DWI signal $z(\mathbf{x})$ scores each voxel against the whole brain: HIE lesions are hyperintense on $b=1000$ and stand out brain-wide rather than only locally, so a global reference suffices.

$$z(\mathbf{x}) = \frac{b_{1000}(\mathbf{x}) - \text{median}_{\text{brain}}(b_{1000})}{\max(\text{std}_{\text{brain}}(b_{1000}), 1)}. \quad (4.3)$$

Only positive z contributes to the combination.

Combined mask. The two criteria are combined by union, intersected with the brain mask. The final mask is stored as a binary volume where 1 marks voxels included in the registration cost and 0 those excluded:

$$\text{mask}(\mathbf{x}) = \text{brain}(\mathbf{x}) \wedge \neg(ADC(\mathbf{x}) < 1 \mu\text{m}^2/\text{ms} \vee c(\mathbf{x}) > \tau_c), \quad (4.4)$$

where τ_c is the confidence threshold. We set $\tau_c = 0.3$ for all masking in our pipeline. The value was tuned on the annotated D1.5T-U cohort based on the trade-off between flagged lesioned and healthy voxels across thresholds, and may need adjustment for cohorts with markedly different contrast.

GT-aware lesion mask. When ground-truth annotations are available, they replace the heuristic flags and combine with the same brain mask:

$$\text{mask}(\mathbf{x}) = \text{brain}(\mathbf{x}) \wedge \neg(GT(\mathbf{x}) > 0.5), \quad (4.5)$$

where $GT(\mathbf{x})$ is the binary lesion label. The 0.5 threshold is redundant for clean binary labels but is kept for consistency with Phase 2, where warping produces fractional edge values. The convention matches the GT-free mask: 1 included in the registration cost, 0 excluded.

4.2.3 Phase 1: Popdramms Atlas Construction

As introduced in Section 2.3.2, Popdramms is DRAMMS’s iterative groupwise registration framework for atlas-building [27], also used by [4]. In our pipeline, Popdramms registers on each subject’s $b=0$ volume, while masking out per-subject masks.

We initially tried to reproduce the pipeline of [4] directly, but using only ADC (their $b=0$ and $b=1000$ DWI are not released, Section 3), our maps were substantially less precise than theirs, even when registering to their published atlas.¹ The main cause appeared to be registration accuracy: HIE lesions are often small, so even voxel-level error displaces them from their true atlas-space location, degrading both atlas construction and per-patient scoring. We therefore developed our own variant.

A key deviation was to register on the $b=0$ DWI scan rather than the ADC map. The $b=0$ scan carries richer structural T2-like contrast across the whole brain, with clearer gray/white differentiation, cortical structures, and ventricle boundaries, whereas ADC is comparatively homogeneous. Since deformable registration is driven by intensity gradients between corresponding regions, this richer contrast provides more anatomical landmarks to align on. Atlas-space ADC statistics are then derived in Phase 2 by warping each subject’s $b=0$ and $b=1000$ to atlas space and computing ADC there from the warped pair.

Internal Popdramms Registration Framework Popdramms was used largely unmodified. We refer to the original paper [27] for the rigid, affine, and deformable atlas-building schedule and its SSD-based outlier rejection.² From the masked $b=0$ cohort, Popdramms returns a population-level $b=0$ atlas template and a per-subject deformation field, mapping each subject into a common atlas space. There, Phase 2 builds the ADC mean and std atlases separately, warping each subject’s $b=0$ and $b=1000$ via these fields and computing ADC.

Subjects dropped by SSD outlier rejection do not contribute to template-building but are still registered to the final atlas by standalone pairwise DRAMMS. This yields an equivalent deformation field, so every subject retains a native-to-atlas mapping. The inverse mapping (atlas $\rightarrow$ native) is computed from each forward deformation field and used in Phase 3 to warp per-patient Z_{ADC} maps back to native space.

4.2.4 Phase 2: Voxel-wise ADC Statistics in Atlas Space

Phase 2 produces per-voxel ADC μ and σ atlases from each subject’s $b=0$ and $b=1000$ DWI pair, using the atlas template and

deformation fields from Phase 1. As in earlier phases, lesion masks suppress contaminated voxels. The per-subject masks follow the mode-specific setup from Phase 0 (Section 4.2.2), but now in atlas space: each subject’s DWI pair is warped to atlas space via the Phase 1 deformation fields, and used to recompute ADC (Eq. C.1, Appendix C). The mask is then rebuilt from these warped volumes rather than warped directly, since warping a binary mask blurs lesion borders (Eq. 4.4 for GT-free, Eq. 4.5 for GT-aware). The iterative cohort-level outlier-rejection step then acts as a second-pass filter, removing lesion voxels that escape the masks before the final per-voxel μ and σ are aggregated.

Cohort-level Statistical Aggregation As introduced above, the final per-voxel statistics are computed in two steps: iterative outlier rejection across masked subjects, then aggregation over the survivors. At each atlas voxel $\mathbf{x}$, the cohort supplies one ADC value per subject, $x_i(\mathbf{x})$. Outliers are flagged by a modified Z_{score} of $x_i(\mathbf{x})$ relative to the cross-subject median ($\tilde{\mu}(\mathbf{x})$) and MAD ($\widetilde{MAD}(\mathbf{x})$) at that voxel, giving the modified Z_{score} :

$$Z_{mod}(\mathbf{x}, i) = \frac{0.6745 \cdot (x_i(\mathbf{x}) - \tilde{\mu}(\mathbf{x}))}{\widetilde{MAD}(\mathbf{x})}. \quad (4.6)$$

The resulting $|Z_{mod}|$ map is smoothed with a 5 mm FWHM Gaussian so that flagged deviations are spatially coherent rather than isolated noise. At each voxel, subject exceeding $|Z_{mod}(\mathbf{x}, i)| > 3.5$ are dropped from its contribution. The procedure runs for two iterations, recomputing the $\tilde{\mu}(\mathbf{x})$ and $\widetilde{MAD}(\mathbf{x})$ after each. The surviving sub-group is the aggregated at voxel $\mathbf{x}$ to form the final atlases: μ_{atlas} is the per-voxel pseudo-healthy mean, and σ_{atlas} the per-voxel root-mean-squared deviation from it.

The iterative step compliments the per-subject masking: the mask removes the clearly contaminated voxels, while the rejection catches those that escape it but still deviate at their atlas coordinate. It is most beneficial in GT-free mode, where the mask misses more lesion voxels, but it also helps GT-aware mode by flagging outliers or annotator-missed lesions.

4.2.5 Phase 3: Per-Patient Z_{ADC} Map Computation

Phase 3 produces per-patient Z_{ADC} maps in native space from the Phase 1 deformation fields, the per-patient atlas-space ADCs, and the cohort atlases from Phase 2. From this part of the pipeline, every step is mode-agnostic. Each patient’s voxel-wise Z-score is

$$Z(\mathbf{x}) = \text{clip}\left(\frac{ADC_{patient}(\mathbf{x}) - \mu_{atlas}(\mathbf{x})}{\max(\sigma_{atlas}(\mathbf{x}), 10^{-3})}, -10, +10\right), \quad (4.7)$$

where μ_{atlas} and σ_{atlas} are the Phase 2 mean and standard-deviation atlases. The floor on σ_{atlas} guards against division by zero where the whole cohort was excluded from aggregation, and the clip bounds numerical blow-ups. Each Z_{ADC} map is then warped back to native space via the patient’s inverse deformation field.³

¹We contacted the authors for further details that might explain the discrepancy but did not receive a response.

²To prevent distortion of the SSD outlier thresholds, Popdramms’s default cohort-wide intensity rescaling was disabled; see Appendix C.2

³This round-trip interpolates twice; warping the atlas to patient space directly (one warp) was tested, but interpolating the atlas proved worse than interpolating the patient map twice.

4.2.6 Phase 4: Mode-agnostic Post-processing

The Phase 3 Z_{ADC} maps are noisy from interpolation and registration error accumulated in earlier phases. Two native-space, mode-agnostic post-processing steps counter this before the maps are used downstream. Since post-processing is in native-space, we now use the user-provided skull-stripped ADC. These are cleaner, due to having proper brain-stripping that the internal $b_0 > 10$ validity check of the internally computed ADC might miss. The Phase 4 brain mask is simply $\text{brain} = (\text{ADC} > 0)$.

Confidence-Based Guided Correction Registration drift blurs voxel Z -scores, which is tolerable for severe lesions but catastrophic for small ones. To counter this, we re-anchor confident lesion voxels to a sensible Z -score. The multi-signal confidence score $c(\mathbf{x})$ is recomputed in native space (Eq. 4.1, using native $b=0$, $b=1000$, and ADC), but without its absolute-ADC criterion, which would also drag many healthy voxels down. The multi-signal $c > \tau_c$ is a much tighter lesion indicator, so the correction fires almost exclusively on true lesions. We reuse the same threshold $\tau_c = 0.3$ as used in the heuristic mask (Eq. 4.4). The correction itself improves the maps substantially, while the exact threshold mattered less in our tests (Appendix C.3), so we kept it consistent. However, it may still warrant tuning on new cohorts. Eligible voxels are those flagged with high confidence that still sit above the target $Z = -2.5$:

$$\text{corrected}(\mathbf{x}) = (c(\mathbf{x}) > \tau_c) \wedge (Z(\mathbf{x}) > -2.5) \wedge \text{brain}(\mathbf{x}). \quad (4.8)$$

The target is set just below the conventional -2 threshold for high deviation from the norm, since clearer lesions typically fall between two and three standard deviations below normal tissue. Voxels already at $Z \leq -2.5$ are left untouched. Rather than hard-assigning a lesion Z , the correction is a confidence-weighted nudge toward the target, so low- c voxels shift only slightly and high- c voxels shift more:

$$Z_{\text{new}}(\mathbf{x}) = (1 - c(\mathbf{x})) \cdot Z(\mathbf{x}) + c(\mathbf{x}) \cdot (-2.5). \quad (4.9)$$

Omitting this step reduces F1 and Dice by roughly 15%.

Brain-Median Recalibration After the guided correction, a Z_{ADC} map can still carry a small global offset, with healthy tissue sitting slightly off $Z = 0$ due to registration bias. We normalize this with a per-patient shift: the median of Z over brain voxels with $Z > -2.5$, excluding likely lesion voxels. Including them would pull the median down in lesion-heavy patients, so subtracting a more negative shift would push healthy tissue further from 0 rather than toward it. The exclusion keeps the recalibration a clean translation by a mostly healthy-tissue offset.

4.3 Training and Evaluation

4.3.1 Training Pipeline Overview

The full project-pipeline is visualized in Fig. 4.7, and uses three datasets — D3T-B, D1.5T-B, D1.5T-U — to evaluate within-scanner, cross-scanner, and cross-domain generalization. It

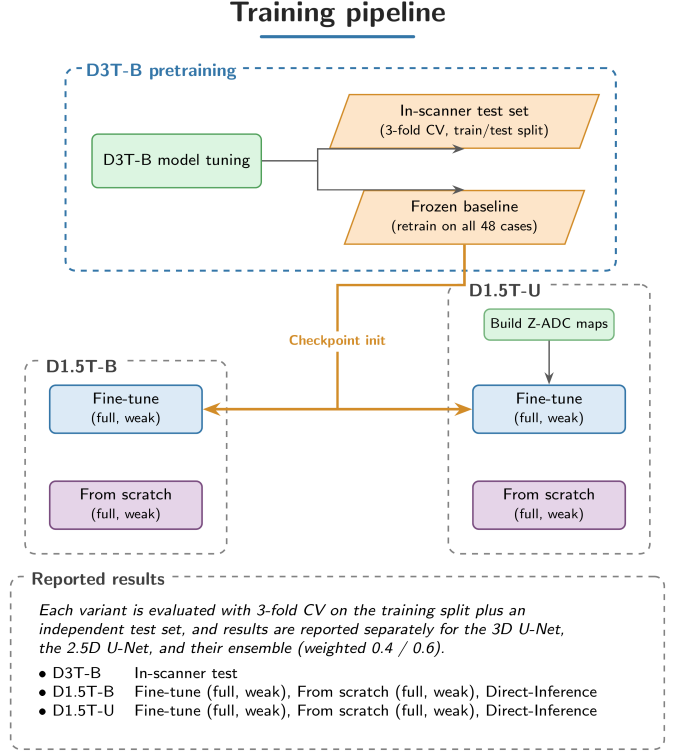


Figure 4.7: Overview of the training pipeline. The D3T-B baseline is trained in two phases: 3-fold cross-validation for hyperparameter tuning, followed by full 48-case training to produce a pre-trained checkpoint. This checkpoint then serves as the starting point for full- and weak-supervision fine-tuning on D1.5T-B and D1.5T-U.

starts from full-supervision pre-training, continuing on to both full- and weak-supervision fine-tuning. Across all training and testing, a three-channel input is used consisting of the skull-stripped ADC scans, the Z_{ADC} maps, and thresholded binary masks with $Z < -2$. For the Boston datasets we use the provided per-patient Z_{ADC} maps, while for the Utrecht dataset we generate our own using the pipeline described in Section 4.2. All full-supervision training uses the provided ground-truth annotations directly, while in weak-supervision experiments these are first converted into scribble-based annotations.

Downstream fine-tuning is evaluated on two cohorts, each targeting a distinct generalization question. The Boston 1.5T subset shares acquisition site with the pre-training data but differs in scanner field strength, isolating cross-scanner robustness. The Utrecht cohort differs in site, scanner, protocol, and treatment status, providing a test of cross-domain generalization. Additionally, on both datasets we evaluate the effectiveness of weak-supervision training using the frozen D3T-B checkpoint as a starting point.

Full subject-level split assignments per cohort are listed in Appendix D. Across all conditions, results are reported for the 2.5D model, the 3D model, and their probability-based ensemble. The evaluation metrics and ensembling procedure are defined in Section 4.4.

Initial tuning of model architecture, trainer design, and hyperparameters is performed on D3T-B through 3-fold CV. De-

sign decisions were made based on validation-set performance, while within-scanner generalization is reported on the held-out test folds.

Once these design choices were frozen for both the 2.5D and 3D models, we trained the final D3T-B baseline on all 48 cases without held-out testing for 200 epochs. This was the empirically optimal convergence point observed in the earlier validation runs.

4.3.2 Baseline: D3T-B

The first stage of our pipeline uses the D3T-B dataset and is carried out in two phases. We first split the dataset into a 3-fold train, validation, and test split. This allows us to tune the model architecture, trainer design, and hyperparameters based on validation-set performance, while within-scanner generalization is reported on the held-out test-splits of each fold.

Subsequently, with the design choices fixed for both the 2.5D and 3D models, we train the final D3T-B baseline on all 48 cases without validation or held-out testing. Both models run for 200 epochs. The resulting baseline serves as the pre-trained checkpoint for the next stages. It is the starting point for fine-tuning on D1.5T-B and for full-supervision fine-tuning on D1.5T-U. Most importantly, it enables weak-supervision training on the same D1.5T-U dataset, where the pre-trained weights provide a crucial prior for the structural knowledge that is harder to learn without full annotations.

4.3.3 Fine-Tuning Experimental Setup

Both the D1.5T-B and D1.5T-U cohorts exhibit distributional shifts relative to D3T-B in Z-dimension spacing, voxel spacing, and in-plane resolution. The cropping approach (Sections 4.1.3, 4.1.2) fixes the sampling region to the smallest brain bounding box across the cohort, rounded to the nearest training-friendly resolution (64, 96, 128, 196, or 256 voxels), meaning the crop size differs between pre-training and fine-tuning. D1.5T-B is additionally affected by high brain-size heterogeneity, causing the largest crop to introduce substantial zero-padding for smaller brains.

For D1.5T-B specifically, we explored resampling as a way to homogenize the spacing. The first was uniform resampling of all pretraining and fine-tuning volumes to isotropic $2 \times 2 \times 2$ mm spacing, homogenizing the cohorts. However, it resulted in significant smearing or wiping-out of the small lesions that characterize our data, while the tick-sliced volumes were particularly sensitive to such artifacts. The second approach used SynthSR to synthesize the missing slices, with its T1-weighted $1 \times 1 \times 1$ mm output serving as a coordinate grid into which the original ADC values were resampled. However, projecting ADC onto a T1-derived grid caused artifacts, and small lesions still suffered from smearing. In both cases the costs outweighed the benefits, and we retained native spacing across both cohorts. The crop-size mismatch was similarly left unaddressed. In both cases we hypothesize that the models are robust enough to adapt during fine-tuning.

Both downstream cohorts use a three-fold train/test split (Appendix D), fixed across all conditions for direct comparabil-

ity. Three experimental conditions are evaluated:

First, from-scratch training for 200 epochs serves as the lower-bound reference, establishing what the model can learn from the new data alone. No additional hyperparameter tuning is performed, as design choices were frozen during the D3T-B tuning stage.

Second, direct-inference transfer applies the pre-trained D3T-B checkpoint directly to the new cohorts without any fine-tuning, quantifying how much generalization is achieved from pre-training alone.

Third, fine-tuning initializes from the pre-trained checkpoint and trains for 100 epochs. This is assumed to be long enough to adapt to the new distribution while being short enough to preserve the structural prior learned during pre-training. On small datasets like D1.5T-B and D1.5T-U, training beyond this point risks fully overwriting the initial weights, effectively reducing fine-tuning to from-scratch training.

4.3.4 Cross-Scanner Robustness: D1.5T-B

D1.5T-B exhibits substantial data heterogeneity, with wide variation in both lesion size and acquisition parameters (Section 3.1). Initial experiments showed high performance fluctuation that was difficult to interpret. After further inspection, we identified a small subset of cases that dominated the variance: volumes with no lesions received zero Dice scores when the model predicted even a few spurious voxels, while very small lesions could disproportionately penalize an otherwise informative prediction if the predicted region was slightly too large. These edge cases made results highly sensitive to minor design choices or inter-run noise.

Given the small dataset, we did not redesign the pipeline for these cases: any such adjustment trades general robustness for a few challenging cases and risks overfitting to cohort-specific artifacts. Instead, training was kept uniform across cohorts, and we report metrics both with and without the zero-lesion cases where needed, highlighting the misleading effect a small subset of cases can have on the cohort-level metrics. The issue is partly one of limited representation and would likely diminish with more data.

4.3.5 Cross-Domain Generalization: D1.5T-U

The Utrecht cohort differs from D3T-B in acquisition site, scanner, imaging protocol, and treatment status, making it a useful cross-domain generalization test. Unlike the Boston datasets, no per-patient Z_{ADC} maps are provided, so these are generated using the pipeline described in Section 4.2 before any training or evaluation.

For this cohort, we additionally train all models on single-channel ADC input alongside the standard three-channel input. Unlike the Boston data, the raw ADC signal in D1.5T-U is sufficiently informative for lesion detection on its own (Section 3.1), making single-channel training a meaningful baseline. Comparing it against three-channel performance directly tests whether our Z_{ADC} map generation pipeline adds meaningful signal, where a consistent improvement would validate the

pipeline.

Additionally, for this cohort we compare against nnU-Net under both single- and three-channel input. nnU-Net failed on D3T-B, where lesion and healthy ADC overlap heavily. However, the non-TH-treated D1.5T-U has far clearer intensity separation (Section 3.3), so its patch-based, local-intensity design is more likely to work here. This makes it a suitable reference point, and the three-channel variant tests whether the Z_{ADC} maps help a general-purpose model as much as our own.

4.3.6 Weak-Supervision Experimental Setup

Full supervision requires fully annotated ground-truth labels, which are scarce in the field because annotation is time-consuming. Weak supervision addresses this by requiring only sparse labels, with three requirements for it to be a useful addition. First, weakly supervised models should perform significantly better than pure ADC- or Z_{ADC} -based thresholding. Second, the annotation effort should be substantially less than full labeling, justifying any drop in quality. Third, Z_{ADC} map generation should remain at a competitive level without needing full ground-truth labels, which we address through heuristics (Section 4.4)

The goal is not to match full-supervision quality but to produce results useful as a prior in clinical settings. Such priors can be used either to assist radiologists directly or, more importantly, as a starting point for annotators building full labels. If weak supervision can reduce the initial annotation burden by turning it into mostly a correction task, it could significantly lower the barrier to creating new fully annotated datasets. In turn, larger annotated datasets would further improve pre-trained models, with each step reinforcing the next.

For the weak-supervision experiments, the GT-free Z_{ADC} maps are used for D1.5T-U, while the provided maps are used for D1.5T-B. We compare weak supervision against two full-supervision references. To isolate the effect of weaker supervision alone, we first train and evaluate a full-supervision model on the same GT-free maps. In doing so, we set the performance target attainable with these maps. Then, the primary comparison is against the full-supervision runs built on GT-aware masks, which represent the upper bound of the full pipeline.

Weak-Supervision Label Generation Since the goal is to minimize annotation burden, we experiment with increasingly sparse label configurations to find the minimum annotation needed to still improve significantly over direct-inference performance.

Full supervision trains on a dense ground-truth mask where every voxel is labeled either lesion or healthy, with the loss

$$\mathcal{L} = 0.5 \text{Dice} + 0.5 \text{BCE}. \quad (4.10)$$

Weak supervision replaces this with a sparser labeling that assigns each voxel one of three labels: hard positive, hard negative, or ambiguous. Like in full-supervision, hard positives and hard negatives supervise the loss directly. Ambiguous voxels are left to a teacher to assign, as described below. Accordingly, the loss

consists of a hard term over the confident labels and a soft term over the teacher-resolved ambiguous voxels,

$$\mathcal{L} = \alpha [\text{Dice} + \text{BCE}_w]_{\text{hard}} + \lambda(t) [\text{Dice} + \text{BCE}]_{\text{soft}} \quad (4.11)$$

where α weights the confident labels and $\lambda(t)$ decays over training to largely fade out the teacher. The mapping from label type to loss contribution is summarized in Table 4.7.

Table 4.7: Weak-supervision voxel labels and their loss contribution. t denotes the teacher probability.

Voxel type	Origin	Contrib.	Loss
Scribble dot	confident	POS	hard
Outside lassos	brain	NEG	hard
Non-brain	—	NEG	hard
Lasso, $t > 0.7$	ambiguous	soft POS	soft
Lasso, $t < 0.3$	ambiguous	soft NEG	soft
Lasso, $0.3-0.7$	ambiguous	excl.	—

We generate these labels from the ground-truth masks by simulating what a clinician would produce in a quick annotation pass. The core idea is that an annotator should not need to label every voxel. They mark the lesions they are confident about, draw a loose boundary around each, and leave genuinely unclear tissue for later. We attempt to artificially reproduce each of these steps deterministically.

Clusters and subregions. Clusters are formed by connected-component grouping of the ground-truth annotations in each slice. Large lesions are then split, since an annotator might not treat a big spread-out lesion as one confident connected unit. We achieve this by dividing any cluster above 150mm^2 into subregions of roughly 50mm^2 through watershedding of the different intensity regions within the lesion. These thresholds follow the cohort’s cluster-size distribution: 150mm^2 is the ~ 90 th percentile, so only the largest clusters, holding the bulk of total lesion volume, are split. The subregion size of 50mm^2 (~ 75 th percentile, ~ 100 voxels) is a more typical cluster size that remains large enough for reliable intensity statistics. Smaller clusters stay whole. By creating these subregions, larger lesions can be partially confident and partially ambiguous, which is more realistic for a clinical setting where clinicians may be unsure about some areas of a lesion while confident about others.

In a clinical setting, annotators commit first to lesions that are less ambiguous, largely through contrast against nearby tissue in both ADC and DWI. To mimic this, we score each subregion’s ambiguity from local ADC and DWI values. Specifically, we take a 5 mm ring of healthy tissue around each lesion and score the subregion as

$$(\text{ADC}_{\text{ring}} - \text{ADC}_{\text{sub}})/\sigma + (\text{DWI}_{\text{sub}} - \text{DWI}_{\text{ring}})/\sigma.$$

Through this rough approach, we treat the top fraction with the most contrast as confident regions, while the rest are set as ambiguous. Scribbles are added only to the confident regions.

Scribbles. An annotator, in our hypothetical workflow, starts by placing as many quick dot-based scribbles in confident

lesions as possible, rather than directly tracing boundary outlines. We represent this by placing dots in the confident regions, with the number per cluster set by

$$N = \text{clip}(\text{round}(\text{area}/50 \text{ mm}^2), 1, 5). \quad (4.12)$$

Larger clusters are assigned a few more scribbles, so dots are assigned based on the size of the parent cluster, with small clusters getting a single dot and larger ones up to five. Dots are placed at the ground-truth cluster’s k-means centroids, with any dot landing in an ambiguous part of the parent cluster moved to the closest non-ambiguous voxel. As a result, each dot is a true positive, which is further dilated to a 3×3 in-plane click to increase hard-positive signal. Clusters with no confident subregion have no dot scribbles.

Lassos. Next, the annotator is asked to draw a large lasso around every confident or ambiguous subregion, with the goal of categorizing the remaining healthy voxels outside. The lasso mimics a quick hand-drawn boundary, extended non-uniformly to include tissue that looks abnormal, determined as tissue with intensity values similar to the lesion. To begin with, all voxels within a minimum distance of the lesion are included: 4 mm for confident regions and 5 mm for ambiguous ones. Beyond this base margin, the band grows only where the tissue still resembles the lesion, based on whether both its ADC and DWI stay similar to lesion intensities. This extension reaches up to 9 mm for confident regions and 11 mm for ambiguous ones. As such, the healthy voxels within a lasso are included based on tissue similarity rather than a fixed or random dilation, which is more clinically realistic.

The distances are set empirically, aiming to stay precise where possible while leaving more room in ambiguous regions. Ambiguous regions are given slightly larger margin to reflect that an annotator would leave more room where they are less certain of the boundary.

In a post-processing step, we tighten the lasso by excluding all non-brain voxels and obviously bright voxels, through

$$\text{ADC}(x) > \mu_{\text{lesion}} + 3 \sigma_{\text{lesion}}.$$

This catches regions an annotator would never include in a lasso. Small holes within lassos are filled and the edges are smoothed with an in-plane Gaussian, so the union of lassos has a hand-drawn look. Figure 4.8 illustrates this across three cases, with the lasso tight where the lesion is sharply distinct and looser where it blends into surrounding tissue.

There is a deliberate bias toward over-inclusion, reflecting that in our setting we would rather capture healthy tissue than miss lesion voxels, since the model can handle ambiguous regions later. We expect an annotator would rather trim an over-inclusive lasso than reconsider the scan afterwards for clusters that were missed entirely. Due to the bright-voxel exclusion, lassos sometimes sit very close to the lesion boundary, which is realistic, since an annotator could also draw tightly in such more obvious areas.

Resulting labels. This weak annotation produces the three label types of Table 4.7. Hard positives are the dilated

scribbles in confident regions. Hard negatives are all voxels outside every lasso, together with non-brain tissue. Ambiguous voxels are the lasso interiors without a scribble, handled by the teacher during training. Since we want to test whether a few grounded hard positives benefit results, we keep scribbles and scribbled lassos as separate regions, rather than marking all lasso tissue as ambiguous.

Annotation sweep. The confident lasso fraction is swept from 0 to 100% in eleven steps (lf0 to lf100) to test the importance of these added hard scribbles. At lf0, every subregion is ambiguous, with lasso interiors resolved by the teacher. At lf100, every cluster contains scribbles. Figure 4.9 shows this on a single case, where lassos flip from ambiguous to confident as the fraction rises. The results highlight the trade-off between annotation effort and meaningful improvement over direct-inference, compared also against the full-supervision upper bound.

Training. The teacher supplies positive signal inside the lassos and resolves the ambiguous region through the soft term introduced above. It is influential early in training, then fades to a guard against the model collapsing to all-zero predictions. The teacher is imperfect, but higher confidence indicates a more reliable label, and the thresholds governing its contribution are those in Table 4.7. They balance the need for additional positive signal against the risk of poisoning training with unreliable teacher predictions. Initially, we tested separate teacher weights for the confident and ambiguous lassos. However, we found the thresholds strict enough that treating both identically worked well. This allowed for the inclusion of more of the ambiguous region, which proved beneficial. The effect may not hold for true annotations, which are not yet available to test.

The teacher weight $\lambda(t)$ decays linearly over the first 100 epochs and then stays constant, while α is fixed throughout. The two models use different ranges: 0.25 to 0.05 for the 3D model and 0.5 to 0.2 for the 2.5D model. The 2.5D model keeps the higher range because it is more prone to collapsing under low teacher influence. The 3D model is more stable and less prone to collapse, so giving its supervision a slightly larger weight is not harmful. In both cases this keeps the teacher influential early, when the model needs positive signal to avoid collapse, and reduces its influence as the model learns to segment reliably on its own.

The teacher predictions are precomputed from pre-trained teacher models and fixed before training starts. Only the weight given to them changes over training, not the predictions. The 3D model has a lower ceiling, but is also less prone to collapsing, as it relies heavily on global information that is less sensitive to the type of annotation. It is therefore trained first, using the frozen Boston model both as initialization and as a teacher. Its predictions are then recomputed with the weak labels as input, and this click-aware 3D model serves as the teacher for the 2.5D model, whose initialization weights still come from the pre-trained Boston model. This lets us exploit the higher ceiling of the 2.5D model while reducing the risk of collapse to all-zero predictions.

To increase the chance of the model predicting as many lesion clusters as possible, which can be refined later, we add the

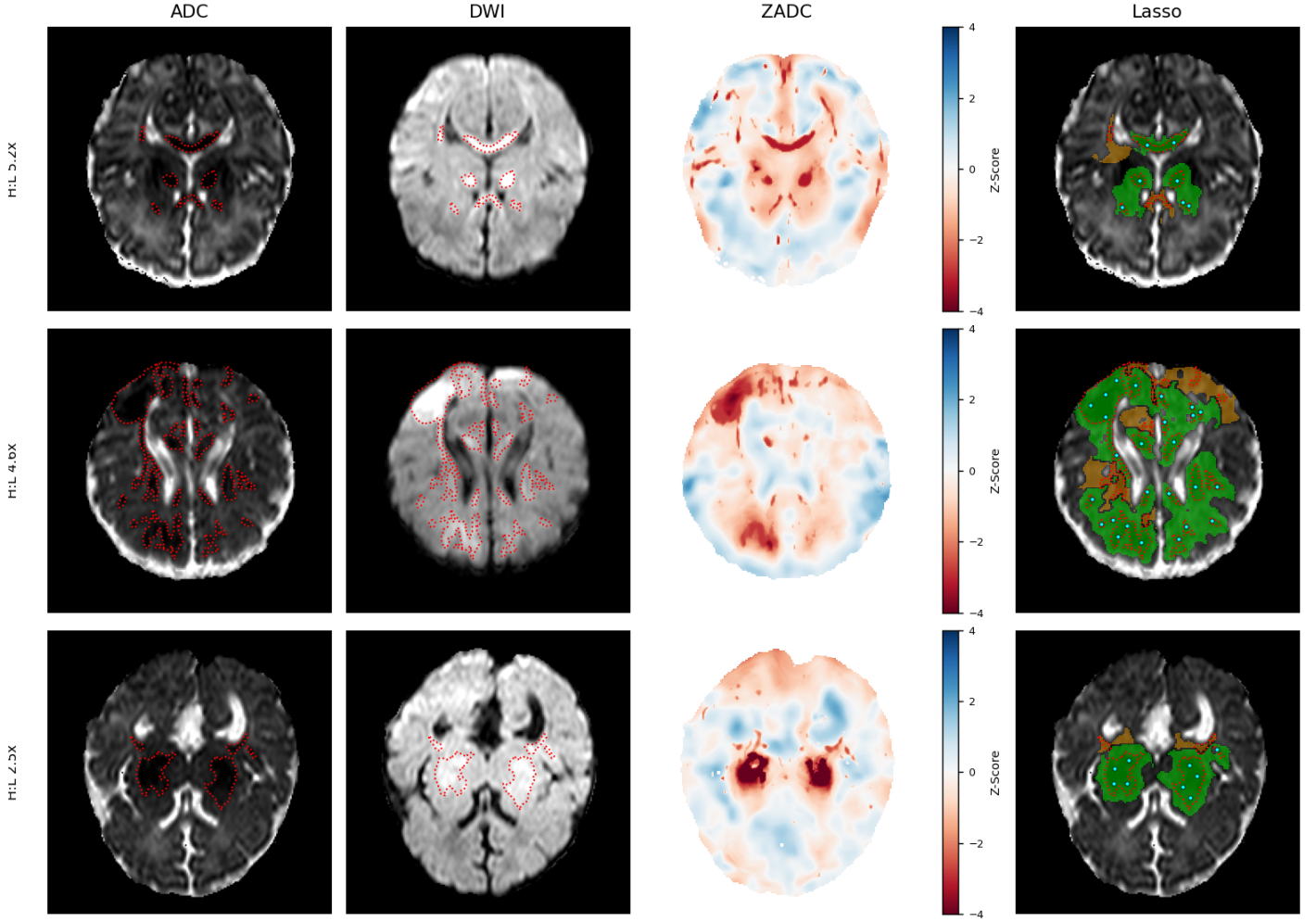


Figure 4.8: Generated lassos for three cases, shown next to their ADC and DWI inputs. The cases were chosen to span different ratios of healthy (H) to lesion (L) tissue inside the lasso (H:L). The boundary follows ADC and DWI similarity to the lesion rather than its true edge, staying tight where the lesion stands out and loosening where it blends into nearby tissue. This resembles the loose, imprecise outline of a quick annotation pass.

hard lesion scribbles as an extra input channel in both models. The pretrained first convolution is zero-padded to accept this added channel. During testing we revert to the original full-supervision channels in the standard label-free test setting. In the next paragraph, we describe a label-aware setting, which does this differently.

Human-in-the-loop Approach Our main goal is to combine computer-assisted lesion detection with limited annotation effort, turning a sparse prior into full ground-truth annotations. To this end, we also experiment with providing weak annotations at test time, giving the model a stronger prior when predicting lesions. We do this by training a single model on all 20 cases without a held-out test set. After which, testing is done on each case with the same model, which has seen all cases during training.

The scribbles are added as an additional input channel. After inference the prediction is restricted to the annotated region. Concretely, the final mask combines the model prediction with the annotated clicks, keeps only what falls inside the lasso re-

gion, and removes annotated hard negatives:

$$\hat{Y}_{\text{final}} = (\hat{Y}_{\text{pred}} \cup C) \cap L \setminus N, \quad (4.13)$$

where $\hat{Y}_{\text{pred}}$ is the binarized model prediction after inference, $\hat{Y}_{\text{final}}$ is the final post-processed lesion mask used for evaluation, C is the set of click voxels, L is the union of all lasso regions, and N represents the annotated hard negatives.

This setup intentionally provides the weak prior at test time, which is a form of leakage, but an intended one in our setting. The model only ever sees the clinician’s sparse annotations, never full ground-truth masks. This reflects the intended use, where the model refines a clinician’s annotations for the patient being annotated rather than generalizing to an unseen one. Training on all cases gives the model as much data as possible, and it outperforms the setting with held-out test cases, highlighting the benefit of the added training data.

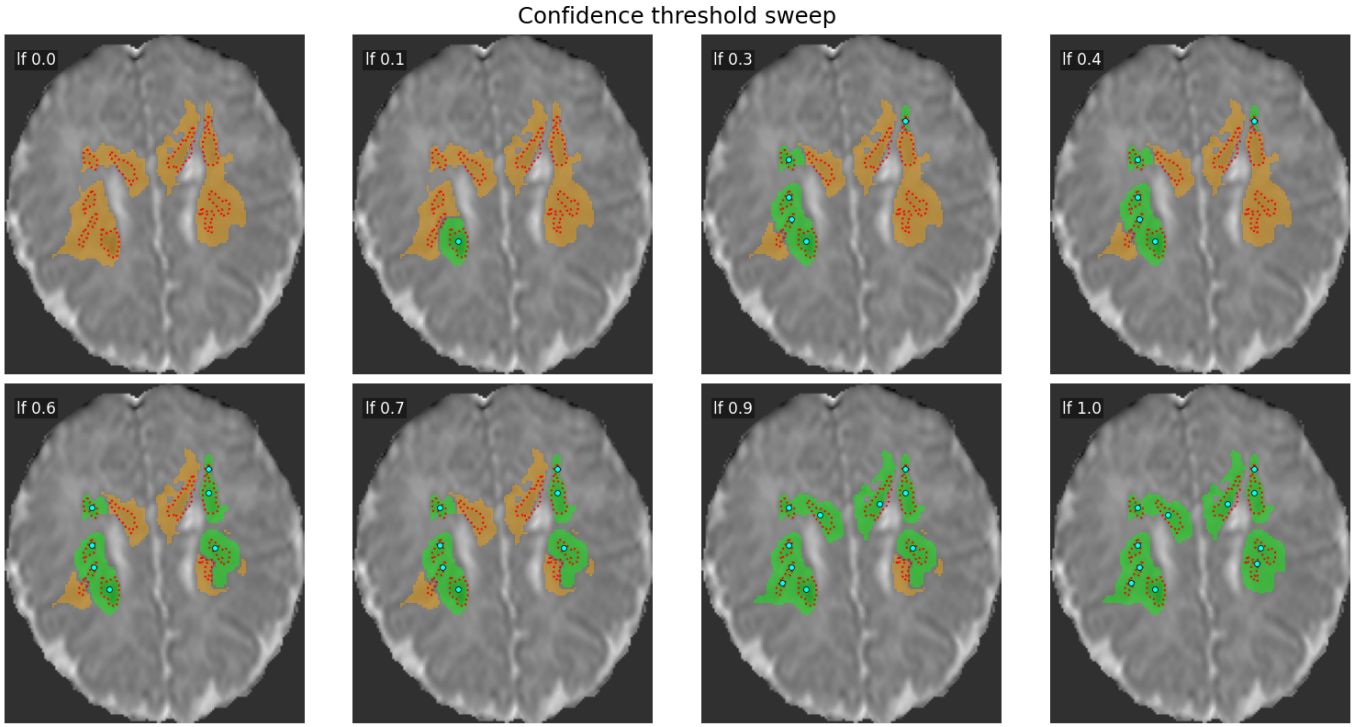


Figure 4.9: Different levels of sparse supervision on a single multifocal subject, shown for 8 of the 11 sweep steps from lf0 to lf100. As the confident fraction rises, lesion lassos flip from ambiguous (orange) to confident (green). The number of resulting scribbles (blue dots) is the main message, since it sets the amount of hard supervision.

4.4 Evaluation Metrics

Segmentation performance of our models is quantified through several complementary metrics. The primary metric is the Dice Similarity Coefficient, a widely used measure of spatial overlap between predicted and ground-truth segmentations. However, Dice alone does not fully capture topological and boundary-level detail.

To address these limitations, we also report surface-based metrics. Following the evaluation protocol of the *BONBID-HIE Challenge* [4], we use the Mean Average Surface Distance (MASD) and the Normalized Surface Distance at a tolerance of 2mm (NSD@2mm). MASD measures the average distance, in millimeters, between the predicted and ground-truth lesion boundaries. NSD instead reports the fraction of the boundary that lies within a fixed tolerance of 2mm of the reference surface. NSD, like Dice, ranges from 0 to 1 (higher is better), but is computed over surfaces rather than volumes. Because deviations within the tolerance are not penalized, NSD reflects clinically acceptable boundary disagreement. We additionally report the 95th-percentile Hausdorff Distance (HD95): the largest boundary deviation after discarding the top 5% of distances, which makes it robust to isolated voxel noise. Together, these are complementary: MASD captures typical boundary error, NSD the fraction of boundary within tolerance, and HD95 large local discrepancies. Finally, voxel-level counts (True Positive (TP), False Positive (FP), False Negative (FN), True Negative (TN)) are reported for a more detailed assessment of model behavior.

Following the convention of [4] and other work in the field, we report the mean for each metric. However, the conventional

$\pm$ std spread is problematic here: metrics are bounded (Dice, NSD in $[0, 1]$; MASD, $\text{HD95} \geq 0$), and large outliers make the spread skewed rather than symmetric, so mean $\pm$ std can fall outside the valid range, for example implying a negative distance. We therefore also report the median and the 25th–75th percentile (IQR) range alongside the mean. The median gives an outlier-robust measure of typical performance, while the IQR stays within bounds and highlights the skewed distribution.

4.4.1 Ensembles

The 3D U-Net and 2.5D SegResNet have complementary strengths: the 3D model’s full-volume processing supports stronger global awareness, while the 2.5D model’s higher in-plane resolution supports stronger learning of local patterns. We therefore combine them by averaging their predicted probabilities, weighting the 2.5D SegResNet by ω and the 3D U-Net by $1 - \omega$, and final thresholding at $p = 0.5$. We set $\omega = 0.6$, chosen on validation performance to reflect the 2.5D model’s more consistent behavior across our cohorts, with the 3D U-Net regularizing the ensemble where slice-wise processing is more error-prone. The optimal ω may differ on new cohorts. We discuss it, together with the pipeline’s other hyperparameters and how they might be tuned for unseen cohorts, in Section 6.1.6.

Results

5.1 Baselines

5.1.1 State-of-the-Art

A direct comparison to the state of the art automated HIE-segmentation performance is not possible. The only published HIE benchmarking dataset, BONBID, withholds its test set [4], and that set differs from our setup in ways that would make any comparison misleading. It mixes 1.5T and 3T patients, and retains cases we exclude throughout: patients scanned more than 7 days after birth, and easier non-TH-treated cases that we drop from D3T-B. Additionally, the challenge models were trained on far more data: an 85-case training set, versus a largest training set of 32 in our setup.

For context only, the top reported BONBID-2023 Dice scores fall around 0.62–0.67. That said, due to the above discrepancies we report our own results without benchmarking against these numbers.

A different reference point for segmentation performance comes from the inter-annotator agreement published for D1.5T-U [26]. They report a median Dice of 0.56 (range 0.22–0.83) between the experienced annotator, whose annotations we use as ground truth in our experiments for D1.5T-U, and an inexperienced second rater,¹ giving a clinically grounded reference against which our model’s agreement with the expert can be judged.

5.1.2 Thresholding

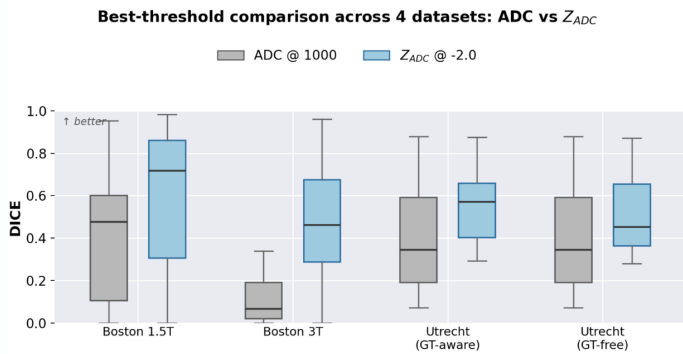


Figure 5.1: Threshold-based Dice across cohorts. Reported for ADC@1000 and the Z -score threshold $Z < -2$, across D3T-B, D1.5T-B, and D1.5T-U (GT-aware and GT-free Z_{ADC} maps).

Additionally, we report Dice scores at a global ADC threshold of ADC@1000, and our best-tested Z -score threshold of $Z < -2$.

¹They report inter-rater agreement as median with min–max, whereas our model metrics use the median with 25th–75th percentile range. The two spreads are not directly comparable but give a good sense of the range of annotation differences.

Results are shown for D3T-B, D1.5T-B, and D1.5T-U under GT-aware and GT-free Z_{ADC} maps (Figure 5.1).

The Z_{ADC} threshold outperforms ADC thresholding on Dice within every dataset. ADC thresholding fails completely on D3T-B and is more variable on the other cohorts, where the FP:TP ratio and intensity separation play a role, as discussed in Section 3.1. These thresholds set the floor that any learned model must exceed.

5.1.3 GT-free Masking Heuristic

The effectiveness of the two GT-free masking criteria, the absolute-ADC threshold and the confidence score, is evaluated against the ground-truth annotations at a range of ADC and τ_c thresholds. The confidence-score sweep is also relevant to the Phase 4 guided correction, which is mode-agnostic and reuses the same score in both GT-aware and GT-free Z_{ADC} post-processing. We also report the union of the two criteria, sweeping τ_c with the ADC threshold fixed at its production value, to show why we combine them rather than relying on either one alone. The sweep is reported in Table 5.1.

Table 5.1: GT-free masking heuristic on D1.5T-U (n=20). Per-patient median [Q1, Q3]. Production settings in bold.

Criterion	Recall (%)	Precision (%)
<i>Absolute ADC threshold</i>		
ADC<900	33.4 [7.7, 62.4]	81.1 [57.1, 90.7]
ADC<950	49.0 [15.3, 77.2]	62.4 [33.1, 84.9]
ADC<1000	60.9 [26.9, 86.7]	41.1 [22.6, 73.9]
ADC<1050	73.4 [38.5, 92.9]	31.1 [14.6, 60.5]
<i>confidence score</i>		
$\tau_c > 0.1$	61.4 [45.5, 70.7]	45.1 [28.5, 62.5]
$\tau_c > 0.2$	52.2 [37.1, 59.6]	58.5 [42.5, 74.0]
$\tau_c > 0.3$	45.6 [29.1, 50.8]	69.1 [54.3, 80.9]
$\tau_c > 0.5$	33.8 [17.9, 40.6]	79.3 [65.2, 88.6]
<i>Combination ($\vee$ ADC<1000)</i>		
$\tau_c > 0.1$	73.3 [61.9, 88.2]	33.0 [19.0, 53.2]
$\tau_c > 0.2$	68.9 [53.8, 88.0]	36.8 [26.1, 63.0]
$\tau_c > 0.3$	66.7 [48.3, 88.0]	41.6 [29.4, 66.9]
$\tau_c > 0.4$	65.3 [45.8, 88.0]	44.1 [30.5, 69.0]
$\tau_c > 0.5$	64.2 [43.8, 88.0]	45.0 [30.6, 70.4]

Every threshold is a recall–precision trade-off, with more lenient settings raising recall at the cost of precision. The two criteria sit at opposite ends of this trade-off. The absolute-ADC threshold has the highest recall, since most lesion voxels fall below a low ADC value, while having low precision, as it flags any voxel below the cut-off regardless of context. The confidence score is the inverse: substantially more precise, but with lower recall, missing lesion voxels that are not locally anomalous on both ADC and DWI.

The combination improves on either alone. At the production setting ($\text{conf} > 0.3 \vee \text{ADC} < 1000$), recall (66.7%) exceeds both the ADC threshold (60.9%) and the confidence score (45.6%), as

the score adds locally contrasted lesion voxels that ADC alone misses. The added voxels are mostly true positives, so precision barely drops relative to $\text{ADC} < 1000$, while the lower tail improves. The confidence threshold τ_c was set to 0.3 from the confidence-score sweep, where it balances recall and precision. In union with the ADC threshold τ_c has a positive but limited effect, with no single optimal value standing out, so 0.3 is a balanced choice but a nearby value would be equally defensible.

The ADC threshold is the more sensitive of the two settings. $\text{ADC} < 950$ misses many lesion voxels, while $\text{ADC} < 1050$ includes large healthy regions of low local ADC. As such, $\text{ADC} < 1000$ was chosen as the balance between both. This works well on D1.5T-U but would likely need re-tuning for cohorts with different ADC scales or contrast

Finally, the confidence score is also reused in the Phase 4 guided correction, where it selects the voxels to re-anchor. The absolute-ADC criterion is dropped here, as its low precision (Table 5.1) drags too many healthy voxels into the correction. The confidence trigger improves segmentation substantially over no correction. However, like before, the exact value of τ_c has limited effect, so we keep the same constant value of 0.3. The full trigger ablation is reported in Appendix C.3.

5.2 nnU-Net

As discussed in Section 4.1.1, initial testing was done on D3T-B with the nnU-Net model. Further testing was done on D1.5T-U, where the clearer intensity separation of its non-TH-treated cohort (Section 3.3, Table 3.2) suits nnU-Net’s patch-based, local-intensity design. We tested both a single-channel ADC setup and the three-channel input with the GT-aware Z_{ADC} maps, to assess the effectiveness of our Z_{ADC} pipeline. The GT-free maps were not run on nnU-Net for compute reasons. Instead, their benefit over raw ADC is shown on our own models (Section 5.6 Table 5.8).

The results are reported in Table 5.2. On D3T-B, ADC-only training was run early and disregarded for barely learning, so we report only the 3-channel Z_{ADC} version to highlight the reason for switching models. On D1.5T-U we ran and report both input-channel variants, as described above. D1.5T-B was not run, as further training of nnU-Net was deemed too costly in time and compute for a model that was already outperformed by ours.

Even with Z_{ADC} maps, nnU-Net struggles to segment robustly on D3T-B. Its mean dice sits below even pure thresholding at $Z < -2$, while the low end of the distribution is especially poor: the low Q1 metrics reveal failure on a large portion of the patients. On D1.5T-U it performs far better. ADC-only training already matches the inter-annotator agreement on this cohort, so even the single-channel ADC model exceeds an inexperienced human annotator. Adding the our Z_{ADC} -based channels improves every metric, demonstrating the effectiveness of the pseudo-normative spatial priors and pushing the model well clear of the inexperienced-rater reference.

5.3 Baseline Development D3T-B

The three-fold results on D3T-B are shown in Table 5.3. The results far exceed nnU-Net’s performance on the same dataset: the IQR sits much higher, with strong upper quartiles and relatively stable lower ones. On cohort-level metrics, the 3D model outperforms the 2.5D model in all aspects, while the 60/40 ensemble in favor of the 2.5D model proves detrimental. The IQRs are very similar across the three models, yet their means differ more, pointing to a few extreme outliers beyond the quartile range. These are confirmed by the finite MASD numbers, which reveal 5–6 cases consistently misclassified, either as FPs in zero-lesion cases or as zero-lesion predictions in lesioned patients.

The HD95 results are poor relative to other metrics. Lesions occupy only a small fraction of the brain (Section 3.1), and FP clusters tend to lie far from the true lesion boundaries. The high standard deviation reflects this variance: some cases achieve near-perfect borders while others fare much worse.

Predictions for two zero-lesion cases warrant additional inspection, as a single misclassified voxel shifts the mean Dice by approximately 0.03. At the voxel level, the picture is more nuanced: as shown in Table 5.4, the FP predictions are negligible relative to total brain size.

Table 5.4: FP predictions on the two zero-lesion cases in D3T-B. Voxel counts; final column is FP as a percentage of brain volume.

Case	Brain	2.5D FP	3D FP	% brain
MGHNICU_329	42,854	33	22	0.05–0.08
MGHNICU_432	28,117	1	0	0.004

Restricting evaluation to the 46 lesioned patients confirms that the two zero-lesion cases have a disproportionate effect on the cohort-level metrics rather than reflecting a genuine large gap between models. The 2.5D model misclassifies both, while the 3D model handles one, so excluding these cases leaves the models much more similar, with the ensemble overtaking both on mean Dice and $\text{NSD}@2\text{mm}$. This matches the earlier discrepancy between IQR and mean: removing the outliers brings the means into line. The full metrics are reported in Appendix E.1 (Table E.1).

Apart from the healthy cases, Table 5.5 shows three further failure modes. First, several lesioned cases are predicted as empty, nearly all involving tiny lesions. The exception is MGH-NICU_234, a large lesion case for which the 3D model predicts zero lesion voxels, likely because 31 of the 32 training cases seen across its CV folds had lesion fractions below 5%. Second, high-disagreement cases ($\Delta\text{Dice} > 0.20$) reflect genuine ambiguity between architectures on very small lesions. Third, MGH-NICU_439 achieves low Dice across all modes, again on a tiny lesion.

The difficulty of this cohort is also evident at the voxel level. As with the two zero-lesion cases, small lesions in these same difficult cases (Table 5.5) mean that a handful of FP or FN voxels can determine a case’s Dice almost entirely, with small isolated FP clusters inflating MASD and HD95 more than in cohorts with larger lesions. This effect operates in both directions: a few FPs can substantially reduce an otherwise reasonable pre-

Table 5.2: Fully-supervised nnU-Net (ResEnc-M, 200 ep), 3-fold CV, reported as mean and median [Q1, Q3] (25th–75th percentile) per test set. D1.5T-U uses 1 channel (ADC) or 3 channels (ADC, Z_{ADC} , binary mask at $Z < -2$); D3T-B uses the same 3 channels. Distances (MASD, HD95) are in mm. On D3T-B, 6/48 cases gave empty predictions, excluded from MASD/HD95 ($n = 41$ with finite surface distances).

Test set	Dice $\uparrow$		MASD (mm) $\downarrow$		HD95 (mm) $\downarrow$		NSD@2mm $\uparrow$	
	mean	median [Q1, Q3]	mean	median [Q1, Q3]	mean	median [Q1, Q3]	mean	median [Q1, Q3]
D3T-B (3-ch)	0.37	0.37 [0.13, 0.57]	6.39	3.24 [1.97, 8.84]	21.67	17.77 [8.94, 34.37]	0.48	0.51 [0.24, 0.72]
D1.5T-U (1-ch)	0.63	0.66 [0.55, 0.76]	1.74	1.18 [0.73, 2.35]	10.61	9.34 [4.25, 16.57]	0.75	0.82 [0.65, 0.88]
D1.5T-U (3-ch)	0.69	0.69 [0.62, 0.76]	1.36	1.09 [0.76, 1.92]	9.63	5.42 [4.39, 16.19]	0.81	0.84 [0.74, 0.89]

Table 5.3: D3T-B test-set metrics under 3-fold CV, reported as mean and median [Q1, Q3] (25th–75th percentile) per architecture. Distances (MASD, HD95) are in mm; their statistics are over cases with a finite boundary metric only. The number of cases with a finite Dice vs MASD (n_{fin}) is 43/48 for the 2.5D and 3D models and 42/48 for the ensemble.

Arch	Dice		MASD (mm)		HD95 (mm)		NSD@2mm	
	mean	median [Q1, Q3]	mean	median [Q1, Q3]	mean	median [Q1, Q3]	mean	median [Q1, Q3]
2.5D	0.53	0.55 [0.39, 0.78]	4.73	2.43 [0.98, 4.88]	20.18	20.45 [7.38, 31.23]	0.62	0.67 [0.49, 0.87]
3D	0.57	0.60 [0.41, 0.80]	3.43	1.90 [0.87, 4.00]	17.45	12.88 [4.64, 27.62]	0.66	0.71 [0.52, 0.89]
Ens.	0.56	0.59 [0.42, 0.79]	3.79	2.05 [1.00, 4.88]	18.55	13.64 [6.89, 30.23]	0.63	0.71 [0.50, 0.87]

Table 5.5: D3T-B: difficult cases — Dice across models. *GT vox* is the ground-truth lesion voxel count; *2.5D*, *3D*, and *Ens.* are the CV Dice scores.

Case	Type	GT vox	2.5D	3D	Ens.
MGHNICU_329	healthy	0	0.000	0.000	0.000
MGHNICU_432	healthy	0	0.000	1.000	0.000
MGHNICU_234	lesion miss	36459	0.965	0.000	0.966
MGHNICU_331	lesion miss	14	0.000	0.000	0.000
MGHNICU_377	lesion miss	14	0.000	0.353	0.000
MGHNICU_448	lesion miss	11	0.000	0.000	0.000
MGHNICU_453	lesion miss	13	0.000	0.000	0.000
MGHNICU_370	Dice ($\Delta > 0.20$)	24	0.309	0.095	0.283
MGHNICU_447	Dice ($\Delta > 0.20$)	21	0.000	0.593	0.000
MGHNICU_452	Dice ($\Delta > 0.20$)	13	0.483	0.870	0.600
MGHNICU_439	Low Dice	106	0.294	0.215	0.299

diction, while a missed lesion of only a few voxels reduces Dice to 0. The sole exception is MGHNICU_234, whose missed large lesion represents full model failure. The full per-case TP/FP/FN voxel breakdown is provided in Appendix E.2 (Table E.2).

5.4 Cross-Scanner Generalization and Finetuning

The three-fold CV results on D1.5T-B are shown in Table 5.6. Results are better than for D3T-B across all training types. Patients are on average more severely injured, so lesions are easier to detect and individual FP/FN errors are less prominent. The higher lesion burden leads to improved boundary metrics, as FP clusters are more likely to fall near a true lesion. From-scratch training performs worst, except for the ensembled boundary metrics. The 2.5D model outperforms the 3D model across the board, with the larger lesions reducing the importance of the global spatial context that was critical in D3T-B.

The median Dice and NSD@2mm are comparable across all three training types, yet the means diverge. From-scratch and even fine-tuning sit slightly below direct-inference on mean Dice and NSD@2mm, while the boundary metrics show the opposite measurement, with both trained settings ahead of direct-inference.

This split points to a tail of a few cases pulling the means down rather than a broad performance gap. A missed lesion drops Dice to zero but leaves MASD and HD95 unaffected, since these are computed only over cases with a finite boundary metric. A handful of missed very-small or zero-lesion cases can therefore significantly affect mean Dice and NSD@2mm without touching the boundary means. The pretrained features are suited to handling these cases, as they are close to the pretraining distribution of D3T-B and under-represented in D1.5T-B. They score near-perfect under direct-inference, and that gain outweighs fine-tuning’s broader improvements on the mean.

Similar to D3T-B, removing the three zero-lesion cases confirms this and changes the ranking (Appendix E.3, Table E.3): the means come back into line with the median and quartile patterns, and from-scratch becomes fully comparable to fine-tuning and direct-inference, with better boundary metrics. The mismatch in sampling distribution between D3T-B pretraining and D1.5T-B fine-tuning, together with spacing differences across cohorts, is more apparent here. From-scratch performs better than Table 5.6 indicates, while fine-tuning improves boundary precision over direct-inference.

5.5 Cross-Domain Generalization and Finetuning

The segmentation performance on D1.5T-U is quantified in Table 5.7. It is better than for both Boston cohorts across all

Table 5.6: D1.5T-B test-set metrics, reported as mean and median [Q1, Q3] (25th–75th percentile) per architecture. Distances (MASD, HD95) are in mm; their statistics are over cases with a finite boundary metric only. The number of cases with a finite Dice vs MASD (n_{fin}) is 25/28 for *from-scratch*, 26/28 for *fine-tuned*, and 27/28 for *direct-inference* (26/28 for the 3D direct-inference performance).

Arch	Dice		MASD (mm)		HD95 (mm)		NSD@2mm	
	mean	median [Q1, Q3]	mean	median [Q1, Q3]	mean	median [Q1, Q3]	mean	median [Q1, Q3]
<i>From scratch</i>								
2.5D	0.61	0.76 [0.38, 0.85]	1.93	1.04 [0.57, 2.19]	10.38	4.57 [3.39, 14.27]	0.66	0.80 [0.48, 0.91]
3D	0.59	0.72 [0.36, 0.85]	2.04	1.12 [0.62, 2.24]	10.73	5.63 [2.72, 12.59]	0.66	0.80 [0.46, 0.89]
Ens.	0.61	0.76 [0.39, 0.85]	1.73	1.06 [0.53, 1.86]	9.60	4.57 [3.28, 12.15]	0.67	0.81 [0.54, 0.91]
<i>Fine-tuned</i>								
2.5D	0.66	0.76 [0.41, 0.90]	1.87	1.02 [0.53, 2.19]	10.09	4.65 [2.82, 14.52]	0.71	0.81 [0.56, 0.92]
3D	0.61	0.74 [0.39, 0.84]	1.95	1.13 [0.57, 2.43]	9.97	5.47 [3.64, 12.56]	0.68	0.82 [0.48, 0.90]
Ens.	0.66	0.77 [0.44, 0.91]	2.09	0.99 [0.50, 1.94]	10.30	5.32 [2.63, 11.54]	0.71	0.83 [0.52, 0.93]
<i>Direct-inference</i>								
2.5D	0.68	0.76 [0.45, 0.95]	2.19	1.12 [0.33, 2.69]	11.39	6.25 [2.30, 18.66]	0.73	0.83 [0.58, 0.95]
3D	0.61	0.74 [0.34, 0.91]	2.43	1.14 [0.50, 3.00]	11.18	7.59 [2.67, 16.24]	0.65	0.74 [0.42, 0.93]
Ens.	0.68	0.76 [0.42, 0.95]	2.29	1.21 [0.31, 2.61]	11.37	6.44 [2.30, 16.38]	0.72	0.80 [0.56, 0.95]

Table 5.7: Utrecht test-set metrics, reported as mean and median [Q1, Q3] (25th–75th percentile) per architecture. From-scratch training is denoted by *cv*, fine-tune by *cv_ft*. Distances (MASD, HD95) are in mm; their statistics are over cases with a finite boundary metric only. All 20 cases have a finite Dice vs MASD ($n_{\text{fin}} = 20/20$ in every condition).

Arch	Dice		MASD (mm)		HD95 (mm)		NSD@2mm	
	mean	median [Q1, Q3]	mean	median [Q1, Q3]	mean	median [Q1, Q3]	mean	median [Q1, Q3]
<i>From scratch (cv)</i>								
2.5D	0.72	0.72 [0.66, 0.79]	1.25	1.07 [0.76, 1.33]	8.17	6.12 [4.61, 10.90]	0.82	0.86 [0.81, 0.88]
3D	0.64	0.65 [0.55, 0.70]	2.04	1.61 [1.01, 3.00]	13.35	10.92 [5.26, 21.14]	0.74	0.78 [0.63, 0.85]
Ens.	0.72	0.73 [0.67, 0.79]	1.18	0.96 [0.74, 1.26]	7.37	5.16 [4.21, 7.89]	0.83	0.86 [0.81, 0.89]
<i>Fine-tuned (cv_ft)</i>								
2.5D	0.69	0.70 [0.62, 0.79]	1.58	1.23 [0.79, 2.30]	10.96	5.79 [4.62, 18.51]	0.80	0.82 [0.76, 0.89]
3D	0.62	0.64 [0.54, 0.71]	2.52	2.27 [1.14, 3.55]	15.71	13.90 [7.08, 24.12]	0.71	0.74 [0.63, 0.81]
Ens.	0.70	0.72 [0.63, 0.79]	1.51	1.15 [0.75, 1.81]	10.06	6.02 [4.22, 16.64]	0.81	0.83 [0.76, 0.89]
<i>Direct-inference</i>								
2.5D	0.58	0.59 [0.43, 0.68]	3.34	2.57 [1.48, 4.85]	20.05	21.65 [8.20, 31.17]	0.67	0.73 [0.50, 0.79]
3D	0.57	0.59 [0.41, 0.68]	3.47	2.58 [1.60, 5.25]	20.22	22.23 [8.03, 31.61]	0.66	0.70 [0.50, 0.78]
Ens.	0.58	0.60 [0.42, 0.68]	3.32	2.55 [1.48, 4.78]	19.98	21.63 [8.20, 30.98]	0.67	0.73 [0.50, 0.79]

training types. The IQR distribution is also more compact, with few low outliers, reflected by the mean and median sitting close together. This makes sense at the data level: D1.5T-U has no extreme tiny-lesion cases like D3T-B and D1.5T-B, the lowest FP/TP ratio at its optimal ADC threshold, and clearer intensity separation between lesion and non-lesion tissue (Table 3.3). All cases share the same spacing, with no very thick Z-slices and high in-plane resolution.

Even though the pretraining data is different and more heterogeneous, direct-inference still performs adequately, matching the inexperienced annotator’s level and showing that cross-domain transfer is viable. That said, from-scratch works better: the cohort mismatch is visible in the results, but a degree of generalization clearly transfers. The strong direct-inference performance measured here matters, since weak-supervision training relies strongly on it.

The 3D model falls further behind the 2.5D model than on the other cohorts, but ensembling still improves on pure 2.5D and gives the best boundary metrics (MASD, HD95). In contrast to D1.5T-B, both fine-tuning and from-scratch substantially beat direct-inference, and fine-tuning is slightly outperformed by from-scratch for both architectures.

These results also sit slightly above the 3-channel nnU-Net performance on D1.5T-U (Table 5.2), confirming our models stay ahead even where nnU-Net’s patch-based design works well.

5.6 Weak-Supervision

This section reports weak-supervision training results. Baselines establish the performance ceiling and are reported for both

Table 5.8: Baselines on D1.5T-U full-supervision, reported as mean and median [Q1, Q3] (25th–75th percentile) for the 2.5D+3D ensemble at 0.6/0.4 weighting. All results are either direct-inference or 3-fold CV. Distances (MASD, HD95) are in mm, over cases with a finite boundary metric. From-scratch is initialized with random weights, while the fine-tuned/direct-inference models use the D3T-B pre-trained model for initialization.

Z_{ADC}	Dice		MASD (mm)		HD95 (mm)		NSD@2mm	
	mean	median [Q1, Q3]	mean	median [Q1, Q3]	mean	median [Q1, Q3]	mean	median [Q1, Q3]
<i>From scratch ADC only</i>								
N/A	0.64	0.67 [0.58, 0.76]	1.67	1.52 [0.83, 1.96]	10.28	8.75 [4.93, 12.75]	0.75	0.75 [0.68, 0.87]
<i>Direct-inference</i>								
GT-free	0.54	0.47 [0.39, 0.66]	3.98	3.78 [2.04, 5.87]	21.41	21.88 [14.85, 31.49]	0.61	0.61 [0.51, 0.71]
GT-aware	0.58	0.60 [0.42, 0.68]	3.32	2.55 [1.48, 4.78]	19.98	21.63 [8.20, 30.98]	0.67	0.73 [0.50, 0.79]
<i>Fine-tuned</i>								
GT-free	0.68	0.66 [0.59, 0.79]	1.58	1.13 [0.81, 1.80]	8.77	7.21 [5.01, 10.65]	0.78	0.81 [0.69, 0.88]
GT-aware	0.70	0.72 [0.63, 0.79]	1.51	1.15 [0.75, 1.81]	10.06	6.02 [4.22, 16.64]	0.81	0.83 [0.76, 0.89]
<i>From scratch</i>								
GT-free	0.69	0.69 [0.61, 0.81]	1.53	1.00 [0.70, 1.82]	8.68	6.09 [4.55, 11.71]	0.79	0.84 [0.69, 0.89]
GT-aware	0.72	0.73 [0.67, 0.79]	1.18	0.96 [0.74, 1.26]	7.37	5.16 [4.20, 7.89]	0.83	0.86 [0.81, 0.89]

GT-Free and GT-Aware settings, across direct-inference, fine-tuning, and from-scratch approaches.

For weak-supervision, results follow a stepwise sweep of segmentation performance under increasing annotation-levels: lasso surface area is fixed while scribble annotation density within lassos varies. The same sweep is reported for direct-inference and the trained weak-supervision model, both with human-in-the-loop test-time annotation.

Finally, all approaches are compared on detection performance for 2D-disconnected lesion clusters, measuring weak-supervision effectiveness as a prior for human-in-the-loop annotation. The gap between the best GT-Aware full-supervision result and the best weak-supervision result quantifies the trade-off between annotation effort and segmentation accuracy.

5.6.1 Baselines

Table 5.8 reports the baseline performance ceiling for D1.5T-U with GT-aware and GT-free Z_{ADC} maps, across direct-inference, fine-tuning, and from-scratch approaches. GT-aware maps slightly outperform GT-free ones across the board, and for both map types fine-tuned models sit just below their from-scratch counterparts. The GT-aware from-scratch model gives the highest test-time independent segmentation performance on D1.5T-U.

The mean and quartile ranges are very similar between GT-aware and GT-free maps. GT-aware is consistently a touch better, as its pseudo-normative atlas is less noisy: the GT-free masks miss more lesioned tissue and remove more healthy tissue, giving a slightly less representative pseudo-normative spatial prior. Even so, the GT-free maps still beat ADC-only training and further surpass the inexperienced annotator. This gain over ADC-only is achieved fully unsupervised, without any separate healthy cases.

5.6.2 Results Annotation Sweep

Results are reported for weak-supervision segmentation performance across annotation levels. Lassos are fixed throughout the sweep, with only the assignment of ambiguous vs. lesion-containing lassos varying from $L_f=0.0$ (all lassos set to ambiguous) to $L_f=1.0$ (all lassos set to lesion-containing). Higher L_f levels mean more lesion-containing clusters, which receive added hard-positive scribbles, increasing annotator supervision.

First, the results are reported for pure direct-inference segmentation performance using HIL test-time scribbles and lasso filtering in Table 5.9.

At $L_f = 0.0$, direct-inference already improves significantly compared to pure direct-inference without HIL (Table 5.8), from a mean Dice of 0.54 to 0.62. At the boundary level, MASD and HD95 drop substantially. Since the model itself produces the same predictions as pure direct-inference, this gain is purely due to the removal of FP predictions outside the bounding boxes. Switching ambiguous lassos to lesion-containing lassos yields a constant increase in segmentation accuracy through two factors. First, for every additional lesion-containing region, the model gains a few free TP voxels that are added to the final segmentation output. Second, the way lassos are engineered means that ambiguous lassos are slightly less tight than their lesion-containing counterparts to represent boundary uncertainty. Consequently, lassos systematically tighten during the sweep, excluding more boundary FP predictions. Therefore, the improvement stems not from the pretrained model improving but from increasingly thorough post-processing. In short, the table shows what pure HIL post-processing already achieves at test time, with the all-lesioned setting at $L_f = 1.0$ approaching the Z_{ADC} GT-free full-supervision ceiling (Table 5.8).

The same sweep results but with weak-supervision fine-tuning are shown in Table 5.10. At each level of additional supervision, results can be compared to their direct-inference counterpart in Table 5.9 to highlight the effect of post-processing combined with additional weak-supervision learning. In this setting, clicks are provided as an input channel at inference,

Table 5.9: Direct-inference performance on D1.5T-U for the 2.5D+3D ensemble at 0.6/0.4 weighting, with click + lasso HIL post-processing, reported as mean and median [Q1, Q3] (25th–75th percentile). Results are swept as a progression of increasing confident-lesion fraction. The pre-trained model was trained on D3T-B. Distances (MASD, HD95) are in mm.

Lesion frac.	Dice $\uparrow$		MASD (mm) $\downarrow$		HD95 (mm) $\downarrow$		NSD@2mm $\uparrow$	
	mean	median [Q1, Q3]	mean	median [Q1, Q3]	mean	median [Q1, Q3]	mean	median [Q1, Q3]
0.0	0.62	0.61 [0.51, 0.72]	1.56	1.30 [1.01, 1.78]	7.67	6.70 [5.30, 8.06]	0.74	0.76 [0.69, 0.82]
0.2	0.63	0.62 [0.52, 0.72]	1.41	1.26 [0.98, 1.65]	6.65	5.84 [5.00, 7.07]	0.75	0.77 [0.70, 0.83]
0.4	0.64	0.63 [0.53, 0.73]	1.27	1.14 [0.92, 1.44]	5.75	5.10 [4.50, 5.87]	0.77	0.79 [0.71, 0.84]
0.6	0.65	0.64 [0.55, 0.73]	1.13	1.05 [0.84, 1.32]	5.12	4.62 [4.26, 5.27]	0.78	0.80 [0.72, 0.85]
0.8	0.66	0.65 [0.57, 0.74]	1.02	0.96 [0.78, 1.20]	4.53	4.35 [4.06, 4.85]	0.80	0.81 [0.75, 0.86]
1.0	0.67	0.66 [0.59, 0.75]	0.93	0.86 [0.73, 1.11]	4.16	4.12 [4.00, 4.24]	0.82	0.83 [0.77, 0.87]

Table 5.10: Weak-supervision trained performance on D1.5T-U for the 2.5D+3D ensemble at 0.6/0.4 weighting, reported as mean and median [Q1, Q3] (25th–75th percentile). Clicks are provided as an additional hard-positive channel. As train and test share the same cohort, the HIL effect enters through lasso internalisation during training and the click channel at test time. Results are swept as a progression of increasing confident-lesion fraction. The pre-trained model was trained on D3T-B. Distances (MASD, HD95) are in mm.

Lesion frac.	Dice $\uparrow$		MASD (mm) $\downarrow$		HD95 (mm) $\downarrow$		NSD@2mm $\uparrow$	
	mean	median [Q1, Q3]	mean	median [Q1, Q3]	mean	median [Q1, Q3]	mean	median [Q1, Q3]
0.0	0.63	0.63 [0.52, 0.73]	1.46	1.18 [0.93, 1.64]	6.87	5.50 [4.40, 7.76]	0.75	0.78 [0.70, 0.84]
0.2	0.66	0.65 [0.59, 0.74]	1.32	1.26 [0.93, 1.55]	7.85	7.22 [5.56, 9.10]	0.78	0.78 [0.74, 0.84]
0.4	0.68	0.67 [0.60, 0.73]	1.17	1.19 [0.87, 1.36]	6.62	5.74 [4.40, 8.24]	0.80	0.80 [0.75, 0.85]
0.6	0.69	0.68 [0.63, 0.75]	0.96	0.98 [0.73, 1.08]	5.42	4.89 [4.24, 6.02]	0.83	0.83 [0.80, 0.88]
0.8	0.71	0.71 [0.65, 0.75]	0.81	0.77 [0.64, 0.98]	4.26	4.10 [3.95, 4.67]	0.86	0.87 [0.81, 0.90]
1.0	0.72	0.72 [0.66, 0.76]	0.71	0.68 [0.57, 0.85]	3.49	3.69 [2.90, 4.02]	0.88	0.88 [0.84, 0.91]

with the lassos acting as dense outside-region negative supervision during training. As the interactive setting trains and tests on the same cohort, the lasso’s spatial prior is internalized in the weights, meaning the results shown in this table are identical with and without the HIL post-processing step. The HIL effect itself is still present, with the same influence as in the direct-inference HIL result, but due to a different part of the pipeline.

Table 5.10 shows that weak-supervision learning adds on top of pure HIL-based direct-inference. At every level the learned approach outperforms its direct-inference counterpart, already exceeding the direct-inference $L_f = 1.0$ Dice by the low-to-mid sweep (Figure 5.2). Boundary metrics follow more slowly but pass the direct-inference results around mid-sweep. Relative to direct-inference performance, localization becomes more precise, with fewer spurious clusters predicted and within-lasso boundary precision increases. The HIL approach is effective on the Z_{ADC} GT-free maps, where filtering of the noisier predictions leads weak supervision to overtake the GT-free full-supervision ceiling around mid-sweep (Figure 5.2). At $L_f = 1.0$, the weak-supervision interactive setting surpasses the GT-aware full-supervision baseline on all metrics.

The progressions for direct-inference and weak-supervision performance across the annotation levels are combined into four panels, one per metric, in Figure 5.2. The green dashed line marks the GT-aware full-supervision ceiling and the grey dashed line the direct-inference baseline. As discussed, the boundary metrics (MASD, HD95) cross the full-supervision ceiling between the beginning and mid-sweep, while weak-

supervision reaches it even in Dice at the high-supervision end. For NSD@2mm, weak-supervision surpasses the baseline clearly over the later half of the sweep. Both curves are roughly linear, but weak-supervision rises more steeply and stays above the direct-inference line throughout, except for HD95 in the late sweep where direct-inference dips slightly lower.

5.6.3 Separate clusters

Beyond the general metrics, we are also interested in detection performance at the level of disconnected 2D lesion clusters. As a prior, a HIL segmentation model is more useful when it detects a high fraction of separate lesion clusters, since an annotator can resolve boundary discrepancies far more easily than find clusters the model missed entirely. We measure this through per lesion-cluster recall and precision, reported in Table 5.11.

Among the baselines, direct-inference and full-supervision performance baselines show varying recall and precision, with GT-aware full-supervision highest on both. Full-supervision detects more clusters than direct-inference, while direct-inference, particularly with GT-free maps, trades precision for recall through its noisier slight over-prediction.

Comparing precision across methods, the FP-cluster detection behaviour shows high variation. The non-HIL baselines produce far more FP clusters than the HIL and weak-supervision variants, and direct-inference HIL produces more than weak supervision. This is partly due to the post-processing, which discards predictions outside the lassos, and partly the teacher-

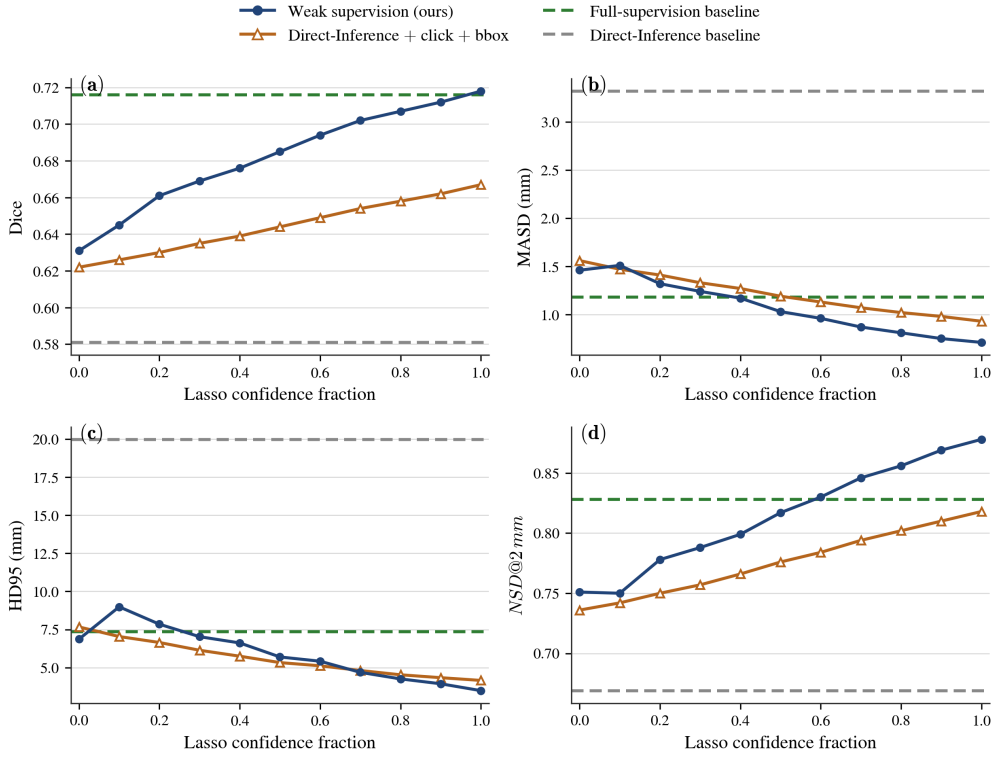


Figure 5.2: Supervision-performance trade-off on D1.5T-U (2.5D+3D ensemble): (a) Dice, (b) MASD, (c) HD95, (d) NSD@2mm versus lasso confidence fraction. Each point is the cohort mean over the 20 cases. Dashed lines mark the full-supervision (green) and direct-inference (grey) baseline means. MASD and HD95 in mm.

Table 5.11: Detection of separate 2D lesion clusters (defined by 8-connectivity per slice) on Utrecht ($n=20$). Recall and precision are reported as the per-patient median [Q1, Q3]. * = from scratch.

Run	Recall	Precision
<i>Baselines (3-fold ensemble)</i>		
Direct-inference (GT-free)	63.8 [56.9, 71.9]	42.7 [24.5, 58.0]
Direct-inference (GT-aware)	61.1 [55.1, 73.1]	57.3 [42.9, 74.8]
Full-sup. (GT-free)	73.7 [59.7, 88.2]	71.3 [57.5, 76.3]
Full-sup. (GT-free)*	75.7 [67.2, 89.0]	68.9 [54.3, 75.9]
Full-sup. (GT-aware)	77.2 [65.7, 85.1]	74.3 [56.9, 82.9]
Full-sup. (GT-aware)*	80.1 [68.4, 87.7]	71.1 [53.2, 84.1]
<i>Weak supervision: click_sup (all-20)</i>		
lf 0.0	72.6 [59.5, 77.6]	75.3 [65.9, 85.8]
lf 0.2	58.6 [52.6, 68.2]	98.2 [94.7, 100]
lf 0.4	68.2 [62.4, 72.6]	97.4 [95.2, 100]
lf 0.6	74.4 [71.3, 79.7]	99.2 [94.6, 100]
lf 0.8	85.8 [84.0, 90.4]	97.2 [94.8, 98.5]
lf 1.0	100.0 [100, 100]	97.4 [95.6, 99.0]
<i>Direct-inference + click + bbox (all-20)</i>		
lf 0.0	63.8 [56.9, 71.9]	78.9 [68.4, 82.9]
lf 0.2	68.4 [59.5, 74.0]	80.6 [72.1, 87.5]
lf 0.4	72.3 [66.9, 79.5]	82.9 [74.4, 90.7]
lf 0.6	79.9 [76.2, 84.7]	85.0 [78.1, 91.4]
lf 0.8	88.5 [84.1, 91.6]	86.6 [80.7, 91.9]
lf 1.0	100.0 [100, 100]	88.1 [82.1, 92.8]

student scheme, which only supervises the most confident voxels. As a result, the learning-based weak-supervision models have almost no separate FP clusters. Together with

the similar mean and median Dice reported in Table 5.10, this indicates that their remaining errors are boundary FP voxels on correct clusters or FN predictions on fully missed clusters, rather than scattered separate false detections.

At recall, pure direct-inference HIL already becomes competitive with the baselines from mid-sweep. It achieves this, while staying highly precise throughout, as the lasso filter removes any FP clusters outside the annotated regions. Its recall IQR is tight. Untrained, it misses the most ambiguous clusters but reliably catches the easier ones. Recall also stays well above the scribbled-cluster fraction at every step, showing that the high un-scribbled catch rate comes from the pre-trained model itself and not only from the HIL annotations.

Weak supervision behaves differently across the sweep, with the recall and precision curves revealing the effects of our approach. At $l_f=0.0$, with no scribbles, the model is the least conservative. It over-predicts separate clusters, giving competitive recall but the lowest precision of any weak-supervision setting, similar to the baselines. The L_f trained model reaches near-baseline cluster-detection recall with almost no annotations, using only teacher-student lasso-supervision training. Notably, lasso HIL does not affect recall.

Once scribbles enter, the behavior flips. The scribble channel teaches the model that lesion clusters sit near scribbles, making predictions highly conservative concerning distant clusters. Precision immediately jumps to near-perfect, but recall drops below $l_f=0.0$, as the same conservatism that removes FP clusters also suppresses the detection of true lesion clusters far away from any input scribbles. The conservative teacher-student sig-

Table 5.12: D1.5T-U, 3-fold CV weak-supervision sweep for two configurations. The 4-channel HIL block is the CV control for the all-20 deployed setting, confirming its gains stem from the larger training pool rather than leakage. The 3-channel no-HIL block is the fully standalone ceiling, with no click channel or lasso post-processing. Reported as mean and median [Q1, Q3] (25th–75th percentile); confident-lesion fraction in steps of 0.2, best row per metric in bold. Distances (MASD, HD95) in mm. The two intermediate component configurations are in Appendix E.5 (Table E.4).

Lesion frac.	Dice $\uparrow$		MASD (mm) $\downarrow$		HD95 (mm) $\downarrow$		NSD@2mm $\uparrow$	
	mean	median [Q1, Q3]	mean	median [Q1, Q3]	mean	median [Q1, Q3]	mean	median [Q1, Q3]
<i>4-channel, lasso HIL post-processing (deployed)</i>								
0.0	0.63	0.62 [0.51, 0.71]	1.53	1.28 [0.91, 1.76]	7.45	6.22 [4.40, 8.44]	0.74	0.76 [0.69, 0.84]
0.2	0.62	0.58 [0.55, 0.69]	1.57	1.57 [1.13, 1.79]	8.42	8.45 [5.66, 10.47]	0.72	0.74 [0.65, 0.78]
0.4	0.65	0.63 [0.58, 0.70]	1.28	1.30 [0.94, 1.54]	6.85	6.71 [4.96, 7.88]	0.76	0.76 [0.70, 0.84]
0.6	0.67	0.66 [0.59, 0.71]	1.06	1.05 [0.80, 1.23]	5.53	4.93 [4.42, 6.08]	0.80	0.80 [0.77, 0.86]
0.8	0.69	0.69 [0.61, 0.73]	0.89	0.91 [0.73, 1.00]	4.63	4.28 [4.00, 5.15]	0.83	0.83 [0.81, 0.87]
1.0	0.71	0.70 [0.63, 0.75]	0.76	0.72 [0.54, 0.91]	3.76	3.69 [3.44, 4.02]	0.86	0.86 [0.83, 0.91]
<i>3-channel, no HIL post-processing (standalone)</i>								
0.0	0.59	0.57 [0.47, 0.69]	2.31	2.05 [1.39, 2.75]	12.90	11.48 [6.31, 18.28]	0.68	0.72 [0.60, 0.80]
0.2	0.60	0.56 [0.51, 0.74]	2.06	1.69 [1.17, 2.53]	10.55	9.55 [6.78, 12.23]	0.68	0.68 [0.59, 0.78]
0.4	0.62	0.59 [0.53, 0.75]	1.82	1.34 [1.06, 2.32]	9.19	6.97 [5.44, 10.69]	0.71	0.74 [0.61, 0.83]
0.6	0.62	0.59 [0.52, 0.75]	1.88	1.64 [0.99, 1.99]	9.38	7.90 [5.58, 10.22]	0.71	0.73 [0.66, 0.82]
0.8	0.63	0.60 [0.55, 0.74]	1.85	1.45 [0.99, 1.95]	9.12	7.18 [5.08, 10.40]	0.72	0.78 [0.66, 0.83]
1.0	0.62	0.62 [0.54, 0.75]	1.87	1.58 [1.07, 1.84]	9.83	8.57 [5.52, 13.07]	0.72	0.75 [0.69, 0.82]

nal, which only supervises high-confidence voxels, reinforces this. Recall then climbs steadily back up as more hard-positive scribble signal accumulates, giving the model enough learning-signal to predict distant clusters without abandoning its precision. From mid-sweep there is enough hard-positive signal that the conservatism becomes worth it, with recall matching or exceeding $l_f=0.0$ while precision stays near-perfect. Throughout, the weak-supervision approach recovers a large share of un-scribbled clusters, detecting them without direct supervision.

At $l_f=0.7$, where weak supervision first matches the best full-supervision baseline’s recall, the per-case results provide a more detailed understanding (Figure E.1, Appendix E). The cohort-level TP means are dominated by a few high-lesion subjects, on which the HIL variants recover far more clusters than the full-supervision baseline, while the two are more comparable across the cases with small lesions. As seen in the cohort-wide metrics, FP clusters separate the approaches most clearly: weak supervision stays near zero on almost every subject, whereas the full-supervision baseline and direct-inference HIL both produce scattered high-FP cases.

5.6.4 Robustness to Leakage and Interactive HIL Components

The results in Tables 5.10–5.11 present segmentation performance of models trained and tested on the same cohort. This was an intentional, HIL-specific choice, with HIL further present through the injection of weak annotations via both the scribble channel and the lasso post-processing filter. To check for the effects of same-case leakage and to quantify the influence of the HIL component, the sweep was repeated under 3-fold CV. Table 5.12 reports the two configurations: the 4-channel HIL setting (the CV control for the deployed model) and the 3-channel standalone setting with no HIL signal. Two intermediate configurations are reported in Appendix E.5

(Table E.4).

First, we check for effects of train-test leakage by comparing the 4-channel model trained under CV against the all-20 deployed setting. The CV test curves (Table 5.12, top block) track the all-20 results to within roughly 0.01–0.02 Dice. The earlier reported results are therefore not driven by leakage. Instead, the small all-20 advantage is consistent with improvement expected from a slightly larger training pool.

The lasso filter contributes little once the click channel is present, consistent with the cluster-detection results (Table 5.11). As shown there, the scribble channel makes the model highly conservative: detected clusters arise from either very high-confidence un-scribbled regions or the scribbles themselves, with few FPs left for the lasso to remove. This conservatism, combined with better accuracy, strengthens as scribbles accumulate, so the lasso post-processing matters even less at higher L_f . The per-component decomposition is reported in Appendix E.5.

The 3-channel standalone configuration, with no interactive signal at all, saturates near 0.63 Dice (Table 5.12, bottom block). This is above the 0.54 direct-inference baseline on the same GT-free maps but below the interactive HIL settings, with the gap coming from a higher FP rate that lowers Dice and the boundary metrics. As highlighted in the 4-channel $l_f=0.0$ cluster analysis, without scribble supervision the model is less conservative and produces more FP clusters. Even fully non-HIL, it still achieves comparable performance to the inexperienced second rater (median Dice 0.56), showing the method retains value without any HIL component.

Discussion

6.1 Interpretation of the Results

In this chapter we discuss the baseline and in-scanner performance, the cross-scanner and cross-domain transfer of the models, the weak-supervision results, and how the Z_{ADC} priors influence performance across all of these settings, especially on small lesions and under weak supervision. Additionally, we discuss hyperparameter choices and how they might be tuned for unseen cohorts. Finally, we discuss limitations of this work and directions for future research.

6.1.1 Baseline and In-Scanner Performance

The first goal of this thesis was to develop baseline models capable of robust segmentation and adaptation to new cohorts. Tuning and in-scanner testing of these baseline models was done on D3T-B. As it is restricted to only TH-treated cases, it reflects the difficulty to be expected in modern clinical practice. Due to the TH-treatment, most patients have tiny and often ambiguous lesions, while patient data is additionally highly heterogeneous in voxel spacing and image size (Table 3.3, Table 3.2). These characteristics make it the most challenging cohort in this thesis.

Consequently, on D3T-B the initial multi-purpose segmentation model, nnU-Net, failed (Table 5.2), with performance dropping even below threshold-based segmentation on the pseudo-normative priors. The patch-based design of nnU-Net favors learning local intensity patterns over global context. This conflicts with the lower tail of tiny-lesion cases that have poor local intensity separation, reflected in its collapse in performance at Q1.

In contrast, our 3D and 2.5D models rely more on global context and on the pseudo-normative priors, reflected in their improved segmentation performance on D3T-B (Table 5.3). Their performance is comparable to each other, with differences driven by the noisy influence of a few extremely challenging tiny-lesion cases. Relative to nnU-Net, the IQR shifts up substantially, and overall performance improves over threshold-based segmentation. That said, like nnU-Net, our models systematically miss a few very small cases. There, a handful of voxels leaves little signal to learn from, while the pseudo-normative priors are least reliable, since even a small registration drift can displace lesions of this size.

Although the ensemble weighting proved detrimental here relative to the single 3D model, the noise and occasional full failures of both models, often on different cases, make combining their predictions the more robust choice for generalizability. The goal was not to maximize performance on D3T-B but to build models and a pipeline that is robust across cohorts, supporting all cross-scanner, cross-domain, and weak-supervision experiments. These results provide a stable platform for what follows.

6.1.2 Cross-Scanner Performance

The cross-scanner experiments on D1.5T-B test the general robustness of the tuned architecture and trainer structure for a new cohort within the same hospital, specifically through fine-tuned and direct-inference segmentation performance using the D3T-B pretrained weights. Corresponding segmentation performance is shown in (Table 5.6).

Cross-scanner performance is shaped by a few key differences and similarities between D3T-B and D1.5T-B. Both share similar levels of heterogeneity, with the thick-sliced, tiny-lesion cases being extremely challenging to segment precisely. However, D1.5T-B has a much easier upper tail, due to a subgroup of large-lesion patients. This wide range of lesion burden is reflected directly in the results: the lower IQR tail matches D3T-B, but the large-lesion cases lift the mean and median segmentation performance well above it.

The potential strength of using cross-scanner pretrained weights is shown by the direct-inference performance on a few outlier tiny-lesion cases in D1.5T-B that do not match its general shape: these are picked up well by the pretrained model but missed without pretrained weights, since they match the D3T-B distribution much better. In contrast, inconsistencies in voxel and in-plane spacing directly influence boundary precision on other cases. For these cases, from-scratch training is most effective, as it is best tailored to the specific data characteristics. With enough representation of lesion-severity in larger datasets, from-scratch is likely better, but since available datasets are small and hard to come by, these results are promising for the potential of combining knowledge across multiple varying datasets.

6.1.3 Cross-Domain Performance

Cross-domain experiments on D1.5T-U test the general robustness of the tuned architecture and trainer structure for a new cohort at a new hospital, with the correspondingly large potential differences in acquisition, treatment, and data handling. The effectiveness of the cross-domain pretrained weights is critical to the success of our corresponding weak-supervision experiments, which rely on them as both initialization and teacher.

The domain difference between D1.5T-U and D3T-B is clearest in the fact that D1.5T-U is highly homogeneous, in contrast to D3T-B, while also being fully non-TH-treated. This makes it a relatively less complex dataset to train and segment on, as the extreme tiny-lesion outlier cases that characterize the Boston datasets are absent. D1.5T-U also has more contrast between lesion and non-lesion tissue, possibly due to the treatment status, making both pure-ADC training and the nnU-Net model viable options.

Interestingly, pure-ADC training of our model already outperforms the inter-annotator agreement of median 0.56

Dice (Table 5.8). That said, adding the pseudo-normative priors improves segmentation across the board, highlighting that the region-aware prior contributes real signal beyond raw ADC, even where ADC-learning is already strong. Additionally, developing per-patient pseudo-normative priors is essential for using the pretrained D3T-B weights, which depend on them.

Cross-domain results using the priors are reported in Section 5.5. Segmentation performance is systematically more robust than on D1.5T-B and D3T-B. The difference in cohort lesion-severity is reflected in the much more compact IQR, as expected from the data, with fewer outliers in either direction. This means that, while mean and median performance is comparable to D1.5T-B, it is far more robust across all cases.

From-scratch training performs better than both direct-inference and fine-tuning from the pretrained model. This makes sense: there are enough cases to train from scratch, and the differences between cohorts are large. The pretrained prior that helped most on the small, underrepresented cases in D1.5T-B has little to act on here, since such cases are absent. That said, the direct-inference performance of the pretrained models remains strong, being comparable to the inter-annotator median.

The results on this cohort are strong. However, the properties that make it the least challenging of our datasets also make it less representative of modern clinical practice, where TH-treated patients with smaller lesions dominate.

6.1.4 Importance of Pseudo-Normative Priors

Our pipeline for building pseudo-normative priors delivers clear segmentation gains over raw-ADC training (Table 5.8), showing that accounting for normal regional intensity variation is crucial. Even though the normative reference is built from a lesioned cohort, the resulting spatial-normative priors prove effective. The cost of a less representative atlas is visible in the GT-aware versus GT-free comparison: both help, but GT-aware is better. This indicates that healthy neonatal scans should be used when available, while confirming that the approach of using a (partially) lesioned cohort still beats having no spatial-normative prior at all. The contribution is therefore already significant, but these results could improve further with the right data.

6.1.5 Weak-Supervision

The robustness of our pretrained models, together with the ability to build fully unsupervised pseudo-normative priors, is a crucial starting-point for the weak-supervision experiments. Without them, our approach would not have been viable. Critically, all reported performance and analysis is based on artificial labels, which can only be confirmed by training on real labels. The approach is consequently a proof of concept, with the potential to work well with real clinician annotations. Our interest in weak supervision is twofold: testing its potential use case as a prior that speeds up full annotation, turning labeling into a correction task, and measuring what the method achieves on its own without an HIL component.

The main results (Table 5.10) show the trade-off between an-

notation effort and performance, with more hard-positive annotations mostly leading to better results. What the best setup is, remains open: the right balance of recall, precision, and annotation effort depends on what actually saves clinicians time, something that is still being discussed and needs real testing.

The key observation from these results is that feeding the hard-positive scribbles as an input channel makes the models conservative: they learn that lesions sit near scribbles and make predictions accordingly (Table 5.11). This conservatism is what turns the method into a potentially useful prior. It makes the predicted cluster locations highly trustworthy, so an annotator can correct boundaries rather than search for missed lesions. The cost is recall: at low scribble supervision, the same learned conservatism suppresses the prediction of real lesion clusters that lie far from scribbles. This initially drops recall below the $lf = 0.0$ baseline before recovering. From the mid-sweep onward, the model catches more separate clusters than full supervision, while staying precise. Remaining errors are at the boundaries of correctly predicted clusters or from missing clusters altogether.

The no-HIL model occupies the opposite end of this trade-off, behaving like the $lf = 0.0$ case throughout. Without the scribble channel, it never becomes conservative, predicting far more in ambiguous regions and keeping high recall at lower precision. The post-processing reflects this: lasso filtering barely changes the HIL results, since few FP predictions outside lassos remain, yet removes many from the no-HIL predictions. Even so, with almost only lasso supervision the no-HIL model still improves on direct-inference and matches the inexperienced annotator at the median, with a compact lower tail. This shows that the method could potentially be of value without any HIL component.

These two behaviors are potentially complementary. The HIL models provide trustworthy cluster locations to correct from, while the looser no-HIL predictions can flag clusters that the HIL versions missed. Both together, potentially make a stronger prior than either alone. Clinicians we spoke to saw clear potential for saving annotation time with the right implementation, though the best setup still needs several iterations to optimize.

6.1.6 Hyperparameters

This pipeline contains various hyperparameters across the multiple stages: for architecture and training of the segmentation models, in the construction pipeline for the Z_{ADC} maps, and across the weak-supervision scheme. The model hyperparameters were tuned on D3T-B, while the rest were tuned on D1.5T-U. Consequently, there is a level of uncertainty regarding transferability to new cohorts. That said, they vary in how directly we were able to test their transferability.

The model designs are most directly transferrable. The architecture and optimization settings (Tables 4.1, 4.2, 4.4, 4.5) were frozen after the D3T-B ablations (Appendix B) and applied unchanged to D1.5T-B and D1.5T-U. The discussed results highlight that their frozen design generalizes without re-tuning. Nonetheless, a short retuning on new cohorts where possible could lead to more optimal results. Main tunable

options are the learning rate and augmentations. The latter is especially sensitive to cohort size, and to intensity and lesion-size characteristics.

Additionally, for the Z_{ADC} pipeline, the absolute-ADC threshold (masking) and τ_c (masking and post-processing) were each shown to have a range of defensible choices (Table 5.1), from which we picked a robust option. Other values, or further tuning on new cohorts, may be needed or equally defensible, particularly for the absolute-ADC threshold, which is scanner- and protocol-dependent. The guided-correction target of $Z = -2.5$ was a choice, set just below the optimal mask threshold of $K = -2$. Both may need adjustment on a new cohort.

Finally, the remaining choices are likely the least transferable. The ensemble weight $\omega = 0.6$ (favoring the 2.5D model) behaved differently across cohorts. It helped D1.5T-B and D1.5T-U but hurt some small-lesion D3T-B cases (Section 5.3), so it should probably be re-tuned on a new cohort. The weak-supervision constants used to build the weak labels were set largely empirically, but in practice would be replaced by a real clinician’s annotations rather than re-tuned. In contrast, the teacher weighting, learning rate, soft-label thresholds, and hard-label weight would likely need re-tuning once real weak labels are available, and possibly again on datasets with different characteristics or label uncertainty.

6.2 Limitations

6.2.1 Problems With Data Heterogeneity

Table 3.2 highlights a problem seen in many experiments of this thesis: scans collected over many years, as in our datasets, often reflect changes in protocol, leading to different voxel and slice spacings. The clinical setting adds further variation, as time pressure can lead to different scan-quality choices, such as 2 mm versus 6 mm slices. Our current approach comes with the assumption that the models can generalize across these differences in spacing. However, the fine-tuning results for cross-scanner and cross-domain transfer show the limits of that assumption, and D1.5T-U may perform as well as it does partly because it is homogeneous.

We ran extensive ablations to homogenize the volumes, but all hit the same trade-off: homogenization helped in some cases, but also caused catastrophic smearing and interpolation artifacts, worst on the thick slices. Small lesions in particular were often displaced or wiped out, under both nearest-neighbor interpolation and Gaussian smoothing. We also tested SynthSR [14], which uses T1-weighted images to synthesize slices rather than interpolating new ones in the z-direction. Projecting ADC onto that grid recovered some quality, but the trade-off still proved detrimental overall.

We did not find the key factor that would make this re-sampling worthwhile, and retained native spacing throughout, choosing to focus on other areas. It remains a promising direction for future work on cross-scanner and cross-domain training. More homogeneous input would likely improve segmentation, but only if it does not come at the cost of wiping

out the lesion signal in the ADC, ground-truth, and Z_{ADC} maps.

6.2.2 Limitation of Z_{ADC} Pipeline

The normative pipeline, as described in Section 4.2, comes with limitations. The pipeline was tuned and tested on only one cohort, as D1.5T-U is the only dataset with the $b=0$ and $b=1000$ DWI needed for precise registration. The heuristics rely on the relative contrast of DWI and ADC, which might not be universally transferable. Additionally, they may be less effective on the small lesions of a TH-treated cohort like D3T-B, where contrast is weakest. The same holds for the absolute ADC threshold of 1000, which works well here but may not be optimal elsewhere. However, for now, no annotated datasets are available to test this further.

While improving on pure ADC-based training, the Z_{ADC} maps are imperfect. Registration accuracy could be higher, and the pseudo-healthy tissue may not fully match true healthy tissue, biasing the μ and σ estimates. In the GT-aware setting, inter-rater variability adds further error, since missed lesions degrade the maps.

The pipeline also relies on DRAMMS, an older registration framework (Sections 2.3.2 and 4.2). Being non-deep-learning, it is slow, so iteratively rebuilding the maps using the previous iteration’s masks is highly inefficient. Also, in the weak-supervision setting, the weak labels could have replaced the heuristics in the masking and guided-correction steps. However, the cost of rerunning registration meant we did not have the time or compute to test this, let alone iteratively. Consequently, while such additions could improve map quality, integrating them was beyond the scope of this work.

Finally, the homogeneity of D1.5T-U makes atlas construction straightforward, since patient volumes are similar. For heterogeneous cohorts like D1.5T-B and D3T-B this is more challenging, as the atlas must average over different in-plane resolutions and z-spacings.

6.2.3 Limits of Baseline Models

A limitation of our pipeline is that the baseline models, while effective on the current datasets, were heavily tuned on D3T-B. This was done through 3-fold CV with held-out test sets, yet the chosen settings may still need re-tuning on new cohorts. With limited data available, that is not an option to test in this work.

Both models are also fairly standard architectures and trainers, which for now works in their favor. The simple approach exploits the Z_{ADC} maps well, and the global context they capture lets them outperform a widely used model like nnU-Net. For future gains, however, these models may have a ceiling that more complex architectures do not.

6.2.4 Generalization of Results

All results are based on a small set of patients. This is largely unavoidable, given the limited availability of data, but it means that all segmentation performance and further findings must be

read with generalizability in mind. The results are promising, but finding a way to test them further on larger and more varied cohorts is important.

6.3 Future Works

6.3.1 Improving Baseline Models

A promising direction for future work is a middle ground that combines the strengths of all three tested models to improve the baseline for this work. The 3D U-Net uses only simple augmentation, largely because its single-pass design limits the number of cases seen during training, and heavy augmentation washed out the true signal. The 2.5D model and nnU-Net, by contrast, both show that more augmentation can be very useful. Finding a way to combine these factors, possibly with more modern techniques, could raise the baseline.

More broadly, with more data to tune it, a more complex architecture could become stronger and more robust than the standard models used here.

6.3.2 Extension to Other Neonatal Pathologies

A natural extension is to detect neonatal abnormalities beyond HIE, such as neonatal ischemic stroke or other cases that the BONBID-HIE benchmark explicitly excludes (Section 3.1). This is feasible because, although the models are trained on HIE, they rely heavily on the normative Z_{ADC} prior, largely flagging lesions that deviate strongly from healthy tissue. Since other ischemic pathologies produce similar deviations, the same approach could be transferred.

We ran a brief test on two neonatal stroke patients, building the Z_{ADC} maps and applying the from-scratch trained D1.5T-U model. Both the maps and the segmentations gave visually convincing results, even though the atlas itself was built from the pseudo-normative tissue of HIE-lesioned neonates. This suggests the normative reference generalizes beyond the pathology it was constructed from. It also indicates that the core strength of the approach, exploiting the Z_{ADC} maps efficiently, may transfer to other diffuse pathologies. This would widen the model’s applicability and possibly enlarge the cohorts available for atlas construction.

6.3.3 Weak Supervision

The current weak-supervision experiments use artificial weak annotations. While the results are promising, they are not immediately transferable to real-world performance. We are currently working with a group of clinicians from the AMC to create real annotations, which will provide a useful test of and comparison to the results in this study.

To support this, we have started building an annotation tool suited specifically to HIE data and annotations. The initial design supports simple tests for the weak-supervision setup. After this project concludes, we will work with the same AMC team to incorporate their input and experience, determining what is genuinely useful and where the most time can be saved.

Including the weak-supervision software as an extension in an established annotation platform such as ITK-SNAP[44] or 3D Slicer[10] is another avenue worth exploring, as it might lower the barrier to clinical adoption.

Further experiments on where and when to add annotations in a HIL setting could also help. Approaches such as [40] and [41] have the potential to generalize well to our setting and are worth exploring in future work.

6.3.4 Spatio-Temporal Atlas

When setting up our datasets, we removed all cases scanned more than seven days after birth because of the rapid growth changes in the neonatal brain during the first weeks of life [28]. The current atlas pools all cohort ages into a single normative μ and σ per voxel, and restricting it to a seven-day window keeps this reference reasonably consistent. However, a natural extension is a spatio-temporal atlas that conditions the normative statistics on the infant’s age. This would compare each patient against age-matched healthy tissue, sharpening the Z_{ADC} signal.

Conclusion

This thesis asks: *Can we develop a segmentation pipeline that achieves robust HIE lesion segmentation across domains and scanners, while reducing the need for time-intensive full manual annotation?* We present a pipeline that makes such robust segmentation possible.

Our work provides two tuned models: a 3D U-Net and a 2.5D SegResNet. They function as robust standalone segmentation models on unseen HIE cohorts through from-scratch learning. In addition, we provide pretrained frozen weights that serve as a robust initialization for fine-tuning, and can optionally be used for direct-inference. The best choice may depend on the similarity of new data to the cohort used for pretraining and its size, or the time and compute available for full from-scratch training. We evaluate direct-inference, from-scratch, and fine-tuned approaches, finding them to be valuable strategies for cross-scanner and cross-domain transfer.

Furthermore, the pipeline addresses the need for a pseudo-normative spatial prior. It constructs healthy normative reference atlases without requiring a separate healthy cohort, using the pseudo-healthy tissue of lesioned HIE cohorts. It makes this possible both when lesions can be masked with ground-truth annotations and when these are unavailable. When lesion masks are unavailable, we provide a heuristic scheme that flags probable lesion voxels to mask. This GT-free variant is noisier but still performs well above training without a pseudo-normative spatial prior.

The first two additions, the pretrained baseline and the pipeline for constructing pseudo-normative spatial priors when no ground-truths are available, are the crucial backbone for our novel weak-supervision experiments. Through a HIL scheme of clicks and lassos, weak-supervision segmentation is able to approach full-supervision performance on Dice and far exceeds it on the boundary metrics, while substantially reducing annotation effort. It also clearly surpasses naive thresholding. However, the weak annotations are artificial and the results still need testing with real annotations. That said, they show the approach could be viable both as a standalone segmentation method and as a prior that reduces total annotation time. This addresses the need for larger and more fully annotated datasets.

The pseudo-normative prior raised segmentation well above raw diffusion, with mean Dice reaching 0.72 under both full and weak supervision, against the 0.64 ADC-only baseline. The median Dice of 0.73 was substantially higher than the cohort’s inter-rater agreement of 0.56, and the model’s agreement with the experienced annotator was higher than the inexperienced rater’s agreement with the same annotator, under both full and weak supervision.

In conclusion, the results of this research are promising. We develop robust baseline models that perform reliably across cohorts of differing scanner, site, and complexity. This paves the way for a novel weak-supervision approach not yet attempted in HIE lesion-segmentation. Future work includes validating the

methods on larger datasets and real clinical annotations, since the findings in this thesis rest on small cohorts and artificial weak labels.

APPENDIX A

Supplementary Data Analysis

A.1 Per-Patient ADC Distributions

Figure A.1 shows the full per-patient breakdown of healthy and lesion ADC distributions summarized in Table 3.3 of Chapter 3. It illustrates the within-patient lesion/healthy overlap that drives the separability differences between cohorts: clearly offset on D1.5T-U, heavily overlapping on D3T-B.

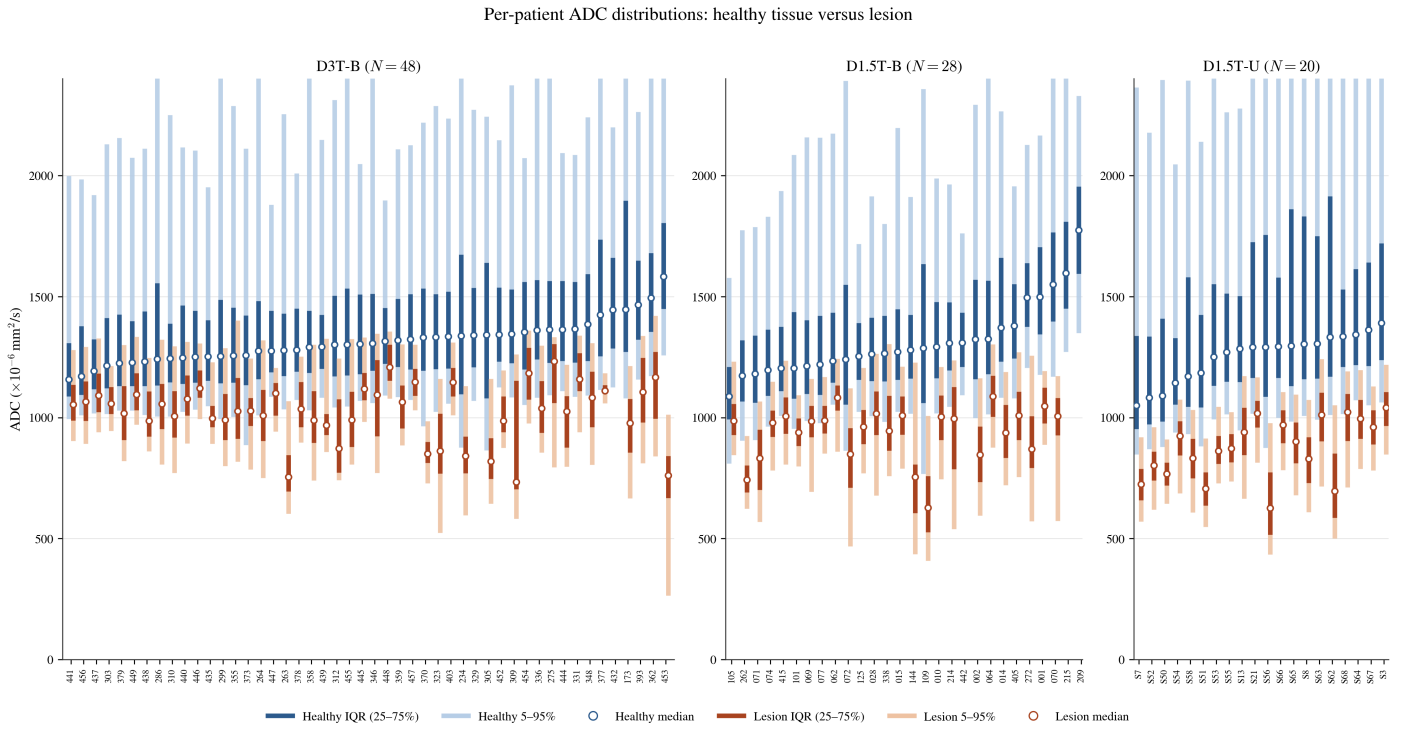


Figure A.1: Per-patient ADC distributions for healthy tissue (blue) and annotated lesion (red), for every patient in each cohort. The dark bar marks the interquartile range (25–75%), the light bar the 5–95% range, and the dot the median. Patients are ordered left-to-right by ascending healthy-tissue median ADC. Within-patient lesion/healthy overlap is substantially smaller on Utrecht (mean 20.5%) than on Boston 3T (mean 36.7%).

APPENDIX B

Ablation Studies

B.1 K-mask Threshold Tuning

Threshold K for the binary mask, swept on D3T-B, with results in Table B.1. Mean validation Dice peaks at $K = -2$ (used in all experiments). Soft sigmoid- k variants replace the hard mask with a sigmoid centered at K . Final test results for $K = -2$ (the dicebce_baseline) are below the final D3T-B performance reported in Chapter 5.3, as other parts of the training and architecture were changed after this ablation was run.

Table B.1: K -threshold ablation (D3T-B, 3-fold CV). **Bold** exceeds the DiceBCE baseline in that column; * marks a collapsed fold. The soft sigmoid- k variants test effect of replacing hard mask with sigmoid centered at K

Config	Validation				Test			
	F0	F1	F2	Mean	F0	F1	F2	Mean
dicebce_baseline	0.6335	0.5878	0.6493	0.6235	0.5125	0.4811	0.5809	0.5248
<i>Hard binary threshold variants</i>								
$k = -1.0$	0.0000*	0.3462	0.5840	0.3101	0.0625*	0.3513	0.3963	0.2700
$k = -1.25$	0.4264	0.0625*	0.4454	0.3114				
$k = -1.5$	0.6436	0.5137	0.5894	0.5822	0.4226	0.4083	0.4719	0.4343
$k = -1.75$	0.6300	0.5761	0.4964	0.5675	0.5112	0.4432	0.5053	0.4866
$k = -2.25$	0.5589	0.5407	0.5998	0.5665	0.4497	0.4372	0.5098	0.4656
$k = -2.5$	0.5371	0.4704	0.4592	0.4889	0.4019	0.4466	0.4874	0.4453
$k = -2.75$	0.4920	0.4819	0.4307	0.4682	0.3775	0.4311	0.4428	0.4171
$k = -3.0$	0.4621	0.4736	0.4117	0.4491	0.3814	0.4183	0.4538	0.4178
no mask	0.5090	0.4905	0.4049	0.4681	0.3679	0.4331	0.4391	0.4134
<i>Soft sigmoid-k variants</i>								
sigmoid, centre = -1.5	0.5578	0.0000*	0.4243	0.3274	0.4096	0.0625*	0.4393	0.3038
sigmoid, centre = -2.0	0.5206	0.4563	0.5218	0.4996	0.3982	0.3370	0.4771	0.4041

B.2 Loss Function

All loss experiments were run on the 3D U-Net, for two reasons. First, it trains far faster than the 2.5D model. Second, it came with no reference loss or trainer, unlike the 2.5D model whose source paper and repository specified both, so it had to be tuned from scratch and was developed first.

We tested two multiple directions together with the simple equal-weight Dice + BCE baseline we started with: a region-sensitive Tversky loss (alone and combined with BCE) motivated by the strong foreground/background imbalance, and the two losses Chandran and Reghunath [7] describe for the 2.5D model, namely BCE + Dice and a Hausdorff-Dice loss (BCE + Dice is baseline). Sweep is shown in Table B.2.

Table B.2: Loss sweep on the 3D U-Net, D3T-B 3-fold CV. **Bold** exceeds the DiceBCE baseline in that column.

Config	Validation				Test			
	F0	F1	F2	Mean	F0	F1	F2	Mean
dicebce_baseline	0.6335	0.5878	0.6493	0.6235	0.5125	0.4811	0.5809	0.5248
tversky ($\alpha=0.3, \beta=0.7$)	0.5889	0.1079	0.5179	0.4049	0.4921	0.0606	0.5645	0.3724
tversky ($\alpha=0.4, \beta=0.6$)	0.5657	0.1015	0.5257	0.3976	0.4707	0.0563	0.5627	0.3632
tversky + BCE (0.3, 0.7)	0.6169	0.5487	0.5489	0.5715	0.5252	0.5122	0.5709	0.5361
tversky + BCE (0.4, 0.6)	0.6236	0.5712	0.6427	0.6125	0.5091	0.4850	0.5467	0.5136
Hausdorff-Dice ($\beta=0.1$)	0.5763	0.5830	0.6076	0.5890	0.4638	0.4191	0.5158	0.4662

Takeaway. No variant beats Dice + BCE: plain Tversky collapses fold F1, the Tversky + BCE versions only match the baseline, and Hausdorff-Dice falls clearly below it. Crucially, this sweep covers *both* losses from Chandran and Reghunath [7], and Dice + BCE is by far the better of the two on our data. It is also the most robust choice here, and the one most likely to generalize to the cross-scanner and cross-domain settings that follow. For these reasons and consistency we kept Dice + BCE for both models.

B.3 Architecture and Trainer (3D U-Net)

Starting from the fixed Dice + BCE loss established above and the plain 3D U-Net baseline, we varied single architecture or trainer components one at a time, keeping all else fixed. The components swept were guided by the widely cited ConvNeXt study [23], which isolates the individual design choices that modernize a convolutional backbone. We tested the standalone effect of those most relevant to our setting.

Table B.3: Architecture/trainer sweep on the 3D U-Net, D3T-B 3-fold CV. Each row changes one component of the DiceBCE baseline. **Bold** exceeds the baseline in that column; * marks a collapsed fold.

Config	Validation				Test			
	F0	F1	F2	Mean	F0	F1	F2	Mean
dicebce_baseline	0.6335	0.5878	0.6493	0.6235	0.5125	0.4811	0.5809	0.5248
dicebce_residual	0.6643	0.6218	0.6846	0.6569	0.4920	0.4985	0.6133	0.5346
lr_3e3	0.6557	0.6234	0.6560	0.6450	0.5088	0.5589	0.5959	0.5545
inv_bottleneck	0.6782	0.6397	0.5698	0.6292	0.5434	0.5070	0.6200	0.5568
wide_bn	0.6389	0.5827	0.6620	0.6279	0.5117	0.4814	0.6122	0.5351
additive_skip	0.6415	0.5892	0.6217	0.6175	0.4952	0.5563	0.5500	0.5338
attention	0.6129	0.5971	0.6409	0.6170	0.5234	0.4429	0.5779	0.5147
se_skip	0.5619	0.5768	0.6024	0.5804	0.4912	0.5023	0.5605	0.5180
convnext_block	0.5795	0.5738	0.5441	0.5658	0.5112	0.4977	0.5754	0.5281
aniso_stem	0.6307	0.5441	0.6112	0.5953	0.4761	0.4381	0.5465	0.4869
lr_3e2	0.6265	0.5514	0.6527	0.6102	0.4620	0.4550	0.5703	0.4958
adamw_cosine	0.4990	0.4903	0.4237	0.4710	0.4284	0.4245	0.4032	0.4187
gelu_ln	0.4664	0.2625	0.5295	0.4194	0.3998	0.2478	0.5465	0.3980
deeper	0.5803	0.0000*	0.6253	0.4019	0.4814	0.0625*	0.5790	0.3743
batch_2	0.6095	0.0000*	0.6356	0.4150	0.4951	0.0625*	0.5977	0.3851
dw_sep	0.0000*	0.5159	0.5065	0.3408	0.0625*	0.4360	0.5552	0.3512
cbam_skip	0.0000*	0.4652	0.3968	0.2873	0.0625*	0.3806	0.4340	0.2924
cosine_warm	0.5801	0.0112*	0.1111*	0.2341	0.5077	0.0048*	0.0000*	0.1708
large_kernel_37	0.5026	0.0000*	0.1111*	0.2046	0.3628	0.0625*	0.0000*	0.1418

Takeaway. Several variants clear the baseline. As discussed, the test folds are sensitive to a few edge cases that swing the mean, so we based design choices on the more robust validation results. On those, we kept the simplest changes that helped on every fold without collapse: residual blocks and the higher learning rate, combined into the residual_lr3e3 configuration used for the final 3D U-Net and the augmentation sweep (Appendix B.4). The other above-baseline variants gave only marginal gains and were dropped to keep the model simple; combining the two winning changes with the inverted bottleneck did not hold up, confirming it was not a robust choice. The rest underperformed or collapsed a fold.

The adamw_cosine row collapsed, but it changed two things at once: the optimizer (SGD $\rightarrow$ AdamW) and the schedule (Poly $\rightarrow$ cosine). The collapse was caused by AdamW at too high a learning rate (10^{-3}), not by cosine. Both final model configurations do include warm-up + cosine, but with the right optimizer and learning rate per model (SGD at 3×10^{-3} for the 3D U-Net, AdamW at 5×10^{-5} for the 2.5D model).

B.4 Augmentation (Stage 2, 3D U-Net)

With the architecture frozen, we swept augmentation recipes to find what kind of augmentation are useful for the single-pass 3D model. The baseline here is the frozen configuration, with both the final loss and architecture fixed. The test means of the baseline in this table are in fact lower than in the earlier sweeps even though validation holds up, which again reflects the inter-run variability of these folds. This variability is what motivated adding EMA weight-averaging for the final runs, smoothing out the late-training fluctuations that the bare checkpoints are subject to.

Table B.4: Augmentation sweep on the 3D U-Net (residual_lr3e3 backbone), D3T-B 3-fold CV. **Bold** exceeds the residual_lr3e3 baseline in that column. A dash marks a fold not run; near-zero entries indicate collapse.

Config	Validation				Test			
	F0	F1	F2	Mean	F0	F1	F2	Mean
residual_lr3e3	0.6737	0.6475	0.6609	0.6607	0.4790	0.4902	0.5840	0.5177
flip_rot90_200e	0.6830	0.6394	0.6946	0.6723	0.5057	0.4813	0.6629	0.5500
flip_rot90	0.6472	0.6414	0.7043	0.6643	0.5409	0.5101	0.6557	0.5689
flip_noise_p1	0.6922	0.6359	0.6678	0.6653	0.5180	0.4977	0.6237	0.5465
flip_rot90_g_p1_n_p1_200e	0.6526	0.6251	0.6853	0.6543	0.5088	0.5505	0.6560	0.5718
flip_gamma_p3	0.6453	0.6305	0.6868	0.6542	0.5404	0.5071	0.6195	0.5557
flip_noise_p2	0.6553	0.6393	0.5815	0.6254	0.5437	0.5529	0.6521	0.5829
flip_gamma_p2	0.6333	0.6222	0.5695	0.6083	0.5426	0.5775	0.6454	0.5885
flip_rot90_noise_p5	0.6542	0.0000*	0.6801	0.4448	0.5298	0.0625*	0.6441	0.4121
flip_rot90_adc_shift	0.5654	0.0000*	0.5281	0.3645	0.5040	0.0625*	0.5474	0.3713

Takeaway. Light recipes (flip, rot90, low-probability gamma/noise) give a small, consistent lift, while stronger recipes do not help and the most aggressive ones (noise_p5, adc_shift) collapse a fold. On validation alone the best config uses only flip and rotation, but since this baseline is meant to be retrained on other cohorts, we favored a recipe that generalizes rather than one peaking on D3T-B: the test column shows the added gamma and noise help there. The final 3D U-Net therefore keeps flip, rotation, and light gamma/noise.

B.5 K -Mask (2.5D SegResNet)

The 2.5D SegResNet was reconstructed from Chandran and Reghunath [7], whose input is three ADC and three Z_{ADC} slices (six channels). We add the binary K -mask as a seventh channel: the prior introduced in the 3D U-Net of Kim [18] and reused in our own 3D model. Table B.5 shows the effect on D3T-B; the K threshold itself is tuned separately in Appendix B.1.

Table B.5: 2.5D SegResNet, K -mask channel, D3T-B 3-fold CV. segresnet_paper reproduces the original six-channel input; segresnet_paper_k adds the K -mask as a seventh channel.

Config	Validation				Test			
	F0	F1	F2	Mean	F0	F1	F2	Mean
segresnet_paper (no K)	0.4735	0.4753	0.4387	0.4625	0.3754	0.4020	0.4447	0.4074
segresnet_paper_k (with K)	0.6200	0.5817	0.5589	0.5869	0.5129	0.4916	0.5559	0.5201

Takeaway. The K channel is decisive, lifting the reconstructed 2.5D model from 0.46 to 0.59 mean validation Dice, and it was retained.

B.6 Final Augmentation Choice

To close, we summarize how these sweeps fed into the final augmentation and training choices reported in Sections 4.1.2 and 4.1.3. The 3D U-Net had no augmentations to inherit, so we built our own and kept the recipe favored by the sweep (Appendix B.4): flips, rotation, and light gamma/noise. The 2.5D SegResNet came with a heavier recipe inherited from Chandran and Reghunath [7], which we complemented with the nnU-Net augmentation pipeline [15] (final configuration in Table 4.6). Although nnU-Net itself failed on the Boston data, we determined this was due to its patch-based design rather than its augmentations (Section 4.1.1), so the augmentations were safe to reuse. This also kept the 2.5D tuning effort low, which suited our goal of a robust baseline rather than a standalone model that must exceed the state of the art.

The ablation sweeps also motivated two additional final trainer choices: the collapsed folds seen in several runs led us to add a linear warm-up, which prevents the early divergence behind those collapses, and EMA weight-averaging, which averages the late-training weights into a stable final model. With EMA final weights are always the EMA shadow at the last epoch.,

Per-patient ZADC Map Construction Details

C.1 ADC Recomputation from DWI

Several stages of the ZADC pipeline recompute ADC from the patient’s own $b=0$ and $b=1000$ DWI rather than using the provided ADC map, both for lesion masking (Section 4.2.2) and for the atlas-space statistics in Phase 2 (Section 4.2.4). For each patient, we build an ADC map from the DWI pair after isotropic Gaussian smoothing at $\sigma = 2$ mm. Writing $b_0(\mathbf{x})$ and $b_{1000}(\mathbf{x})$ for the smoothed $b=0$ and $b=1000$ images at voxel $\mathbf{x}$, the ADC is computed as

$$\text{ADC}(\mathbf{x}) = -\ln\left(\text{clip}\left(\frac{b_{1000}(\mathbf{x})}{b_0(\mathbf{x})}, 10^{-6}, 2.0\right)\right) \cdot \frac{10^6}{B}, \quad (\text{C.1})$$

where $B = 1000 \text{ s/mm}^2$ is the diffusion weighting and the clip bounds the ratio for numerical stability. The ADC is defined only at voxels where $b_0(\mathbf{x}) > 10$ and $b_{1000}(\mathbf{x}) > 0$, and is set to 0 elsewhere. The threshold of 10 on b_0 is a noise-floor margin above pure-zero background, chosen to keep the log-ratio numerically stable.

C.2 Popdramms Intensity Normalisation

In our pipeline, Popdramms receives the per-subject masks alongside N4-corrected, reoriented $b=0$ volumes per subject. Volumes are precomputed to disable Popdramms’s default cohort-wide intensity rescaling, which during our testing distorted the SSD outlier thresholds and led to excessive outlier rejections.

C.3 Guided-Correction Ablation

The guided correction (Section 4.2.6) re-anchors confident lesion voxels toward $Z = -2.5$. Table C.1 ablates the trigger that selects these voxels, evaluated direct-inference ensemble on D1.5T-U GT-aware Z_{ADC} maps.

Any correction lifts the mean Dice and, more importantly, the worst case (min 0.12 to 0.31), while the median barely moves: the correction rescues the hardest cases rather than changing the typical one. The confidence-based triggers are interchangeable across τ_c , whereas the ADC-based trigger gives a worse lower bound (min 0.25), consistent with it dragging in borderline low-ADC voxels. We therefore use a confidence-only trigger and keep the same constant $\tau_c = 0.3$ as in the masking.

Table C.1: Guided-correction trigger ablation on D1.5T-U (GT-aware Z_{ADC} , direct-inference 2.5D). Dice over 20 cases.

Trigger	Median	[Q1, Q3]	[Min, Max]	Mean
None	0.467	[0.36, 0.57]	[0.12, 0.88]	0.472
ADC<1000	0.478	[0.38, 0.66]	[0.25, 0.88]	0.528
conf>0.2	0.472	[0.39, 0.66]	[0.31, 0.88]	0.537
conf>0.3	0.472	[0.39, 0.66]	[0.31, 0.88]	0.536
conf>0.4	0.472	[0.38, 0.66]	[0.31, 0.88]	0.535

APPENDIX D

Cross-Validation Splits

This appendix lists the subject-assignment splits across all experiments. All Boston dataset subject identifiers follow the format MGHNICU_XXX_VISIT_01; for clarity, only the numeric portion (XXX) is shown in the tables below.

D.1 D3T-B Splits

Three split configurations are used for D3T-B: a 3-fold split with a validation set for hyperparameter tuning (Table D.1), the same folds with validation subjects merged into training for final evaluation (Table D.2), and a single all-48 split with no held-out test set, used to train the frozen pretrained baseline (Table D.3).

Table D.1: Boston 3T dataset (with validation set): 3-fold cross-validation splits (48 subjects total).

Fold	Split	Subject IDs
1	Train (23)	263, 286, 305, 309, 310, 331, 346, 348, 358, 359, 362, 373, 378, 403, 432, 435, 438, 439, 441, 449, 452, 455, 456
	Val (9)	234, 264, 299, 312, 377, 393, 445, 454, 457
	Test (16)	173, 275, 303, 323, 329, 336, 355, 370, 379, 437, 440, 444, 446, 447, 448, 453
2	Train (23)	173, 264, 309, 310, 329, 336, 355, 359, 370, 377, 378, 393, 437, 440, 444, 445, 446, 448, 452, 453, 455, 456, 457
	Val (9)	275, 299, 303, 305, 323, 346, 358, 379, 447
	Test (16)	234, 263, 286, 312, 331, 348, 362, 373, 403, 432, 435, 438, 439, 441, 449, 454
3	Train (23)	234, 263, 286, 303, 312, 323, 329, 331, 336, 355, 362, 403, 437, 438, 439, 441, 444, 446, 447, 448, 449, 453, 454
	Val (9)	173, 275, 348, 370, 373, 379, 432, 435, 440
	Test (16)	264, 299, 305, 309, 310, 346, 358, 359, 377, 378, 393, 445, 452, 455, 456, 457

Table D.2: Boston 3T dataset (without validation set): 3-fold cross-validation splits (48 subjects total). The validation subjects from Table D.1 are merged into the training set.

Fold	Split	Subject IDs
1	Train (32)	263, 286, 305, 309, 310, 331, 346, 348, 358, 359, 362, 373, 378, 403, 432, 435, 438, 439, 441, 449, 452, 455, 456, 234, 264, 299, 312, 377, 393, 445, 454, 457
	Test (16)	173, 275, 303, 323, 329, 336, 355, 370, 379, 437, 440, 444, 446, 447, 448, 453
2	Train (32)	173, 264, 309, 310, 329, 336, 355, 359, 370, 377, 378, 393, 437, 440, 444, 445, 446, 448, 452, 453, 455, 456, 457, 275, 299, 303, 305, 323, 346, 358, 379, 447
	Test (16)	234, 263, 286, 312, 331, 348, 362, 373, 403, 432, 435, 438, 439, 441, 449, 454
3	Train (32)	234, 263, 286, 303, 312, 323, 329, 331, 336, 355, 362, 403, 437, 438, 439, 441, 444, 446, 447, 448, 449, 453, 454, 173, 275, 348, 370, 373, 379, 432, 435, 440
	Test (16)	264, 299, 305, 309, 310, 346, 358, 359, 377, 378, 393, 445, 452, 455, 456, 457

Table D.3: Boston 3T dataset (all 48 subjects): single training split with no held-out test set.

Split	Subject IDs
Train (48)	173, 234, 263, 264, 275, 286, 299, 303, 305, 309, 310, 312, 323, 329, 331, 336, 346, 348, 355, 358, 359, 362, 370, 373, 377, 378, 379, 393, 403, 432, 435, 437, 438, 439, 440, 441, 444, 445, 446, 447, 448, 449, 452, 453, 454, 455, 456, 457
Test	—

D.2 D1.5T-B Splits

Table D.4: Boston 1.5T dataset: 3-fold cross-validation splits (28 subjects total).

Fold	Split	Subject IDs
1	Train (18)	002, 010, 015, 028, 064, 069, 071, 072, 077, 101, 109, 125, 209, 214, 262, 272, 405, 415
	Test (10)	001, 014, 062, 070, 074, 105, 144, 215, 338, 442
2	Train (19)	001, 010, 014, 028, 062, 069, 070, 072, 074, 101, 105, 125, 144, 214, 215, 272, 338, 415, 442
	Test (9)	002, 015, 064, 071, 077, 109, 209, 262, 405
3	Train (19)	001, 002, 014, 015, 062, 064, 070, 071, 074, 077, 105, 109, 144, 209, 215, 262, 338, 405, 442
	Test (9)	010, 028, 069, 072, 101, 125, 214, 272, 415

D.3 D1.5T-U Splits

Table D.5: Utrecht dataset: 3-fold cross-validation splits (20 subjects total).

Fold	Split	Subject IDs
1	Train (13)	subj3, subj7, subj8, subj21, subj51, subj52, subj53, subj54, subj56, subj58, subj62, subj65, subj68
	Test (7)	subj13, subj50, subj55, subj63, subj64, subj66, subj67
2	Train (13)	subj3, subj13, subj21, subj50, subj51, subj52, subj55, subj56, subj63, subj64, subj65, subj66, subj67
	Test (7)	subj7, subj8, subj53, subj54, subj58, subj62, subj68
3	Train (14)	subj7, subj8, subj13, subj50, subj53, subj54, subj55, subj58, subj62, subj63, subj64, subj66, subj67, subj68
	Test (6)	subj3, subj21, subj51, subj52, subj56, subj65

APPENDIX E

Supplementary Results

E.1 D3T-B Lesioned-Only Segmentation Performance

Table E.1 reports D3T-B segmentation performance on the 46 lesioned patients only, excluding the two zero-lesion cases analyzed separately in Section 5.3. It supplements the full-cohort results in Table 5.3, confirming that the cohort-level ranking is driven by those two cases rather than a genuine performance gap: once they are removed, the ensemble overtakes both single models on Dice and NSD@2mm.

Table E.1: D3T-B metrics on the 46 lesioned patients only, under 3-fold CV, reported as mean and median [Q1, Q3] (25th–75th percentile) per architecture. Cases where the model misclassified a lesioned patient as empty are kept in this view, as those represent more critical failures. Distances (MASD, HD95) are in mm; their statistics are over cases with a finite boundary metric only. The number of cases with a finite Dice vs MASD (n_{fin}) is 43/46 for the 2.5D model and 42/46 for the 3D model and ensemble.

Arch	Dice		MASD (mm)		HD95 (mm)		NSD@2mm	
	mean	median [Q1, Q3]	mean	median [Q1, Q3]	mean	median [Q1, Q3]	mean	median [Q1, Q3]
2.5D	0.56	0.56 [0.42, 0.78]	4.73	2.43 [0.98, 4.88]	20.18	20.45 [7.38, 31.23]	0.64	0.68 [0.52, 0.87]
3D	0.57	0.60 [0.42, 0.79]	3.51	1.96 [0.92, 4.11]	17.86	13.04 [4.98, 27.68]	0.67	0.71 [0.53, 0.89]
Ens.	0.58	0.61 [0.44, 0.79]	3.79	2.05 [1.00, 4.88]	18.55	13.64 [6.89, 30.23]	0.66	0.74 [0.52, 0.87]

E.2 D3T-B Hard-Case Voxel Breakdown

Table E.2 gives the per-case TP/FP/FN voxel counts for the D3T-B hard cases discussed in Section 5.3, for the 2.5D+3D ensemble.

Table E.2: D3T-B: voxel-count breakdown for hard cases (2.5D+3D ensemble). Total brain voxels, ground-truth lesion voxels, and TP/FP/FN predictions.

Case	Brain vox	GT vox	TP	FP	FN
MGHNICU_329	42,854	0	0	33	0
MGHNICU_432	28,117	0	0	1	0
MGHNICU_234	55,146	36459	34657	666	1802
MGHNICU_331	50,740	14	0	0	14
MGHNICU_377	33,772	14	0	0	14
MGHNICU_448	36,640	11	0	0	11
MGHNICU_453	42,330	13	0	0	13
MGHNICU_370	59,557	24	17	79	7
MGHNICU_447	40,610	21	0	20	21
MGHNICU_452	38,943	13	9	8	4
MGHNICU_439	29,661	106	26	42	80

E.3 D1.5T-B Lesioned-Only Segmentation Performance

Table E.3 reports D1.5T-B segmentation performance on the 25 lesioned patients only, excluding the three zero-lesion cases that dominate the cohort-level variance discussed in Section 5.4. It supplements the full-cohort results in Table 5.6, showing that from-scratch training becomes competitive with fine-tuning and direct-inference once these cases are removed, while fine-tuning retains an edge on boundary precision over direct-inference.

Table E.3: D1.5T-B metrics on the 25 lesioned patients only, reported as mean and median [Q1, Q3] (25th–75th percentile) per architecture, for *from-scratch*, *fine-tuned*, and *direct-inference* training. Distances (MASD, HD95) are in mm; their statistics are over cases with a finite boundary metric only. All 25 cases have a finite Dice vs MASD ($n_{\text{fin}} = 25/25$ in every condition).

Arch	Dice		MASD (mm)		HD95 (mm)		NSD@2mm	
	mean	median [Q1, Q3]	mean	median [Q1, Q3]	mean	median [Q1, Q3]	mean	median [Q1, Q3]
<i>From scratch (cv)</i>								
2.5D	0.68	0.76 [0.45, 0.90]	1.93	1.04 [0.57, 2.19]	10.38	4.57 [3.39, 14.27]	0.74	0.84 [0.57, 0.92]
3D	0.66	0.75 [0.49, 0.88]	2.04	1.12 [0.62, 2.24]	10.73	5.63 [2.72, 12.59]	0.73	0.83 [0.61, 0.92]
Ens.	0.69	0.76 [0.44, 0.90]	1.73	1.06 [0.53, 1.86]	9.60	4.57 [3.28, 12.15]	0.75	0.84 [0.61, 0.92]
<i>Fine-tuned (cv_ft)</i>								
2.5D	0.70	0.76 [0.47, 0.89]	1.94	1.15 [0.60, 2.29]	10.49	4.72 [3.13, 15.20]	0.76	0.83 [0.58, 0.92]
3D	0.65	0.75 [0.41, 0.82]	2.03	1.16 [0.71, 2.47]	10.37	5.63 [4.08, 12.71]	0.72	0.83 [0.55, 0.89]
Ens.	0.70	0.77 [0.45, 0.89]	2.17	1.14 [0.50, 2.02]	10.71	5.96 [2.79, 11.61]	0.76	0.84 [0.56, 0.92]
<i>Direct-inference</i>								
2.5D	0.69	0.75 [0.46, 0.90]	2.36	1.22 [0.48, 2.76]	12.30	7.83 [2.51, 20.06]	0.74	0.82 [0.60, 0.92]
3D	0.64	0.74 [0.37, 0.90]	2.53	1.26 [0.50, 3.07]	11.62	7.60 [2.95, 16.32]	0.69	0.76 [0.45, 0.93]
Ens.	0.68	0.76 [0.43, 0.90]	2.47	1.22 [0.47, 2.85]	12.28	7.42 [2.51, 18.51]	0.73	0.77 [0.58, 0.93]

E.4 Weak-Supervision Cluster Counts

Per-patient cluster counts (Utrecht $n=20$, $lf0.7$)

	True positives			False positives		
	Full-sup.	Weak-sup.	Zero-shot HIL	Full-sup.	Weak-sup.	Zero-shot HIL
subj52 (532)	107	426	430	1	23	60
subj21 (289)	82	204	207	9	2	10
subj64 (114)	90	91	93	11	1	23
subj65 (113)	89	95	97	11	2	9
subj8 (83)	78	68	72	50	6	12
subj55 (80)	65	68	69	22	5	9
subj63 (68)	45	53	55	18	1	15
subj7 (64)	57	52	52	92	0	16
subj13 (63)	55	57	59	20	2	8
subj56 (56)	47	41	45	19	1	0
subj58 (56)	41	44	44	20	1	11
subj68 (55)	38	45	46	15	0	0
subj54 (50)	32	37	37	7	1	2
subj3 (47)	34	42	43	15	7	16
subj62 (44)	37	33	36	40	2	18
subj66 (41)	37	35	35	40	5	14
subj67 (36)	35	30	35	37	1	11
subj50 (35)	29	28	28	4	0	5
subj53 (34)	22	30	30	19	4	7
subj51 (26)	25	22	23	21	0	2

Figure E.1: Per-patient true- and false-positive lesion-cluster counts on Utrecht ($n = 20$, $lf = 0.7$) for full-supervision, weak-supervision, and direct-inference HIL.

E.5 Weak-Supervision HIL Component Ablation

This appendix reports two further weak-supervision configurations that isolate the effect of each interactive component. Both vary one component relative to the main-text settings (Table 5.12): the 4-channel no-HIL block keeps the click channel but removes the lasso post-processing filter, while the 3-channel HIL block removes the click channel but keeps the lasso filter. Comparing them against the main-text configurations shows how much each component contributes on its own.

Table E.4: D1.5T-U, 3-fold CV weak-supervision sweep for the two intermediate configurations: 4-channel (click as input) without lasso post-processing, and 3-channel (click withheld) with lasso post-processing. Reported as mean and median [Q1, Q3] (25th–75th percentile); confident-lesion fraction in steps of 0.2, best row per metric in bold. Distances (MASD, HD95) in mm.

Lesion frac.	Dice $\uparrow$		MASD (mm) $\downarrow$		HD95 (mm) $\downarrow$		NSD@2mm $\uparrow$	
	mean	median [Q1, Q3]	mean	median [Q1, Q3]	mean	median [Q1, Q3]	mean	median [Q1, Q3]
<i>4-channel, no HIL post-processing</i>								
0.0	0.59	0.57 [0.47, 0.69]	2.29	2.03 [1.27, 2.64]	12.26	11.22 [6.41, 17.50]	0.68	0.72 [0.61, 0.80]
0.2	0.61	0.57 [0.53, 0.69]	1.77	1.63 [1.21, 2.09]	9.51	8.74 [6.31, 12.13]	0.70	0.71 [0.62, 0.78]
0.4	0.64	0.60 [0.57, 0.70]	1.47	1.37 [1.05, 1.57]	7.97	7.24 [5.60, 9.31]	0.74	0.75 [0.67, 0.82]
0.6	0.66	0.65 [0.59, 0.71]	1.17	1.07 [0.91, 1.37]	6.54	5.42 [4.68, 8.10]	0.79	0.79 [0.74, 0.85]
0.8	0.68	0.68 [0.61, 0.73]	0.94	0.93 [0.83, 1.04]	4.84	4.38 [4.05, 5.33]	0.83	0.83 [0.79, 0.85]
1.0	0.70	0.70 [0.63, 0.75]	0.78	0.72 [0.57, 0.92]	3.86	3.79 [3.52, 4.08]	0.86	0.86 [0.83, 0.91]
<i>3-channel, click + lasso HIL post-processing</i>								
0.0	0.62	0.62 [0.51, 0.71]	1.53	1.29 [0.91, 1.77]	7.39	6.07 [4.49, 8.16]	0.74	0.76 [0.69, 0.84]
0.2	0.63	0.58 [0.55, 0.75]	1.48	1.46 [1.07, 1.61]	7.57	7.46 [4.85, 9.83]	0.73	0.75 [0.67, 0.81]
0.4	0.66	0.63 [0.58, 0.76]	1.19	1.06 [0.86, 1.45]	5.97	5.11 [4.29, 6.79]	0.77	0.79 [0.72, 0.84]
0.6	0.67	0.63 [0.60, 0.77]	1.03	0.99 [0.78, 1.29]	5.14	4.38 [4.12, 6.08]	0.80	0.80 [0.74, 0.85]
0.8	0.69	0.67 [0.60, 0.76]	0.90	0.84 [0.74, 1.11]	4.52	4.37 [4.00, 5.09]	0.82	0.84 [0.78, 0.86]
1.0	0.69	0.67 [0.62, 0.78]	0.83	0.81 [0.69, 0.95]	3.96	4.00 [3.63, 4.48]	0.83	0.85 [0.81, 0.87]

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